

Transport of Sodium, Chloride, and Potassium

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CHAPTER OUTLINE

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KEY POINTS

- Claudin-10 may modulate the paracellular permeability of sodium along the thick ascending limb. Claudin-10 mutations cause salt-wasting with hypokalemic alkalosis in humans.
- Through modulation of the phosphorylation status of the $\text{Na}^+\text{-Cl}^-$ cotransporter (NCC), the distal convoluted tubule acts as a “sensor” of plasma $[\text{K}^+]$ that may also signal to downstream K^+ -secreting segments to modify their structure.
- The ubiquitin ligase NEDD4-2 exerts effects on ENaC and also mediates proteasomal degradation of NCC, which may be important during potassium restriction.
- Disruption of NHE3 specifically in the renal epithelia in mice revealed that it plays a role in sodium balance and blood pressure maintenance when dietary salt intake is modified. It also points to NHE3 playing a more important role in the gut.
- Data have revealed that uromodulin (also known as Tamm-Horsfall protein), which is the most abundant protein excreted in normal urine, plays a central role in modulating the activities of both NKCC2 and ROMK along the thick ascending limb.
- WNK4 and KS-WNK1, members of the WNK kinase family, are now believed to primarily activate NCC rather than inhibit it, though in some contexts they may exert inhibitory effects.
- The combined activity of SLC4A8 and SLC26A4, expressed in intercalated cells, drives net $\text{Na}^+\text{-Cl}^-$ reabsorption along the cortical collecting duct, permitting increased Na^+ reabsorption without altering K^+ secretion.

SODIUM AND CHLORIDE TRANSPORT

Daily sodium (Na^+) intake for adults in the United States is approximately 180 mmol (4.2 g) for men and 150 mmol (3.5 g) for women (<https://www.cdc.gov/nchs/nhanes/wwc.htm>). As Na^+ is the principal osmole in extracellular fluid, the total body content of Na^+ and chloride (Cl^-), its primary anion, determines the extracellular fluid volume. Renal excretion or retention of salt (NaCl) is thus the major determinant of the extracellular fluid volume, such that genetic loss or gain of function in renal NaCl transport can be associated with relative hypotension or hypertension, respectively. On a quantitative level, at a glomerular

filtration rate of 180 L/day and serum Na^+ of about 140 mmol/L, the kidney filters some 25,000 mmol/day of Na^+ ; this is equivalent to about 1.5 kg of salt, which would occupy roughly 10 times the extracellular space.¹ Small changes in renal NaCl excretion can thus have massive effects on the extracellular fluid volume. In addition, 99.6% of filtered NaCl must be reabsorbed to excrete 40 to 220 mmol/day. Energetically, renal reabsorption of Na^+ consumes 1 molecule of ATP per 5 molecules of Na^+ .¹ This is economical, given that the absorption of NaCl is primarily driven by basolateral $\text{Na}^+\text{-K}^+\text{-ATPase}$, which has a stoichiometry of 3 molecules of transported Na^+ per molecule of adenosine triphosphate (ATP).² This estimate reflects net expenditure

because the cost of transepithelial NaCl transport varies considerably along the nephron, from a predominance of passive transport by thin ascending limbs to the purely active transport mediated by the distal nephron (distal convoluted tubule, connecting tubule, and collecting duct).

As much as 60% to 70% of filtered NaCl transport is reabsorbed along the proximal tubule (PT), and approximately 25% along the loop of Henle (mainly the thick ascending limb, TAL; Fig. 6.1). Whereas the PT can theoretically absorb as many as 9 Na⁺ molecules for each hydrolyzed ATP, paracellular Na⁺ transport by the TAL doubles the efficiency of transepithelial NaCl transport (6 Na⁺ per ATP).^{1,3} By the time filtered fluid reaches the macula densa, more than 90% of filtered Na⁺ has been reabsorbed,⁴ a percentage that varies only slightly when dietary NaCl intake is low or high.⁵ Thus the late segments (distal convoluted tubule, connecting tubule, and collecting duct), while reabsorbing only 5% to 10% of filtered Na⁺, are a primary site of transport regulation. Here, renal NaCl absorption occurs at full cost (3 Na⁺ per ATP) while affording the generation of considerable transepithelial gradients.¹

The nephron thus constitutes a serial arrangement of tubule segments with considerable heterogeneity in the physiologic consequences, mechanisms, and regulation of transepithelial NaCl transport. These are reviewed in this section in anatomic order.

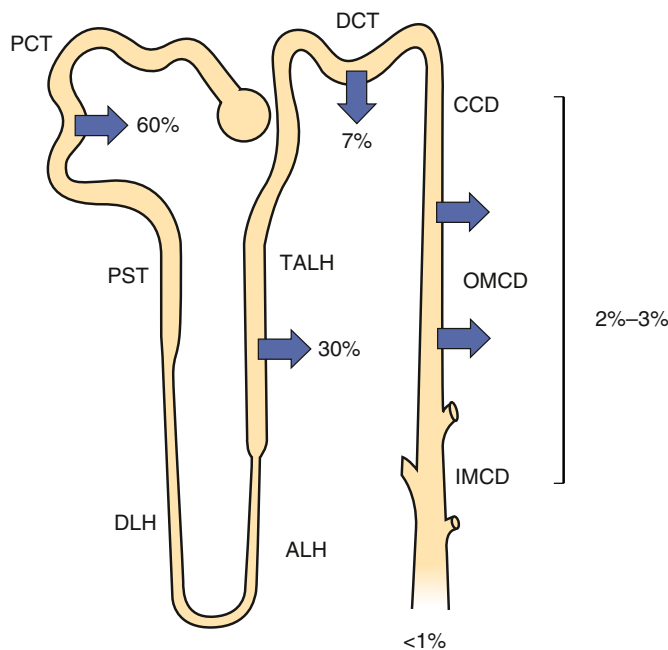


Fig. 6.1 Percentage reabsorption of filtered NaCl along the eu-volemic nephron. ALH, Ascending limb of the loop of Henle; CCD, cortical collecting duct; DCT, distal convoluted tubule; DLH, descending thin limb of the loop of Henle; IMCD, inner medullary collecting duct; OMCD, outer medullary collecting duct; PCT, proximal convoluted tubule; PST, proximal straight tubule; TALH, thick ascending limb of the loop of Henle. (From Moe OW, Baum M, Berry CA, Rector FC Jr. Renal transport of glucose, amino acids, sodium, chloride, and water. In: Brenner BM, ed: *Brenner and Rector's the Kidney*. Philadelphia: WB Saunders; 2004. p. 413–452.)

PROXIMAL TUBULE

The PT performs near-isosmotic reabsorption of two-thirds to three-quarters of the glomerular ultrafiltrate. This encompasses the reabsorption of at least 60% of filtered Na⁺ with accompanying anions (see Fig. 6.1), such that this nephron segment plays a critical role in the maintenance of extracellular fluid volume. The transport characteristics and capacity of early, mid, and late segments of the PT differ considerably, and the volume of transported fluid and solutes declines along the PT. This corresponds to distinct ultrastructural characteristics in the tubular epithelium. Cells of the S1 segment (early proximal convoluted tubule) are characterized by a tall brush border, with extensive lateral invaginations of the basolateral membrane.⁶ Numerous elongated mitochondria are located in lateral cell processes, with a proximity to the plasma membrane that is characteristic of epithelial cells involved in active transport. The ultrastructure of the S2 segment (late proximal convoluted tubule and beginning of the proximal straight tubule) is similar, but with a shorter brush border, fewer lateral invaginations, and less prominent mitochondria. In epithelial cells of the S3 segment (remainder of the proximal straight tubule), lateral cell processes and invaginations are essentially absent, with small mitochondria that are randomly distributed within the cell.⁶ The extensive brush border of proximal tubular cells amplifies the apical cell surface available for reabsorption. This amplification is axially distributed, increasing the apical area 36-fold in S1 and 15-fold in S3.⁷ At the functional level, bicarbonate reabsorption rates decline by at least 80% between the first and last portions of the PT, whereas Cl[−] reabsorption declines by approximately 50%.⁸

There is also considerable axial heterogeneity in the quantitative capacity of the PT for organic solutes such as glucose and amino acids, with predominant reabsorption of these substrates in S1 segments.⁹ The Na⁺-dependent reabsorption of glucose, amino acids, and other solutes in S1 segments results in a transepithelial potential difference (PD) that is initially lumen negative due to electrogenic removal of Na⁺ from the lumen (Fig. 6.2).¹⁰ This is classically considered the first phase of volume reabsorption by the PT.¹¹ The lumen-negative PD serves to drive both paracellular Cl[−] absorption and a backleak of Na⁺ from the peritubular space to the lumen. Paracellular Cl[−] absorption in this setting accomplishes the net transepithelial absorption of a solute such as glucose, along with equal amounts of Na⁺ and Cl[−]; in contrast, backleak of Na⁺ leads only to reabsorption of the organic solute, with no net transepithelial transport of Na⁺ or Cl[−]. The amount of Cl[−] reabsorption driven by this lumen-negative PD thus depends on the relative permeability of the paracellular pathway to Na⁺ and Cl[−]. There appears to be considerable heterogeneity in the relative paracellular permeability to Na⁺ and Cl[−]; for example, whereas superficial proximal convoluted tubules and proximal straight tubules in the rabbit are Cl[−] selective, juxtamedullary PTs in this species are reportedly Na⁺ selective.^{12,13} Regardless, the component of paracellular Cl[−] transport that is driven by this lumen-negative PD is restricted to the early PT.

The second phase of volume reabsorption by the PT is dominated by NaCl reabsorption via paracellular and transcellular pathways.¹¹ In addition to the Na⁺-dependent

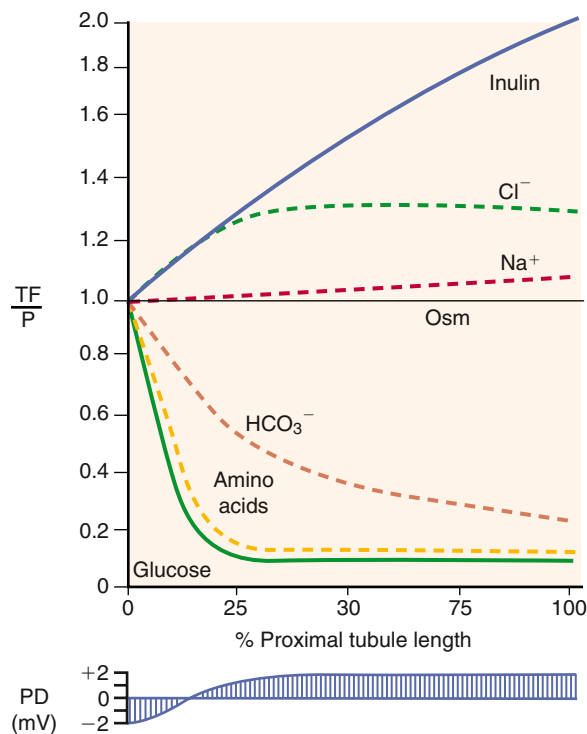


Fig. 6.2 Concentration of solutes along the proximal tubule in relation to the transepithelial potential difference (PD). *Osm*, Osmolality; *TF/P*, ratio of tubule fluid-to plasma-concentration. (From Rector FC Jr. Sodium, bicarbonate, and chloride absorption by the PT. *Am J Physiol.* 1983;244:F461–F471.)

reabsorption of organic solutes, the early PT has a much higher capacity for HCO_3^- absorption via the coupling of apical $\text{Na}^+\text{-H}^+$ exchange, carbonic anhydrase, and basolateral $\text{Na}^+\text{-HCO}_3^-$ cotransport.⁹ As the luminal concentrations of HCO_3^- and other solutes begin to drop, the concentration of NaCl rises to a value greater than that of the peritubular space.¹⁴ This is accompanied by a reversal of the lumen-negative PD to a lumen-positive value generated by passive Cl^- diffusion (see Fig. 6.2).¹⁵ This lumen-positive PD serves to drive paracellular Na^+ transport, whereas the chemical gradient between the lumen and peritubular space provides the driving force for paracellular reabsorption of Cl^- . This passive paracellular pathway is thought to mediate about 40% of transepithelial NaCl reabsorption by the mid to late PT.¹² Of note, however, there may be heterogeneity in the relative importance of this paracellular pathway, with evidence that active (i.e., transcellular) reabsorption predominates in proximal convoluted tubules from juxtamedullary versus superficial nephrons.¹⁶ Regardless, the combination of passive and active transport of NaCl explains how the PT is able to reabsorb about 60% of filtered NaCl , despite $\text{Na}^+\text{-K}^+\text{-ATPase}$ activity that is considerably lower than that of distal segments of the nephron (Fig. 6.3).¹⁷

The transcellular component of NaCl reabsorption initially emerged from studies of the effect of cyanide, ouabain, luminal anion transport inhibitors, cooling, and luminal-peritubular K^+ removal.¹¹ For example, the luminal addition of SITS (4-acetamido-4'-isothiocyanostilbene-2,2'-disulfonic acid), an inhibitor of anion transporters, reduces volume reabsorption of proximal convoluted tubules perfused with

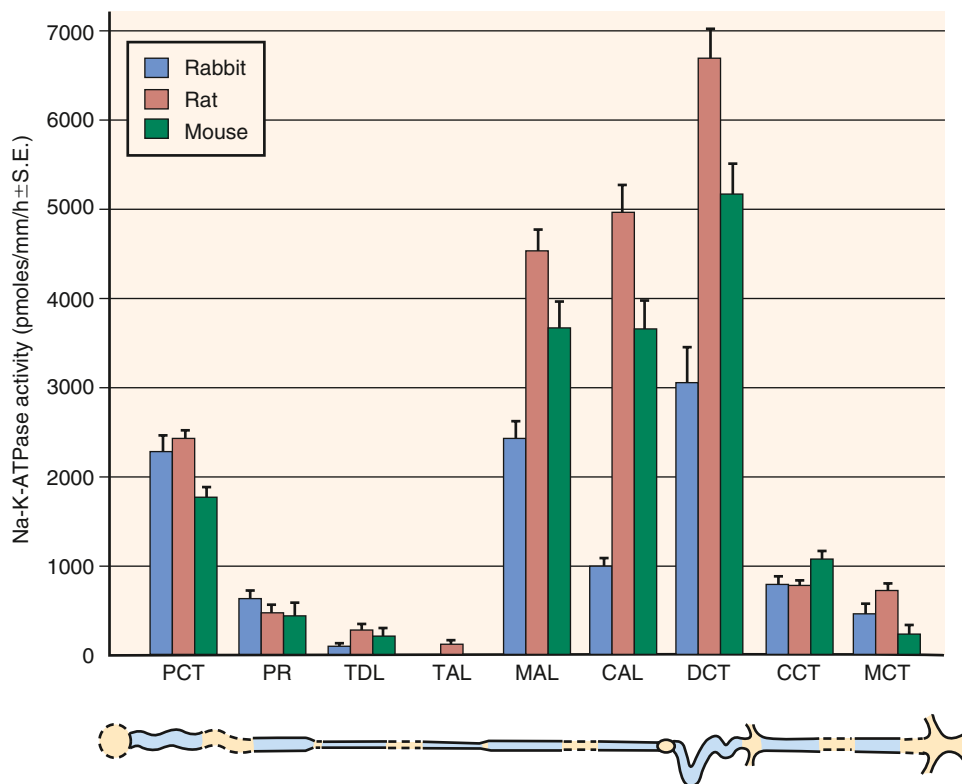


Fig. 6.3 Distribution of $\text{Na}^+\text{-K}^+\text{-ATPase}$ activity along the nephron. *cTAL*, Cortical thick ascending limb; *CCD*, cortical collecting duct; *DCT*, distal convoluted tubule; *mTAL*, medullary thick ascending limb; *MCD*, medullary collecting duct; *PCT*, proximal convoluted tubule; *PST*, proximal straight tubule; *tAL*, thin ascending limb of the loop of Henle; *tDL*, descending thin limb of the loop of Henle. (From Katz AI, Doucet A, Morel F. Na-K-ATPase activity along the rabbit, rat, and mouse nephron. *Am J Physiol.* 1979;237:F114–F120.)

a high Cl^- /low HCO_3^- solution that mimics the luminal composition of the late PT; this occurs in the absence of an effect on carbonic anhydrase.¹⁴ This transcellular component of NaCl reabsorption is clearly electroneutral. In the absence of anion gradients across the perfused PT, there is no change in transepithelial PD after the inhibition of active transport by ouabain, despite a marked reduction in volume reabsorption.¹⁸ Transcellular NaCl reabsorption is accomplished by the coupling of luminal $\text{Na}^+\text{-H}^+$ exchange or $\text{Na}^+\text{-SO}_4^{2-}$ cotransport with a heterogeneous population of anion exchangers, as reviewed later.

PARACELLULAR NaCl TRANSPORT

Several factors serve to optimize the conditions for paracellular NaCl transport by the mid to late PT. First, the PT is a low-resistance, so-called leaky epithelium, with tight junctions that are highly permeable to both Na^+ and Cl^- .^{12,13} Second, these tight junctions are preferentially permeable to Cl^- over HCO_3^- , a feature that helps generate the lumen-positive PD in the mid to late PT.¹⁴ Third, the increase in luminal NaCl concentrations in the mid to late PT generates a chemical driving force for paracellular reabsorption of Cl^- .¹⁵ This increase in luminal NaCl is the direct result of the robust reabsorption of HCO_3^- and other solutes by the early S1 segment, combined with the isosmotic reabsorption of filtered water.^{9,19}

Morphologically, the apical tight junction of proximal tubular cells is shallow and has as few as one junctional strand (Fig. 6.4), contrasting with the deeper junctions with extensive junctional strands seen in high-resistance tight epithelia.²⁰

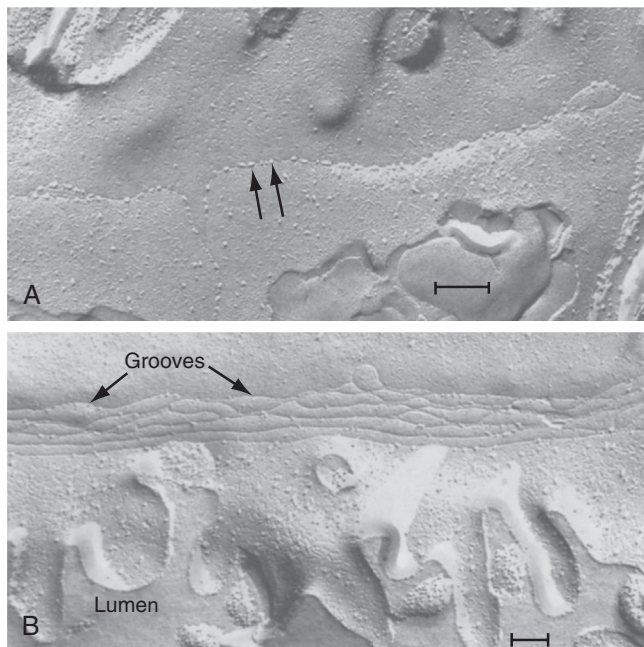


Fig. 6.4 Freeze-fracture electron microscopy images of tight junctions in mouse proximal and distal nephron. (A) Proximal convoluted tubule, a “leaky” epithelium; the tight junction contains only one junctional strand, seen as a groove in the fracture face (arrows). (B) Distal convoluted tubule, a “tight” epithelium. The tight junction is deeper and contains several anastomosing strands, seen as grooves in the fracture face. (From Claude P, Goodenough DA. Fracture faces of zonulae occludentes from “tight” and “leaky” epithelia. *J Cell Biol.* 1973;58:390–400.)

Tight junctions of epithelia function as the paracellular barrier. The charge and size selectivity of tight junctions appears to be conferred in large part by the claudins, a large (>20) gene family of tetraspan transmembrane proteins.^{21–23} The repertoire of claudins expressed by proximal tubular epithelial cells likely determines the high paracellular permeability of this nephron segment. At a minimum, proximal tubular cells coexpress claudin-2, claudin-10, claudin-12, and perhaps claudin-17.^{14,24,25}

Overexpression of claudin-2 lowers the resistance²⁶, increases Na^+ permeability, and increases water flux in epithelial cell lines²⁷, suggesting that claudin-2 directly modulates paracellular Na^+ and water permeability. Consistent with this, targeted deletion of claudin-2 in mice reduced PT Na^+ , Cl^- , and fluid absorption²⁸ by lowering paracellular cation permeability with a secondary reduction in transepithelial Cl^- transport.²⁸ Despite this difference in solute reabsorption along the PT, renal sodium handling is normal because solute transport is increased along more distal segments to compensate. This requirement for more transcellular solute transport along the loop of Henle contributes to lower medullary oxygen tension and increased susceptibility to renal ischemia.²⁹ Terminal differentiation of proximal tubular claudin-2 expression requires the integrin β_1 -subunit, since its disruption in mice converts the PT to a tight epithelium expressing low levels of claudin-2.³⁰

One study demonstrated that claudin-10a is the major paracellular anion-selective pathway in the PT.³¹ RNA sequencing and Western blotting of isolated PT and immunofluorescence on whole kidney sections from wild type and claudin-10a knockout mice confirmed claudin-10a expression along PT. Electrophysiology showed that claudin-10a disruption led to loss of paracellular anion movement, with claudin-10a showing a permeability of $\text{Cl}^- > \text{HCO}_3^-$. Claudin-17 has also been found to generate a predominantly anion-selective paracellular conductance in vitro and has been detected in the PT.²⁵

The reabsorption of HCO_3^- and other solutes from the glomerular ultrafiltrate would be expected to generate an osmotic gradient across the epithelium, resulting in a hypotonic lumen. This appears to be the case, although the absolute difference in osmolality between the lumen and peritubular space has been a source of considerable controversy.¹⁹ Water transport from this hypotonic lumen is primarily transcellular, predominantly via aquaporin-1³², with possibly some paracellular reabsorption through claudin-2.³³ The relative importance of diffusional versus convective (solvent drag) transport of NaCl across the paracellular tight junction remains a matter of debate; convective transport of NaCl with water would seem to play a lesser role than diffusion, given the evidence that the transcellular pathway is the dominant transepithelial pathway for water in the PT.^{12,19,32,34} However, while the evidence does not support a role for convective movement of NaCl or Cl^- alone,³⁵ convective movement of Na^+ alone, through claudin-2, is likely.^{27,35,36}

TRANSCELLULAR NaCl TRANSPORT

APICAL MECHANISMS

Apical $\text{Na}^+\text{-H}^+$ exchange plays a critical role in the transcellular and paracellular reabsorption of NaCl by the PT. Furthermore, $\text{Na}^+\text{-H}^+$ exchange plays a key role in the

functional “absorption” of HCO_3^- by the early PT (HCO_3^- does not actually move across the apical membrane but is generated within cells with H^+ via carbonic anhydrase activity see Chapter 9). As the movement of Na^+ and HCO_3^- drives osmotic water movement, it increases the luminal concentration of Cl^- , driving passive paracellular transport of Cl^- .³⁷ Increases in luminal Cl^- also help drive the apical uptake of Cl^- during transcellular transport. Na^+ - H^+ exchange is predominantly mediated by the NHE proteins, encoded by the nine members of the *SLC9* gene family; NHE3 in particular plays an important role in proximal tubular physiology.³⁸ NHE3 is expressed at the apical membrane of S1, S2, and S3 segments,³⁹ as are other Na^+ -dependent H^+ transporters including NHE8.^{38,40} NHE8 predominates over NHE3 in the neonatal PT, with subsequent induction of NHE3 and downregulation of NHE8 in mature, adult nephrons.³⁸ The primacy of NHE3 in mature PTs is illustrated by the renal phenotype of NHE3 knockout mice, which have a 62% reduction in proximal fluid absorption and a 54% reduction in baseline chloride absorption.^{41,42} Although the severe salt-wasting phenotype seen in global NHE3 knockout mice results largely from its deletion in intestine, NHE3 disruption in the kidney showed that it plays a role in maintaining blood pressure at baseline and plasma Na^+ in response to increased or reduced dietary

NaCl , with compensatory upregulation of the distal Na^+ reabsorptive pathways (NaCl transport, Na^+ - Cl^- -cotransporter [NCC], and epithelial Na^+ channel, ENaC).⁴³

Evidence for the involvement of an apical anion exchanger in transepithelial NaCl transport by the PT first came from the use of anion transport inhibitors. DIDS (4,4'-diisothiocyanostilbene-2,2'-disulfonic acid), furosemide, and SITS all reduce fluid absorption from the lumen of PT (PT) segments perfused with solutions containing NaCl .¹⁴ Determining the identity of this anion exchanger has been challenging. Early studies using isolated PT membrane vesicles suggested the existence of a Cl^- - OH^- exchanger,⁴⁴ but other evidence pointed toward a more dominant Cl^- -formate exchanger activity in the absence of significant Cl^- - OH^- exchange.⁴⁵ Formate transport stimulated by a pH gradient (H^+ -formate cotransport or formate- OH^- exchange) is saturable, consistent with a carrier-mediated process rather than diffusion of formic acid across the apical membrane.⁴⁶ Cl^- -oxalate and SO_4^{2-} -oxalate exchange mechanisms have also been detected in PT membrane vesicles.^{37,47} On the basis of differences in the affinities and inhibitor sensitivity of the Cl^- -oxalate and Cl^- -formate exchange activities, two separate apical exchangers, a Cl^- -formate exchanger and a Cl^- -formate-oxalate exchanger, may exist (Fig. 6.5). In isolated perfused rabbit,

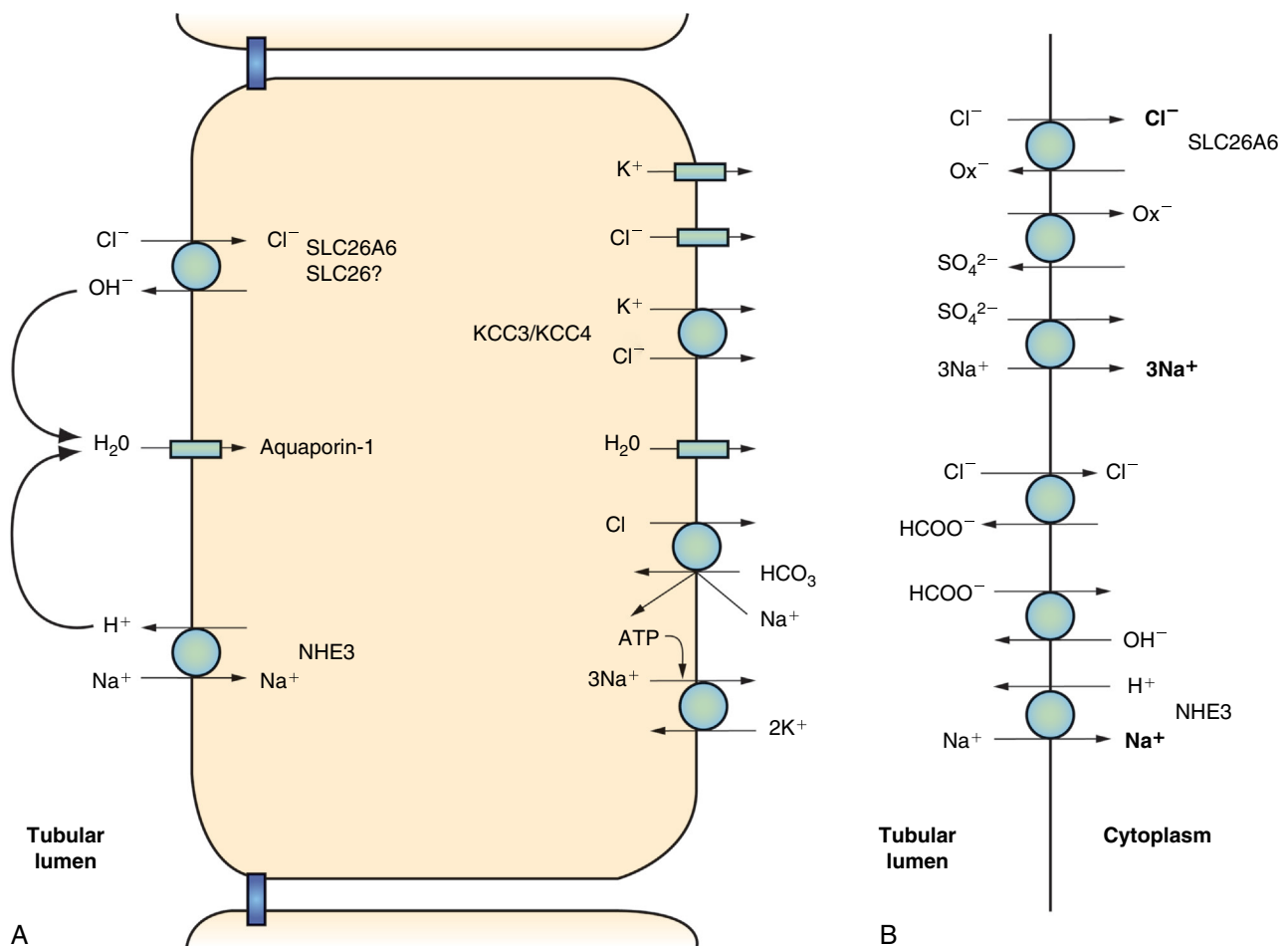


Fig. 6.5 Transepithelial NaCl transport in the proximal tubule. (A) In the simplest scheme, Cl^- enters the apical membrane via a Cl^- - OH^- exchanger, coupled to Na^+ entry via NHE3. (B) Alternative apical anion exchange activities that couple to Na^+ - H^+ exchange and Na^+ - SO_4^{2-} cotransport. (C) Alternative apical anion exchange activities that couple to Na^+ - H^+ and oxalate (Ox^-) or formate (HCOO^-) exchange. See text for details.

rat, and mouse PTs, both formate and oxalate significantly increased fluid transport.⁴² This increase in fluid transport was inhibited by DIDS, suggesting involvement of the DIDS-sensitive anion exchanger(s) detected in earlier vesicle studies. The oxalate- and formate-dependent anion transporters in the PT are likely coupled to distinct Na^+ entry pathways ($\text{Na}^+\text{-SO}_4^{2-}$ cotransport and $\text{Na}^+\text{-H}^+$ exchange, respectively).⁴⁸ Coupling of Cl^- -oxalate exchange and $\text{Na}^+\text{-SO}_4^{2-}$ cotransport requires the additional presence of SO_4^{2-} -oxalate exchange, which has been demonstrated in membrane vesicle studies.⁴⁹ The formate effect on Cl^- transport, but not the oxalate effect, is abolished in NHE3 knockout mice demonstrating the formate and $\text{Na}^+\text{-H}^+$ connection.⁴² Finally, tubular perfusion data from superficial and juxtamedullary proximal convoluted tubules suggest heterogeneity in the dominant mode of anion exchange along the PT, such that Cl^- -formate exchange is absent in juxtamedullary proximal convoluted tubule (PCTs), in which Cl^- - OH^- exchange may instead be dominant.¹⁴

While the SLC26A4 anion exchanger, also known as pendrin, is capable of Cl^- -formate exchange when expressed in *Xenopus laevis* oocytes,⁵⁰ its expression in the PT is minimal or absent in several species. Furthermore, formate-stimulated NaCl transport in this nephron segment is unimpaired in *Slc26A4* null mice.¹⁴ The role of SLC26A4 in Cl^- transport by the distal nephron is reviewed later in this chapter (see the subhead “ Cl^- Transport” under “Connecting Tubules and the Cortical Collecting Duct” later).⁵¹ Another member of the SLC26 family, SLC26A6, is present at the apical membrane of PT cells. Murine SLC26A6 when expressed in *Xenopus* oocytes, mediates the multiple modes of anion exchange that have been implicated in transepithelial NaCl by the PT including Cl^- -formate, Cl^- - OH^- , Cl^- - SO_4^{2-} , and SO_4^{2-} -oxalate exchange.⁵² However, tubule perfusion experiments in *Slc26A6* null mice did not reveal a reduction in baseline Cl^- or fluid transport,⁵³ and whole animal studies revealed no effects on NaCl homeostasis or blood pressure.⁵⁴ SLC26A6 does appear to be the dominant Cl^- -oxalate exchanger along the PT. *Slc26a6* knockout mice display reduced fractional excretion of oxalate⁵⁴ and loss of oxalate-induced tubular fluid transport and Cl^- -oxalate exchange in membrane vesicles.^{53,55} Other candidates for PT Cl^- transport include SLC26A7 and SLC26A2.^{53,56,57} SLC26A7 has been reported to act as a $\text{Cl}^-/\text{HCO}_3^-$ exchanger,⁵⁸ while SLC26A2 displays $\text{SO}_4^{2-}/\text{Cl}^-$ exchange activity.⁵⁹

SLC26A6 mediates electrogenic Cl^- - OH^- and Cl^- - HCO_3^- exchange, and most if not all the members of this family are electrogenic in at least one mode of anion transport.^{14,52,53,60,61} Electroneutrality of transcellular NaCl transport may be preserved by coexpression of two or more electrogenic SLC26 exchangers in the same membrane. Notably, the stoichiometry and electrophysiology of Cl^- -base exchange differ for individual members of the family. For example, SLC26A6 exchanges one Cl^- for two HCO_3^- anions, whereas SLC26A3 exchanges two Cl^- anions for one HCO_3^- anion.^{14,61} Apical K^+ channels in the PT may also stabilize membrane potential during NaCl absorption.⁶² One channel involved may be Kir4.2, recently identified as a major component of the basolateral K^+ conductance in PT, where it contributes to the driving force for HCO_3^- exit through the $\text{Na}^+\text{-HCO}_3^-$ cotransporter NBCe1.⁶³

Preferential coupling of Cl^- -formate exchange to $\text{Na}^+\text{-H}^+$ exchange mediated by NHE3 (see Fig. 6.5) may involve the scaffolding proteins that serve to cluster different transporters to separate membrane microdomains. Notably, while both SLC26A6 and NHE3 bind to the scaffolding protein PDZK1, distribution of SLC26A6 is selectively impaired in PDZK1 knockout mice.⁶⁴ Petrovic and colleagues⁶⁵ have also reported a novel activation of proximal $\text{Na}^+\text{-H}^+$ exchange by luminal formate, suggesting a direct effect of formate per se on NHE3 that may partly explain the preferential coupling of Cl^- -formate exchange to NHE3. Despite these intriguing observations, the relative importance of transcellular vs passive paracellular Cl^- reabsorption in the PT remains to be established with certainty.

BASOLATERAL MECHANISMS

Basolateral $\text{Na}^+\text{-K}^+$ -ATPase activity establishes the Na^+ gradient for transcellular NaCl transport by the PT and provides the major exit pathway for Na^+ . To preserve the electroneutrality of transcellular NaCl transport, basolateral Na^+ exit must be balanced by an equal exit of Cl^- .¹⁸ Several exit pathways for Cl^- have been identified in PT including $\text{K}^+\text{-Cl}^-$ cotransport, Cl^- channels, and various modalities of Cl^- - HCO_3^- exchange (see Fig. 6.5).

Several lines of evidence support the existence of a swelling-activated basolateral $\text{K}^+\text{-Cl}^-$ cotransporter (KCC) in the PT.⁶⁶ KCC proteins are encoded by four members of the cation-chloride cotransporter gene family (KCCs 1–4). KCC1, KCC3, and KCC4 are all expressed in the kidney with strong expression of KCC3 and KCC4 at the basolateral membrane of the PT.⁶⁷ Basolateral membrane vesicles isolated from the renal cortex reportedly contain $\text{K}^+\text{-Cl}^-$ cotransport activity,⁶⁶ and electroneutral $\text{K}^+\text{-Cl}^-$ cotransport has been detected at the basolateral membrane of proximal straight tubules.^{68,69} Cell swelling in response to apical Na^+ absorption is postulated to activate volume-sensitive basolateral $\text{K}^+\text{-Cl}^-$ cotransporter activity that coordinates transepithelial absorption of NaCl .¹⁴ Activation of apical Na^+ -glucose transport in proximal tubular cells strongly activates a barium-resistant (Ba^{2+}) K^+ efflux pathway that is inhibited by 1 mmol/L furosemide,⁷⁰ a concentration sufficient to inhibit activity of all KCC isoforms. In addition, a volume regulatory decrease (VRD) in Ba^{2+} -blocked PTs swollen by hypotonic conditions is blocked by 1 mmol/L furosemide.⁶⁶ Furthermore, targeted deletion of KCC3 or KCC4 in mice reduces VRD in the PT⁷¹ and perfused PTs from KCC3 knockout mice display lower transepithelial fluid transport.⁷² Together, these data suggest an important role for basolateral $\text{K}^+\text{-Cl}^-$ cotransport in transcellular NaCl reabsorption.

The basolateral Cl^- conductance of mammalian proximal tubular cells is relatively low, suggesting a lesser role for Cl^- channels in transepithelial NaCl transport. Basolateral anion substitutions have significant effects on intracellular Cl^- concentrations but minimal effects on the membrane potential; reciprocal changes in basolateral membrane potential do not affect intracellular Cl^- .^{68,69,73} Similar to basolateral $\text{K}^+\text{-Cl}^-$ cotransport, basolateral Cl^- channels in the PT may be relatively inactive in the absence of cell swelling. Cell swelling activates both K^+ and Cl^- channels at the basolateral membranes of proximal tubular cells.^{14,74,75} Seki and associates⁷⁶ reported the presence of a basolateral Cl^- channel in S3 segments of

the rabbit nephron that is resistant to the KCC inhibitor H74. The molecular identity of these and other basolateral Cl^- channels in the proximal nephron is not known with certainty, although S3 segments have been shown to express mRNA exclusively for the swelling-activated ClC-2 Cl^- channel. Its role, however, in transcellular NaCl reabsorption is not yet clear.⁷⁷

Finally, there is functional evidence for Na^+ -dependent and Na^+ -independent Cl^- - HCO_3^- exchange at the basolateral membrane of proximal tubular cells.^{13,73,78} The impact of Na^+ -independent Cl^- - HCO_3^- exchange on basolateral exit is thought to be minimal.⁷³ In contrast, there is a 75% reduction in the rate of decrease in intracellular Cl^- after the removal of basolateral Na^+ alone.⁷³ The Na^+ -dependent Cl^- - HCO_3^- exchanger may thus play a considerable role in basolateral Cl^- exit, with recycled exit of Na^+ and HCO_3^- via the basolateral Na^+ - HCO_3^- cotransporter NBC1 (see Fig. 6.5).

REGULATION OF PROXIMAL TUBULAR NaCl TRANSPORT

GLOMERULOTUBULAR BALANCE

A fundamental property of the kidney is the phenomenon of glomerulotubular balance, whereby changes in the glomerular filtration rate (GFR) are paralleled by changes in tubular reabsorption, thus maintaining a constant fractional reabsorption of fluid and NaCl (Fig. 6.6). Although the distal nephron is capable of adjusting reabsorption in response to changes in tubular flow, the impact of GFR on NaCl reabsorption by the PT is particularly pronounced (Fig. 6.7).⁷⁹ Glomerulotubular balance is independent of direct neuronal and systemic hormonal control and is thought to be mediated by the additive effects of luminal and peritubular factors.⁷⁹

Fluid shear stress (FSS) within the lumen contributes to glomerulotubular balance by increasing solute and water absorption.⁸⁰ Du and coworkers^{79,81} reported linear flow dependence of fluid and HCO_3^- transport in isolated perfused murine PTs (Fig. 6.8), mediated by NHE3 and the H^+ -ATPase, as discussed later. The estimated microvillus torque, which is a function of tubular flow, exhibits a linear relationship with fluid and HCO_3^- absorption.^{79,81} Consistent with an effect of torque rather than flow per

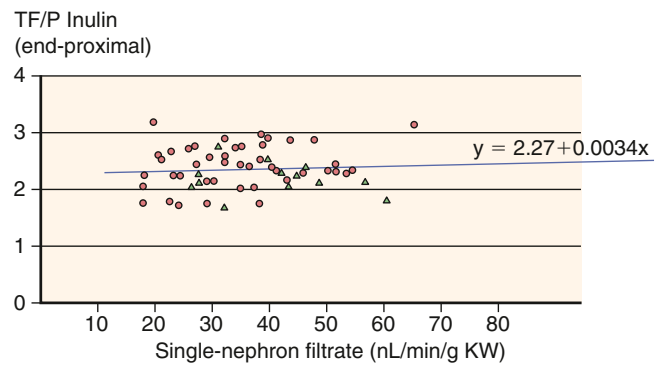


Fig. 6.6 Glomerulotubular balance. The tubular fluid-to-plasma ratio of the nonreabsorbable marker, inulin (TF/P Inulin), at the end of the proximal tubule (PT), which is used as a measure of fractional water absorption by the PT, does not change as a function of single-nephron GFR (SNGFR). Measurements were done during antidiuresis (triangles) and water diuresis (circles). (From Schnermann J, Wahl M, Liebau G, Fischbach H. Balance between tubular flow rate and net fluid reabsorption in the proximal convolution of the rat kidney. I. Dependency of reabsorptive net fluid flux upon proximal tubular surface area at spontaneous variations of filtration rate. *Pflügers Arch.* 1968;304:90–103.)

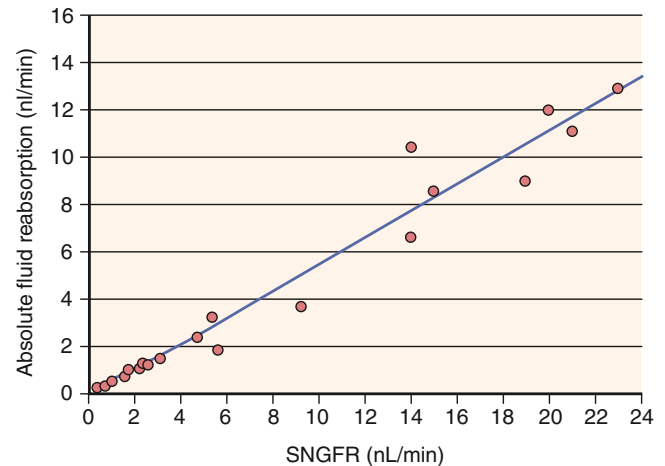


Fig. 6.7 Glomerulotubular balance. Shown is the linear increase in absolute fluid reabsorption by the late proximal tubule as a function of single-nephron GFR (SNGFR). (From Spitzer A, Brandis M. Functional and morphologic maturation of the superficial nephrons. Relationship to total kidney function. *J Clin Invest.* 1974;153:279–287.)

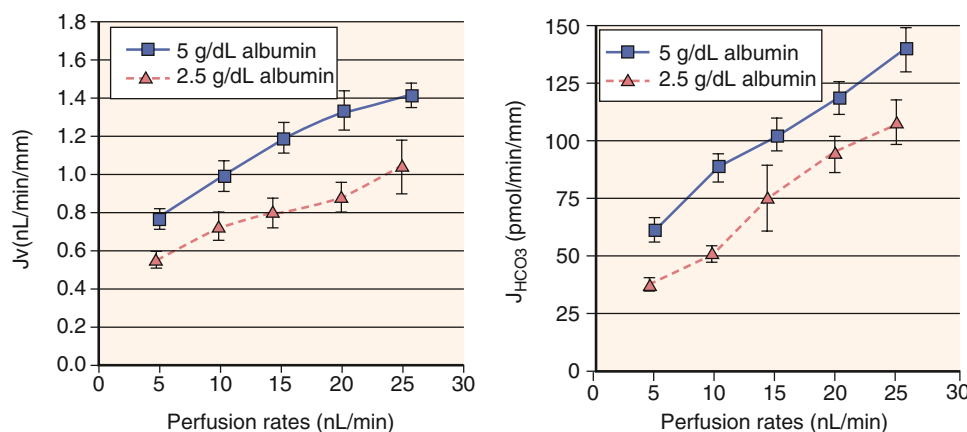


Fig. 6.8 Glomerulotubular balance; flow-dependent increases in fluid (J_v) and HCO_3^- (J_{HCO_3}) absorption by perfused mouse proximal tubules. Absorption also increases when bath albumin concentration increases from 2.5 to 5 g/dL. (From Du Z, Yan Q, Duan Y, et al. Axial flow modulates PT NHE3 and H-ATPase activities by changing microvillus bending moments. *Am J Physiol Renal Physiol.* 2006;1290:F289–F296.)

se, increasing viscosity of the perfusate by the addition of dextran increases the effect on fluid transport.

Pharmacologic inhibition or genetic ablation of NHE3 in mice reveals that tubular flow activates proximal HCO_3^- reabsorption mediated by NHE3 and the apical H^+ -ATPase.^{79,81} Inhibition of the actin cytoskeleton with cytochalasin D reduces the effect of FSS and HCO_3^- transport by impairing movement of NHE3 and the Na^+/K^+ -ATPase, but not the H^+ -ATPase, to the plasma membrane. This suggests flow-dependent movement of microvilli activates these transport proteins via their linkage to the cytoskeleton (Fig. 6.9 for NHE3).⁸² Redistribution of peripheral actin bands and increased formation of tight junctions and adherens junctions, as observed in cultured tubule cells, is hypothesized to maximize flow-activated transcellular salt and water absorption.⁸³

The roles of dopamine, angiotensin II (AngII), and calcium on FSS-induced sodium and bicarbonate transport have been examined. Luminal dopamine completely inhibited the flow-induced increase in Na^+ transport,⁸⁰ mainly through the D_{1A} receptor. Deletion of AT_{1A} receptors in mice also abrogated flow-induced increments in Na^+ transport,⁸⁴ but these effects may be related to the profound basal reductions in NHE3 activity. When AT_1 receptor blockers are employed, flow-induced Na^+ transport remained.^{84,85} Intracellular activation of IP_3 , through a local calcium signal, may mediate the effects of flow on NHE3 activity, although increased calcium influx does not appear to play a role.⁸⁶ Flow and torque do not appear to stimulate convective flow of chloride through the paracellular pathway.

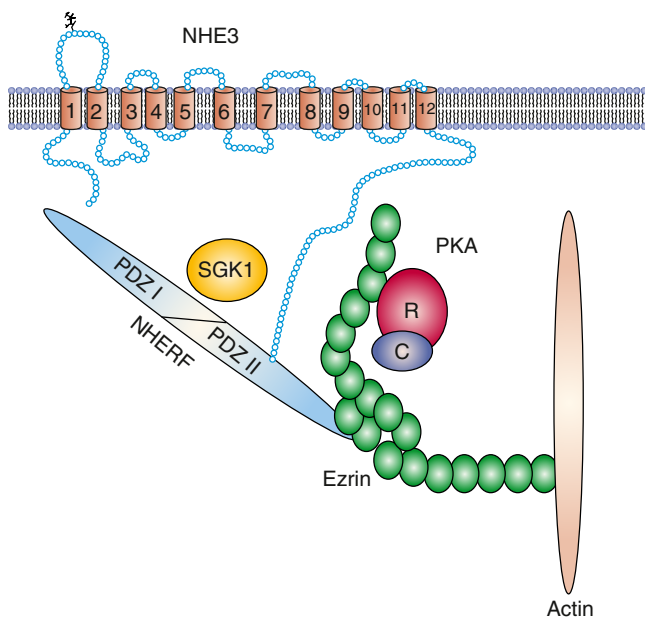


Fig. 6.9 Scaffolding protein NHERF (Na^+ - H^+ exchanger regulatory factor) links the Na^+ - H^+ exchanger NHE3 to the cytoskeleton and signaling proteins. NHERF binds to ezrin, which in turn links to protein kinase A (PKA) and the actin cytoskeleton. NHERF also binds to SGK1 (serum- and glucocorticoid-regulated kinase 1), which activates NHE3. C, Catalytic subunit of PKA; PDZ, domain named for the PSD95, *Drosophila* disk large (*Drosophila*), and ZO-1 proteins; R, regulatory subunit of PKA.

Another mechanism for glomerulotubular balance operating from the luminal side involves limiting solute concentration. Solutes, such as bicarbonate, amino acids, and glucose, that are reabsorbed coupled to sodium will be depleted earlier along the PT when flow is low, thereby limiting reabsorption rates along the segment as a whole.⁸⁷

Peritubular factors also play an important additive role in glomerulotubular balance. Increases in GFR increase filtration fraction, increasing postglomerular protein concentration and thus peritubular oncotic pressure. Changes in peritubular protein concentration have important effects on proximal tubular NaCl reabsorption that have been seen in not only combined capillary and tubular perfusion experiments^{79,88} but also isolated perfused PT segments where the effect of hydrostatic pressure is abolished.⁷⁹ Increases in peritubular protein concentration have an additive effect on the flow-dependent activation of proximal fluid and HCO_3^- absorption, which appears to predominantly affect transcellular transport, though the mechanism for this is still not completely clear.⁷⁹ There are also changes in absorption that correlate with changes in peritubular hydrostatic pressure, as occurs during expansion or contraction of the extracellular fluid volume.⁸⁸

NEUROHUMORAL INFLUENCES

Fluid and NaCl reabsorption by the PT are affected by a number of hormones and neurotransmitters. The major hormonal influences on renal NaCl transport along the nephron are shown in Fig. 6.9. Renal sympathetic tone exerts a particularly important stimulatory influence in the PT, as does AngII; dopamine is a major inhibitor of proximal tubular NaCl reabsorption.

Unilateral denervation of the rat kidney causes a marked natriuresis and reduction in proximal NaCl reabsorption,⁸⁹ while low-frequency electrical stimulation of renal sympathetic nerves increases proximal tubular fluid absorption.⁹⁰ Basolateral epinephrine and/or norepinephrine stimulate proximal NaCl reabsorption via both α - and β -adrenergic receptors. Several lines of evidence suggest that α_1 -adrenergic receptors exert a stimulatory effect on proximal NaCl transport via activation of basolateral Na^+/K^+ -ATPase and apical Na^+/H^+ exchange;⁹¹ the role of α_2 -adrenergic receptors is more controversial.⁹¹ Ligand-dependent recruitment of the scaffolding protein NHERF-1 by β_2 -adrenergic receptors results in direct activation of apical NHE3, bypassing the otherwise negative effect of downstream cyclic AMP (cAMP; see later).^{92,93}

AngII has potent effects on proximal NaCl reabsorption and therefore on blood pressure. Genetic deletion of AT_{1A} receptors from PT cells reduced proximal fluid reabsorption, lowered basal blood pressure, shifted pressure natriuresis, and attenuated the hypertensive response to AngII infused chronically.⁹³ Although basal abundances of NHE3 and $\text{NaPi}2$ were similar in PTs from control and knockout mice, the abundance of NHE3 and $\text{NaPi}2$ was lower following AngII infusion in mice lacking AT_{1A} , suggesting that these effects are mediated, at least in part, through these two prominent Na^+ transport pathways.

Despite this clear stimulatory effect, AngII also has a bimodal (sometimes called “biphasic”) effect on PT Na^+ transport in rats, rabbits, and mice. Stimulation of NaCl reabsorption occurs at 10^{-12} to 10^{-10} M, whereas inhibition

of NaCl reabsorption occurs at concentrations greater than 10^{-7} M (Fig. 6.10).⁹⁴ However, plasma AngII concentrations typically do not exceed 10^{-9} M, even during pathologic states, such as 2-kidney/1-clip Goldblatt hypertension.⁹⁵ Furthermore, this biphasic role of AngII may not hold true for all species, and in human PT obtained from nephrectomy, concentrations up to 10^{-6} M AngII stimulate NaCl reabsorption, primarily owing to a stimulatory effect of the nitric oxide (NO)–cyclic guanosine monophosphate (cGMP) pathway on extracellular signal–regulated kinase (ERK) phosphorylation.⁹⁶ Although the plasma AngII concentration is typically below the inhibitory concentration, it should be noted that the luminal AngII concentration in the PT frequently exceeds that in plasma, and thus plasma concentrations may not be the sole determining factor. Given the substantial effects of proximal AT_{1A} receptor deletion to reduce blood pressure and shift pressure natriuresis,⁹³ AngII is likely to be stimulatory along the PT under most physiologic conditions.

Further complexity of AngII signaling arises from the presence of AT_1 receptors at both the luminal and basolateral membranes in the PT.⁹⁷ AngII application to the luminal or peritubular side of perfused tubules has a similar bimodal effect on fluid transport, albeit with more potent effects at the luminal side.⁹⁸ Experiments using receptor antagonists and knockout mice have indicated

that the stimulatory and inhibitory effects of AngII are both mediated via AT_1 receptors due to signaling at the luminal and basolateral membranes.⁹⁹ However, other work has identified that AT_2 receptors working through NO–cGMP pathway downregulate NHE3 and Na^+ – K^+ –ATPase, leading to natriuresis and reduced blood pressure.¹⁰⁰ Finally, AngII is also synthesized and secreted by the PT, exerting a potent autocrine effect on proximal tubular NaCl reabsorption.¹⁰¹ Though evidence suggests most angiotensinogen is derived from liver,¹⁰² proximal tubular cells express mRNA for angiotensinogen, renin, and angiotensin-converting enzyme,⁹¹ allowing for local generation of AngII. Indeed, luminal concentrations of AngII can be 100- to 1000-fold higher than plasma levels.⁹¹ Evidence suggests that AngII generated by PT cells may exert direct intracellular effects. Mitochondrial-specific delivery of AngII increased NHE3 expression through mitochondrial AT_{1A} receptor activation, an effect countered by activation of mitochondrial AT_2 receptor-mediated NO–cGMP activity.^{103,104} Proximal tubular and systemic synthesis of AngII may be subject to different control. In fact, intrarenal AngII appears to stimulate proximal NaCl and fluid reabsorption even when dietary salt intake is high, thereby helping to prevent rises in glomerular filtration from increasing late proximal flow.¹⁰⁵ It should be noted that in this study, as in many salt-loading studies in rodents, animals received 1% saline as drinking

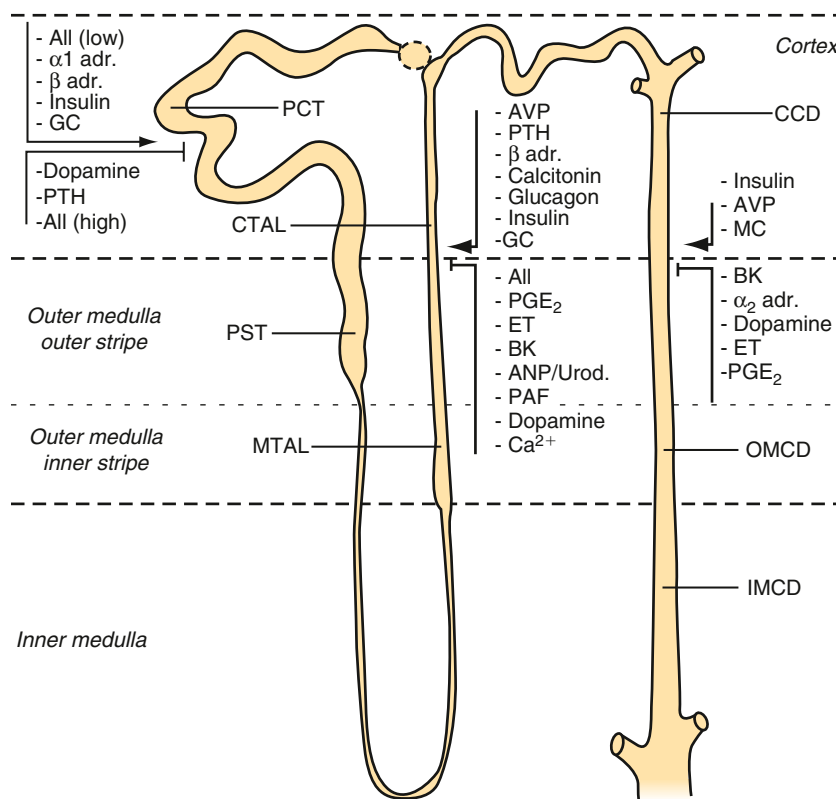


Fig. 6.10 Neurohumoral influences on NaCl absorption by the proximal tubule, thick ascending limb, and collecting duct. Factors that stimulate (→) and inhibit (←) sodium reabsorption are as follows: α_1 adr, α_1 -Adrenergic agonist; All, angiotensin II (low and high referring to picomolar and micromolar concentrations, respectively); ANP/Urod, atrial natriuretic peptide and urodilatin; AVP, arginine vasopressin; β adr, β -adrenergic agonist; BK, bradykinin; CCD, cortical collecting duct; CTAL, cortical thick ascending limb; ET, endothelin; GC, glucocorticoids; IMCD, inner medullary collecting duct; MC, mineralocorticoids; MTAL, medullary thick ascending limb of Henle's loop; OMCD, outer medullary collecting duct; PTH, parathyroid hormone; PAF, platelet-activating factor; PCT, proximal convoluted tubule; PGE_2 , prostaglandin E_2 ; PST, proximal straight tubule. (From Feraille E, Doucet A. Sodium-potassium-adenosine triphosphatase-dependent sodium transport in the kidney: hormonal control. *Physiol Rev*. 2001;81:345–418.)

solution. Thus the high-salt load was accompanied by free water deprivation, a circumstance that leads to stress and inflammation.¹⁰⁶

The PT is also a target for natriuretic hormones; in particular, dopamine synthesized in the PT has negative autocrine effects on proximal NaCl reabsorption.⁹¹ Proximal tubular cells have the requisite enzymatic machinery for the synthesis of dopamine, using L-dopa reabsorbed from the glomerular ultrafiltrate. Dopamine synthesis by proximal tubular cells and release into the tubular lumen is increased after volume expansion or a high-salt diet, resulting in a considerable natriuresis.^{107,108} Luminal dopamine antagonizes the stimulatory effect of epinephrine on volume absorption in perfused proximal convoluted tubules, consistent with an autocrine effect of dopamine released into the tubular lumen.^{107,109} Dopamine primarily exerts its natriuretic effect via D₁-like dopamine receptors. D₁ receptors are expressed at the apical and luminal membranes of PTs.^{97,110} Targeted deletion of the D_{1A} and D₅ receptors in mice leads to hypertension by mechanisms that include reduced proximal tubular natriuresis.^{111,112} PT-specific deletion of aromatic amino acid decarboxylase (AADC), which produces dopamine, clearly demonstrates the role of intrarenal dopamine, leading to upregulation of sodium transporters along the nephron, upregulation of the intrarenal renin-angiotensin axis, decreased natriuresis in response to L-dopa, and reduced medullary cyclooxygenase-2 (COX-2) expression, with reduced urinary prostaglandin levels. These mice also exhibit salt-sensitive hypertension and ultimately a significantly shorter life span compared with wild-type mice.¹¹³

The natriuretic effect of dopamine in the PT is modulated by atrial natriuretic peptide (ANP), which inhibits apical Na⁺-H⁺ exchange via a dopamine-dependent mechanism (Fig. 6.12).¹⁴ ANP appears to induce recruitment of the D₁ dopamine receptor to the plasma membrane of PT cells, thus sensitizing the tubule to the effect of dopamine.¹¹⁴ The inhibitory effect of ANP on basolateral Na⁺-K⁺-ATPase occurs via a D₁-dependent mechanism, with a synergistic inhibition of Na⁺-K⁺-ATPase by the two hormones.¹¹⁴ Furthermore, dopamine and D₁ receptors appear to play critical permissive roles in the *in vivo* natriuretic effect of ANP.¹⁴

Finally, there is considerable crosstalk between the major antinatriuretic and natriuretic influences on the PT. For example, ANP inhibits AngII-dependent stimulation of proximal tubular fluid absorption, presumably via the dopamine-dependent mechanisms discussed earlier.^{14,115} Dopamine also decreases expression of AT₁ receptors for AngII in cultured proximal tubular cells.¹¹⁶ Furthermore, the provision of L-dopa in the drinking water of rats decreases AT₁ receptor expression in the PT, suggesting that dopamine synthesis in the PT resets the sensitivity to AngII.¹¹⁶ AngII signaling through AT₁ receptors decreases expression of the D₅ dopamine receptor, whereas renal cortical expression of AT₁ receptors is in turn increased in knockout mice deficient in the D₅ receptor.¹¹⁷ Similar interactions have been found between proximal tubular AT₁ receptors and the D₂-like D₃ receptor.¹¹⁰

REGULATION OF PROXIMAL TUBULAR TRANSPORTERS

The apical Na⁺-H⁺ exchanger NHE3 and the basolateral Na⁺-K⁺-ATPase are primary targets for the antinatriuretic and

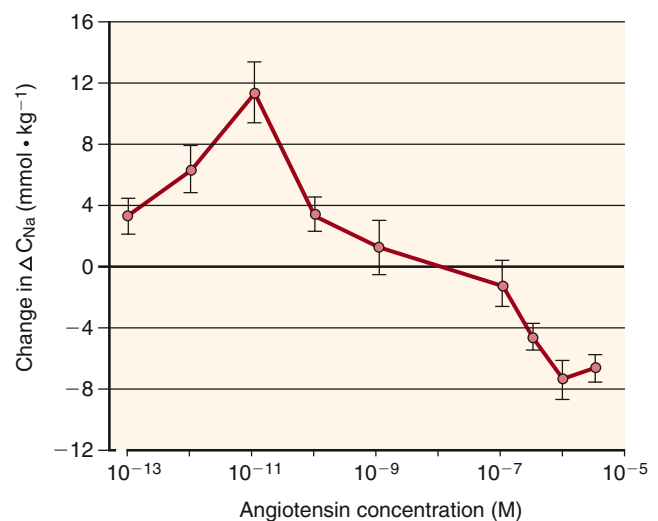


Fig. 6.11 Biphasic effect of AngII on Na⁺ reabsorption in microperfused proximal tubules (PTs). The steady-state transepithelial Na⁺ concentration gradient (peritubular-luminal), ΔC_{Na} , that developed in a stationary split droplet is used as an indication of the rate of active Na⁺ reabsorption. This is plotted as a function of peritubular AngII concentration; low concentrations activate Na⁺ absorption by the PT, whereas higher concentrations inhibit it. (From Harris PJ, Navar LG. Tubular transport responses to angiotensin. *Am J Physiol.* 1985;248:F621-F630.)

natriuretic pathways discussed earlier. Since NHE3 mediates the rate-limiting step in transepithelial NaCl absorption, it is the dominant target for regulatory pathways.⁸¹ NHE3 is regulated by the combined effects of direct phosphorylation and dynamic, C-terminal interaction with scaffolding proteins and signal transduction proteins that primarily regulate NHE3 trafficking to and from the brush border membrane (Fig. 6.11 and Fig. 6.12).^{38,118} Up to 40% of basal activity of NHE3 depends on phosphorylation of serine 719 by C-terminal binding of casein kinase 2; this phosphorylation mediates formation of NHE3-containing protein complexes and alters its distribution to lipid rafts, lowering NHE3 activity.^{119,120}

Increases in cAMP have a profound inhibitory effect on apical Na⁺-H⁺ exchange in the PT. Intracellular cAMP is increased in response to dopamine signaling via D₁-like receptors and/or parathyroid hormone (PTH)-dependent signaling via the PTH receptor, whereas AngII-dependent activation of NHE3 is associated with a reduction in cAMP.¹²¹ PTH potently inhibits NHE3, presumably to promote distal delivery of Na⁺-HCO₃⁻ that promotes distal calcium reabsorption.¹²² Activation of protein kinase A (PKA) by increased cAMP results in direct phosphorylation of NHE3; although several sites in NHE3 are phosphorylated by PKA, serines 552 (S552) and 605 (S605) are the specific residues connected to cAMP-mediated NHE3 inhibition.¹²³ Phospho-specific antibodies, which specifically recognize the phosphorylated forms of S552 and S605, have demonstrated dopamine-dependent increases in the phosphorylation of both residues.¹²⁴ Moreover, immunostaining of rat kidneys has revealed that S552-phosphorylated NHE3 localizes at the coated pit region of the brush border membrane, where the oligomerized inactive form of NHE3 predominates.^{124,125} The cAMP-stimulated phosphorylation of NHE3 by PKA thus causes the transporter to redistribute to an inactive submicrovillar population, similar to the effect dopamine

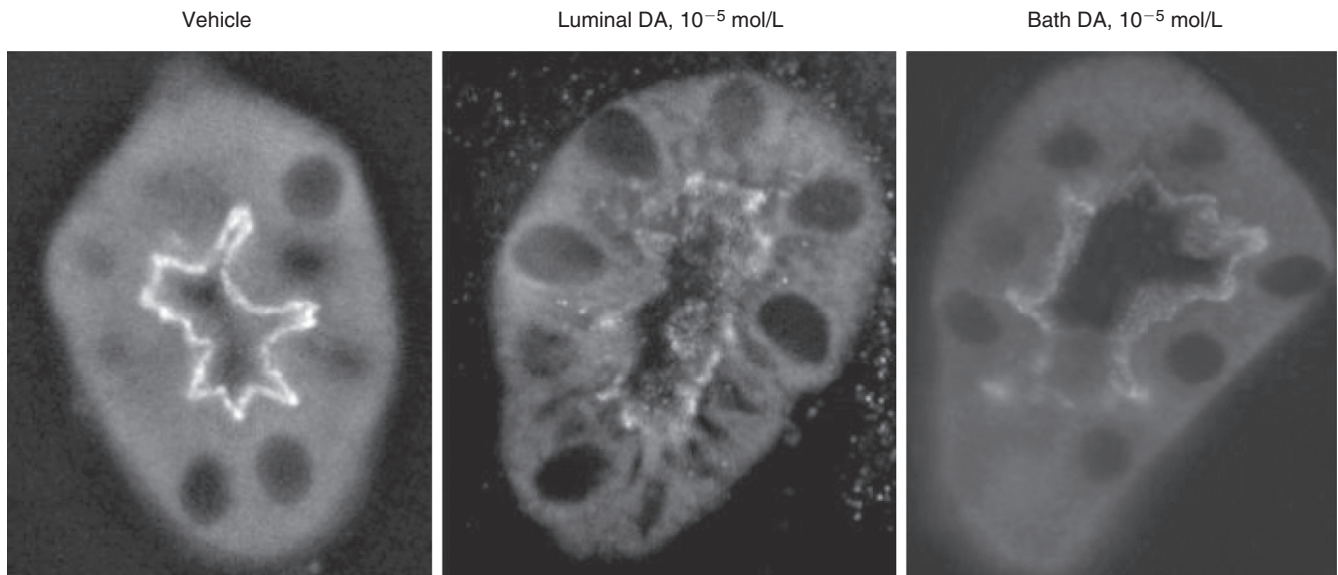


Fig. 6.12 Effect of dopamine on trafficking of the $\text{Na}^+\text{-H}^+$ exchanger NHE3 in the proximal tubule. Microdissected proximal convoluted tubules were perfused for 30 minutes with 10^{-5} mol/L dopamine (DA), in the lumen or bath, inducing a retraction of immunoreactive NHE3 protein from the apical membrane. (From Bacic D, Kaissling B, McLeroy P, et al. Dopamine acutely decreases apical membrane Na/H exchanger NHE3 protein in mouse renal PT. *Kidney Int.* 2003;64:2133–2141.)

(see Fig. 6.11). A number of other regulators of NHE3, including gastrin and uroguanylin, have been found to exert a functional effect through phosphorylation of S552 and/or S605.^{126,127}

Regulation of NHE3 by cAMP also requires scaffolding proteins that contain PDZ domains (see Fig. 6.12). NHE regulatory factor-1 (NHERF-1) was purified as a cellular factor required for the inhibition of NHE3 by PKA.¹²⁸ NHERF-2 also interacts with the C-terminus of NHE3. NHERF-1 and NHERF-2 have similar effects on NHE3 in cultured cells. The related protein PDZK1 interacts with NHE3 and a number of other epithelial transporters and is also required for expression of the anion exchanger SLC26A6 at brush border membranes of the PT.⁶⁴

NHERF-1 colocalizes with NHE3 in microvilli of the brush border of the PT, whereas NHERF-2 is predominantly expressed at the base of microvilli in the vesicle-rich domain.¹²⁸ The NHERFs assemble a multiprotein, dynamically regulated signaling complex that includes NHE3 and several other transport proteins. They also bind to the actin-associated protein ezrin, thus linking NHE3 to the cytoskeleton. This may be particularly important for the mechanical activation of NHE3 by microvillar bending implicated in glomerulotubular balance (see earlier).^{79,81,128} Ezrin also interacts directly with NHE3, binding to a separate binding site within the C-terminus.¹¹⁸ Ezrin anchors PKA, bringing it into close proximity with NHE3 and facilitating its phosphorylation (see Fig. 6.12).¹²⁸ Analysis of knockout mice for NHERF-1 has revealed it is not required for baseline NHE3 activity but is required for cAMP-dependent regulation by PTH.¹²⁸ One long-standing paradox has been that β -adrenergic receptors, which increase cAMP in the PT, activate apical $\text{Na}^+\text{-H}^+$ exchange.⁹¹ This was resolved by the observation that the first PDZ domain of NHERF-1 interacts with the β_2 -adrenergic receptor in an agonist-dependent fashion; this interaction disrupts the interaction between the second PDZ domain and NHE3, resulting in NHE3 activation despite the catecholamine-dependent increase in cAMP.¹²⁸

As discussed earlier, at concentrations higher than 10^{-7} M (see Fig. 6.10), AngII inhibits PT NaCl absorption.⁹⁴ This inhibition is dependent on the activation of brush border phospholipase A_2 , which results in arachidonic acid production.⁹⁸ Metabolism of arachidonic acid by cytochrome P450 mono-oxygenases in turn generates 20-hydroxyeicosatetraenoic acid (20-HETE) and epoxyeicosatrienoic acids (EETs), compounds that inhibit NHE3 and the basolateral $\text{Na}^+\text{-K}^+\text{-ATPase}$.^{91,129} EETs and 20-HETE have also been implicated in the reduction in proximal NaCl absorption that occurs during pressure natriuresis, inhibiting $\text{Na}^+\text{-K}^+\text{-ATPase}$ and removing NHE3 from the brush border membrane.¹³⁰

Antinatriuretic stimuli such as AngII acutely increase NHE3 expression at the apical membrane, partly by inhibiting cAMP generation.¹²¹ Low-dose AngII (10^{-10} M) also increases insertion of NHE3 into the plasma membrane via a mechanism that requires phosphatidylinositol-3-kinase (PI3K).¹³¹ Treatment of rats with captopril thus results in removal of NHE3 and associated proteins from the brush border of PT cells.¹³² Glucocorticoids also increase NHE3 activity by inducing NHE3 gene transcription and acutely stimulating the protein into the plasma membrane.³⁸ Glucocorticoid-dependent exocytosis of NHE3 appears to require NHERF-2, which acts in this context as a scaffolding protein for the glucocorticoid-induced serine-threonine kinase SGK1 (see later, “Regulation of NaCl Transport in the Connecting Tubule and Cortical Collecting Duct: Aldosterone”).¹³³ The acute effect of dexamethasone has thus been shown to require direct phosphorylation of serine 663 in the NHE3 protein by SGK1.¹³⁴

Finally, many of the natriuretic and antinatriuretic pathways that influence NHE3 have parallel effects on the basolateral $\text{Na}^+\text{-K}^+\text{-ATPase}$ (see Feraille and Doucet⁹¹ for a detailed review). Inhibition by dopamine is associated with removal of active $\text{Na}^+\text{-K}^+\text{-ATPase}$ from the basolateral membrane, somewhat analogous to the effect on NHE3 expression at the apical membrane.¹³⁵ This inhibitory effect is primarily mediated by

protein kinase C (PKC), which directly phosphorylates serine 18 of the α_1 -subunit of $\text{Na}^+\text{-K}^+\text{-ATPase}$, the predominant α -subunit in the kidney.⁹¹ This phosphorylation does not affect enzymatic activity of the $\text{Na}^+\text{-K}^+\text{-ATPase}$ but induces a conformational change that enhances binding of PI3K to an adjacent, proline-rich domain. The recruited PI3K then stimulates dynamin-dependent endocytosis of the $\text{Na}^+\text{-K}^+\text{-ATPase}$ complex via clathrin-coated pits.¹³⁵ Evidence suggests that in addition to its pump function, the $\text{Na}^+\text{-K}^+\text{-ATPase}$ has a signaling function involving Src kinase. Along the PT, this is proposed to tonically suppress sodium reabsorption through effects on NHE3 and NBCe1A and may explain the natriuretic effects of endogenous cardiotoxic steroids, which activate $\text{Na}^+\text{-K}^+\text{-ATPase}$ signaling function.¹³⁶

LOOP OF HENLE

The loop of Henle encompasses the thin descending limb, thin ascending limb, and TAL. The descending and ascending thin limbs function in passive absorption of water and NaCl, respectively, whereas the TAL reabsorbs about 30% of filtered NaCl via active transport.^{137,138} There is considerable cellular and functional heterogeneity along the entire length of the loop of Henle, with consequences for the transport of water, NaCl, and other solutes (see Chapter 2).

TRANSPORT CHARACTERISTICS OF THE DESCENDING THIN LIMB

It has long been appreciated that the osmolality of tubular fluid increases progressively between the corticomedullary junction and papillary tip due to active secretion of solutes or passive absorption of water along the descending thin limb.¹³⁹ Outer medullary thin descending limbs display high water permeability in the absence of significant NaCl permeability.¹⁴⁰ However, the permeability properties of descending thin limbs vary as a function of depth in the inner medulla and inclusion in short- versus long-looped nephrons.^{141,142} Transcellular water reabsorption by the thin descending limb is a critical component of the renal countercurrent concentrating mechanism (see Chapter 10).^{137,140} Water permeability of descending thin limbs in the middle part of the inner medulla drops to about 42% that of outer medullary thin descending limbs.¹⁴³ Furthermore, the distal 20% of descending thin limbs have a low water permeability.¹⁴³ These changes in water permeability are accompanied by a progressive increase in NaCl permeability, although the ionic permeability remains considerably less than that of the ascending thin limb.¹⁴²

Consistent with a primary role in passive water and solute absorption, $\text{Na}^+\text{-K}^+\text{-ATPase}$ activity in the descending thin limb is almost undetectable,¹⁷ suggesting no active NaCl transport. The ion transport pathways identified in descending thin limb cells are thought to contribute primarily to cellular volume regulation.¹⁴⁴

NaCl TRANSPORT BY THE THIN ASCENDING LIMB

Fluid entering the thin ascending limb has a high concentration of NaCl due to osmotic equilibration by the

water-permeable descending limbs. Passive reabsorption of this NaCl by the thin ascending limb is a critical component of the passive equilibration model of the countercurrent multiplication system (see Chapters 6 and 10). To achieve this role, the thin ascending limb has a high permeability to NaCl and vanishingly low water permeability.^{142,145} Passive NaCl reabsorption by thin ascending limbs occurs via a combination of paracellular Na^+ transport and transcellular Cl^- transport.^{138,146-150} Inhibition of paracellular conductance by protamine selectively inhibits Na^+ transport across perfused thin ascending limbs, consistent with paracellular transport of Na^+ .¹⁴⁶ Similar to descending limbs, thin ascending limbs display modest $\text{Na}^+\text{-K}^+\text{-ATPase}$ activity (see Fig. 6.3), but only an estimated 2% of Na^+ reabsorption by this nephron segment occurs by active transport.¹⁵¹ Chloride channel blockers reduce Cl^- permeability of the thin ascending limb, consistent with passive transcellular Cl^- transport.¹⁴⁹ Transepithelial transport of Cl^- , but not Na^+ , is activated by vasopressin, with a pharmacologic profile that is consistent with direct activation of thin ascending limb Cl^- channels.¹⁵²

Both apical and basolateral Cl^- transport in the thin ascending limb appear to be mediated by the CLC-K1 Cl^- channel (homologous to the human chloride channel, CLC-Ka) in cooperation with the barttin subunit (see later and also “NaCl Transport by the Thick Ascending Limb: Basolateral Mechanisms”). CLC-K1 is expressed at both the apical and basolateral membranes of thin ascending limbs, such that in contrast to other segments, both the luminal and basolateral Cl^- channels are encoded by the same gene.^{150,153} Targeted deletion of CLC-K1 in mice causes vasopressin-resistant nephrogenic diabetes insipidus.^{137,154} Given that CLC-K1 is also expressed in the TAL, dysfunction of this nephron segment might contribute to this phenotype. However, the related channel CLC-K2 (CLC-Kb) may compensate for the absence of CLC-K1 in TAL in this model.¹⁵⁵

Detailed characterization of CLC-K1 knockout mice has revealed significant impairment of Cl^- but not Na^+ reabsorption by the thin ascending limb,¹³⁸ leading to a marked reduction in the relative permeability of Cl^- to that of Na^+ ($P_{\text{Cl}}/P_{\text{Na}}$), from 4.02 to 0.63 (Fig. 6.13). Protamine, an inhibitor of paracellular Na^+ transport, has a comparable effect on the diffusion voltage of CLC-K1 deficient mice and of wild-type mice that have been treated with 5-nitro-2-(3-phenylpropylamino)-benzoate (NPPB) to inhibit CLC-K1. Therefore the paracellular Na^+ conductance is similar in CLC-K1 knockout mice when compared with littermate controls, confirming the relative independence of paracellular and transcellular conductances for Na^+ and Cl^- , respectively, in thin ascending limbs.¹³⁸

CLC-K1 associates with barttin, an accessory subunit identified via positional cloning of the gene for Bartter syndrome with sensorineural deafness (see later, “NaCl Transport by the Thick Ascending Limb: Basolateral Mechanisms”).¹⁵⁶ Barttin colocalizes with CLC-K1 in thin ascending limbs.^{155,156} Rat CLC-K1 is unique among the CLC-K orthologs and paralogs (CLC-K1/2 in rodents, CLC-Ka/Kb in humans) in that it can generate Cl^- channel activity in the absence of coexpression with barttin; however, its human ortholog CLC-Ka is nonfunctional in the absence of barttin.^{153,155,157} Regardless, barttin coimmunoprecipitates with CLC-K1 and increases expression of the channel

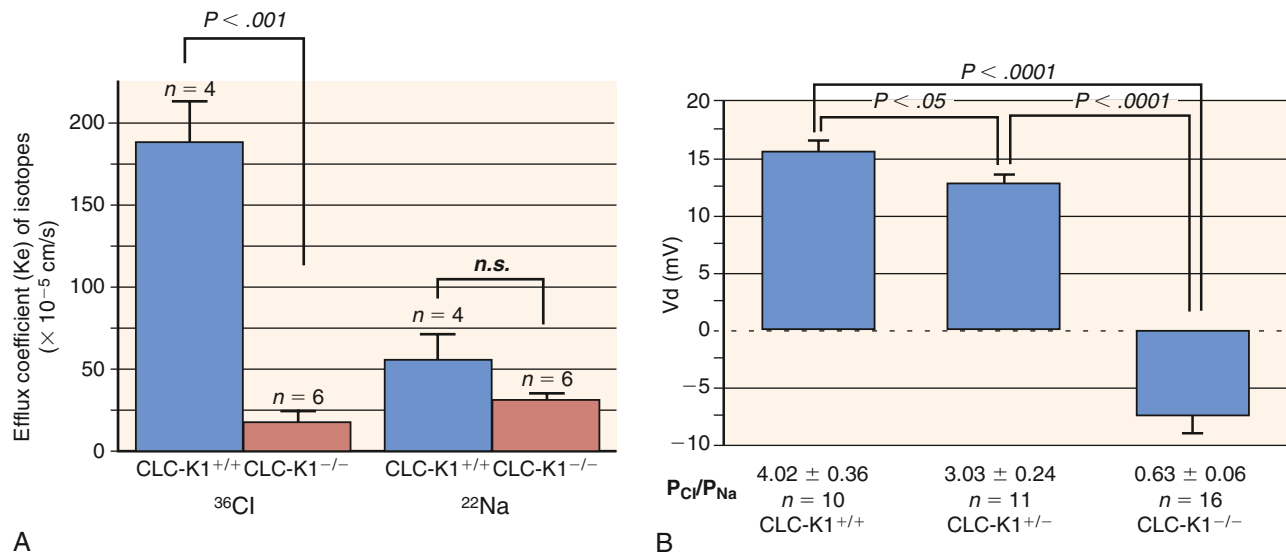


Fig. 6.13 Role of the CLC-K1 chloride channel in Na⁺ and Cl⁻ transport by the thin ascending limbs. Homozygous knockout mice (CLC-K1^{-/-}) are compared with their littermate controls (CLC-K1^{+/+}). (A) Efflux coefficients for ³⁶Cl⁻ and ²²Na⁺ in the thin ascending limbs. Cl⁻ absorption is essentially abolished in the knockout mice, whereas there is no significant effect of CLC-K1 deficiency on Na⁺ transport. (B) The diffusion voltage (V_d), induced by a transepithelial NaCl gradient, is reversed by the absence of CLC-K1, from +15.5 mV in controls to -7.6 mV in homozygous knockout mice. This change in diffusion voltage is due to the dominance of paracellular Na⁺ transport in the CLC-K1-deficient -/- mice, leading to a lumen-negative potential; this corresponds to a marked reduction in the relative permeability of Cl⁻ to that of Na⁺ (P_{Cl}/P_{Na}), from 4.02 to 0.63. (From Liu W, Morimoto T, Kondo Y, et al. Analysis of NaCl transport in thin ascending limb of Henle's loop in CLC-K1 null mice. *Am J Physiol Renal Physiol.* 2002;282:F451-F457.)

protein at the cell membrane and modulates its P_o and unitary conductance.^{155,157}

Vasopressin stimulates Cl⁻ transport by the thin ascending limb, acting as in collecting duct principal cells and TAL through V2 receptors and cAMP.¹⁵² Water deprivation induces a fourfold increase in CLC-K1 mRNA, indicating transcriptional effects of vasopressin or medullary tonicity.¹⁵⁸ In contrast, basolateral calcium inhibits Cl⁻ and Na⁺ transport in the thin ascending limb via activation of the calcium-sensing receptor.¹⁵⁹

NaCl TRANSPORT BY THE THICK ASCENDING LIMB

APICAL NaCl TRANSPORT

The TAL reabsorbs about 30% of filtered NaCl (see Fig. 6.1). In addition to an important role in maintaining extracellular fluid volume, NaCl reabsorption by the water-impermeable TAL is a critical component of the renal countercurrent multiplication system that permits production of either dilute or concentrated urine. In concert with the countercurrent mechanism, NaCl reabsorption by the thin and TALs increases medullary tonicity, facilitating water absorption by the collecting duct (see Chapter 10).

Notwithstanding the morphologic heterogeneity described earlier, cells of the medullary TAL, cortical TAL, and macula densa share the same basic transport mechanisms (Fig. 6.14). NaCl reabsorption by the TAL is thus a secondarily active process, driven by the favorable electrochemical gradient for Na⁺ established by the basolateral Na⁺-K⁺-ATPase.^{14,160} Apical Na⁺-K⁺-2Cl⁻ cotransport is mediated by the furosemide-sensitive cation-chloride cotransporter NKCC2, encoded by *SLC12A1*.¹⁶¹

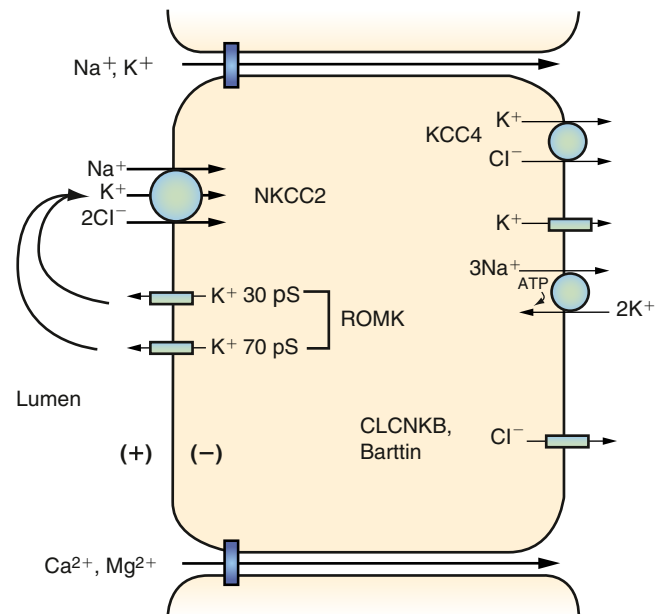


Fig. 6.14 Transepithelial NaCl transport pathways in the thick ascending limb (TAL). CLCNKB, Cl⁻ channel; KCC4, K⁺-Cl⁻ cotransporter-4; NKCC2, Na⁺-K⁺-2Cl⁻ cotransporter-2; ROMK, renal outer medullary K⁺ channel.

This is a member of the cation-chloride cotransporter family of proteins that includes the thiazide-sensitive transporter NCC and the potassium chloride cotransporters. Functional expression of NKCC2 in *Xenopus* oocytes yields Cl⁻- and Na⁺-dependent uptake of Rb⁺ (a radioactive substitute for K⁺) and Cl⁻- and K⁺-dependent uptake of ²²Na⁺.^{94,161-163} As

expected, NKCC2 is sensitive to micromolar concentrations of furosemide, bumetanide, and other loop diuretics.¹⁶¹

Immunofluorescence indicates expression of NKCC2 protein along the entire length of the TAL.¹⁶¹ In particular, immunoelectron microscopy reveals expression in both rough and smooth cells of the TAL.¹⁶⁴ NKCC2 expression in subapical vesicles is particularly prominent in smooth cells, suggesting a role for vesicular trafficking in the regulation of NKCC2 (see later, “Regulation of NaCl Transport by the Thick Ascending Limb”).¹⁶⁴ NKCC2 expression and activity is also present in macula densa cells.^{164,165} This is significant given the role of the macula densa in tubuloglomerular feedback (TGF) and renal renin secretion; luminal loop diuretics block TGF and the suppression of renin release by luminal Cl^- .¹⁴

Alternative splicing of exon 4 of the *SLC12A1* gene yields three variant NKCC2 proteins (NKCC2-A, NKCC2-B, and NKCC2-F) that differ within transmembrane domain 2 and the adjacent intracellular loop.^{161,163} NKCC2-F has a low affinity for Cl^- ($K_m = 113 \text{ mmol/L}$), and NKCC2-B has a high affinity ($K_m = 8.9 \text{ mmol/L}$), whereas NKCC2-A has an intermediate affinity ($K_m = 44.7 \text{ mmol/L}$).^{161,163} These isoforms differ in axial distribution along the tubule, with NKCC2-F expressed in the inner stripe of the outer medulla, NKCC2-A in the outer stripe, and NKCC2-B in cortical TAL.¹⁴ There is thus an axial distribution of the anion affinity of NKCC2 along the TAL, from a low-affinity, high-capacity transporter (NKCC2-F) to a high-affinity, low-capacity transporter (NKCC2-B). In situ hybridization suggests that rabbit macula densa exclusively expresses the NKCC2-B isoform.¹⁴ However, selective knockout of NKCC2-B does not eliminate NKCC2 expression in the murine macula densa, which also seems to express NKCC2-A.¹⁶⁶ The comparative phenotypes of NKCC2-A and NKCC2-B knockout mice are consistent with the relative Cl^- affinity of each isoform. Thus targeted deletion of NKCC2-A selectively reduces TGF responses at the higher range of tubular flow rates (a low-affinity, high-capacity situation), whereas NKCC2-B deletion reduces responses at low flow rates.¹⁶⁷ Loss of NKCC2-A almost abolishes the suppression of plasma renin activity by isotonic saline infusion, which appears more robust in NKCC2-B knockout mice than wild-type littermates.¹⁶⁷

It should be mentioned in this context that NHE3 functions as an alternative mechanism for apical Na^+ absorption by the TAL. There is also evidence in mouse cortical TAL for HCO_3^- absorption mediated via parallel $\text{Na}^+\text{-H}^+$ and $\text{Cl}^-\text{-HCO}_3^-$ exchange, though this mechanism seems less important for transepithelial NaCl transport than in the PT.¹⁶⁸ There is considerable upregulation of both apical $\text{Na}^+\text{-H}^+$ exchange and NHE3 protein in the TAL of acidotic animals, paired with induction of AE2, a basolateral $\text{Cl}^-\text{-HCO}_3^-$ exchanger.^{169,170} A recently generated TAL-specific NHE3 knockout mouse model revealed that under normal conditions, TAL NHE3 has a minor contribution for acid-base balance or can be compensated for by distal segments.¹⁷¹ However, it does appear to play a role in sustained urinary acidification in response to furosemide. NHE3 in the TAL is also upregulated by increased flow, not via shear stress as in the PT, but by the production of endogenous O_2^- and activation of PKC.¹⁷²

APICAL K^+ CHANNELS

The development of a lumen-positive PD plays a critical role in driving paracellular transport of Na^+ , Ca^{2+} , and Mg^{2+} along

the TAL (see Fig. 6.14). This lumen-positive transepithelial PD is generated by the combination of luminal recycling of K^+ via $\text{Na}^+\text{-K}^+\text{-2Cl}^-$ cotransport and apical K^+ channels, along with basolateral depolarization due to Cl^- exit through Cl^- channels.^{14,160}

Apical K^+ channels are required for transepithelial NaCl transport by the TAL.^{14,160} Three types of apical K^+ channels have been identified in the TAL, a channel with a conductance of 30 pS, a channel with a conductance of 70 pS, and a high-conductance, calcium-activated maxi- K^+ channel (see Fig. 6.14).¹⁷³⁻¹⁷⁵ The higher open probability (P_o) and greater density of the 30 pS and 70 pS channels versus the maxi- K^+ channel suggest that these are the primary routes for K^+ recycling across the apical membrane. The 70 pS channel appears to mediate about 80% of the apical K^+ conductance of TAL cells.¹⁷⁶ The low-conductance, 30 pS channel has been identified as the renal outer medullary K^+ channel, ROMK, the prototypic inward-rectifying K^+ channel,¹⁴ since the 30 pS channel is absent from the apical membrane of ROMK knockout mice.¹⁷⁷ In humans, three isoforms of ROMK (ROMK1, 2, and 3) are generated by alternative splicing of the *KCNJ1* gene; ROMK3 has not been detected in rats or mice.¹⁷⁸ ROMK2 has the shortest amino-terminus, ROMK1 has an additional 16 residues, and ROMK3 an additional 22 residues (compared with ROMK2). ROMK1 mRNA is expressed in the mid and late distal tubule and the cortical collecting duct (CCD), as well as in the outer medullary collecting duct (OMCD), but not along the TAL. Consistent with its absence along the TAL, ROMK1-deficient mice do not display a Bartter syndrome phenotype.¹⁷⁹

ROMK2 mRNA is expressed from the medullary TAL through the CCD but is absent from the OMCD.¹⁷⁸ ROMK3 is expressed from the medullary TAL through the distal convoluted tubule (DCT). ROMK protein has been detected at the apical membrane of medullary TAL, cortical TAL, and macula densa,¹⁸⁰ but not all cells in the TAL are labeled with ROMK antibody, suggesting that ROMK might be absent in TAL cells with high basolateral Cl^- conductance and low apical K^+ conductance (see earlier).^{150,181} These cells are thought to correspond to the smooth-surfaced morphologic subtype of TAL cells (S cells)¹⁸¹; however, distribution of ROMK protein by immunoelectron microscopy has not yet been reported.

ROMK clearly plays a critical role in NaCl absorption by the TAL, given that loss-of-function mutations in its gene, *KCNJ1*, are associated with Bartter syndrome.¹⁸² Moreover, pharmacologic ROMK inhibitors function as potent diuretics in vivo, primarily due to inhibition of TAL NaCl transport.^{183,184} However, these observations were difficult to reconcile with the observation that the heterologous expression of ROMK in *Xenopus* oocytes yielded a channel with a conductance of about 30 pS, yet a 70 pS K^+ channel is the dominant conductance at the apical membrane of TAL cells.^{14,176} This paradox was resolved by the observation that the 70 pS channel is absent from the TAL of ROMK knockout mice, indicating that ROMK also functions as a subunit of the 70 pS channel.¹⁸⁵

ROMK activity in the TAL is modulated by association with other proteins that likely generate the 70 pS. ROMK interacts with scaffolding proteins NHERF-1 and NHERF-2 (see earlier, “Regulation of Proximal Tubular NaCl

Transport: Neurohumoral Influences”) via the C-terminal PDZ-binding motif of ROMK; NHERF-2 is coexpressed with ROMK in the TAL.¹⁸⁶ The association of ROMK with NHERFs brings ROMK into closer proximity to the cystic fibrosis transmembrane regulator protein (CFTR).¹⁸⁶ This ROMK-CFTR interaction is required for the native ATP and glibenclamide sensitivity of apical K⁺ channels in the TAL.¹⁸⁷

PARACELLULAR TRANSPORT

The reported transepithelial resistance in the TAL is between 10 and 50 $\Omega \cdot \text{cm}^2$, but water permeability of the TAL is extremely low, less than 1% that of the PT, yet the TAL is not considered a tight epithelium.^{14,160} These hybrid characteristics of relatively low electrical resistance and very low water permeability allow the TAL to generate and sustain NaCl gradients of up to 120 mmol/L.^{14,160} The particular repertoire of claudins expressed in the TAL determines the resistance and ion selectivity and the lack of aquaporins lowers water permeability.

TALs perfused with NaCl develop a lumen-positive, transepithelial PD^{14,160,188-190} that plays a critical role in the paracellular reabsorption of Na⁺, Ca²⁺, and Mg²⁺ (see Fig. 6.14). In the transepithelial transport of Na⁺, the stoichiometry of NKCC2 (1Na⁺:1K⁺:2Cl⁻) is such that other mechanisms are necessary to balance the exit of Cl⁻ at the basolateral membrane; consistent with this requirement, data from mouse TAL have indicated that about 50% of transepithelial Na⁺ transport occurs via the paracellular pathway.^{3,191} In the absence of vasopressin, apical NaCl cotransport is not K⁺ dependent (see “Regulation of NaCl Transport by the Thick Ascending Limb”), reducing the lumen-positive PD; switching to K⁺-dependent Na⁺-K⁺-2Cl⁻ cotransport in the presence of vasopressin results in a doubling of NaCl reabsorption, without an effect on oxygen consumption likely due to passive reabsorption through the cation-permeable paracellular pathway.^{3,192}

Unlike in the PT, the voltage-positive PD in the TAL is generated almost entirely by transcellular transport, rather than by paracellular diffusion.¹⁵ Mouse TAL segments primarily express claudin-14, claudin-16, claudin-19, claudin-3, and the “b” splice form of claudin-10.^{14,193-195} In vasopressin-stimulated mouse TAL, with a lumen-positive PD of 10 mV, the maximal increase in NaCl in the lateral interspace is about 10 mmol/L.¹⁹¹ Tight junctions in the TAL are cation selective,^{160,191} but $P_{\text{Na}}/P_{\text{Cl}}$ ratios can be highly variable in individual tubules, ranging from 2 to 5 in a single study of perfused mouse TAL.¹⁹¹ The claudin profile of cells along the TAL is mosaic, with some cell interfaces expressing the Na⁺-selective claudin 10b, whereas others express claudins 3/16/19.¹⁹⁶ This mosaic pattern suggests the existence of spatially separated paracellular routes for Na⁺ and Ca²⁺/Mg²⁺.¹⁹⁶ Regardless, assuming a net $P_{\text{Na}}/P_{\text{Cl}}$ ratio of about 3, the maximal dilution potential in the mouse TAL is between 0.7 and 1.1 mV, consistent with a dominant effect of transcellular processes on the lumen-positive PD.¹⁹¹

The critical role of claudin-10b in paracellular Na⁺ permeability is clearly illustrated by mutations in the *CLDN10* gene that cause loss of function specifically of the TAL-specific isoform, claudin-10b.^{197,198} Patients with these mutations present with Hypohidrosis-Electrolyte imbalance-Lacrimal gland dysfunction-Ichthyosis-Xerostomia

(HELIX) syndrome. Loss of claudin-10b function leads to hypermagnesemia, hypokalemia with secondary hyperaldosteronism, low blood pressure, and hypocalciuria. The development of hypokalemia is consistent with reduced paracellular Na⁺ reabsorption along the TAL increasing distal Na⁺ delivery (see later, “K⁺ Secretion by the Distal Convolute Tubule, Connecting Tubule, and Cortical Collecting Duct”), while hypermagnesemia and hypocalciuria likely arise from compensatory increases in expression of claudin-16 and claudin-19, which are routes predominantly for paracellular Mg²⁺ and Ca²⁺ reabsorption.^{199,200} The role of claudin-10b is corroborated by the phenotype of claudin-10 knockout mice, which exhibit impaired TAL paracellular Na⁺ transport with increased permeability to Mg²⁺ and Ca²⁺.^{197,200}

BASOLATERAL MECHANISMS

The Na⁺-K⁺-ATPase is the primary exit pathway for Na⁺ at the basolateral membrane of TAL cells. The Na⁺ gradient generated by Na⁺-K⁺-ATPase activity likely drives apical entry of Na⁺, K⁺, and Cl⁻ through NKCC2 because inhibition of Na⁺-K⁺-ATPase with ouabain collapses the lumen-positive PD and abolishes transepithelial NaCl transport in the TAL.^{188,189,192} Basolateral exit of Cl⁻ from TAL cells is primarily but not exclusively electrogenic, mediated by an approximately 10 pS Cl⁻ channel.^{14,160,201} Reductions in basolateral Cl⁻ depolarize the basolateral membrane, whereas decreases in intracellular Cl⁻ induced by luminal furosemide have a hyperpolarizing effect.¹⁴ Intracellular Cl⁻ activity during transepithelial NaCl transport is above its electrochemical equilibrium,¹⁴ with an intracellular negative voltage of -40 to -70 mV that drives basolateral Cl⁻ exit.^{14,160}

While at least two CLC chloride channels, CLC-K1 and CLC-K2 (called CLC-Ka and CLC-Kb in humans), are coexpressed in this nephron segment,^{155,156} evidence suggests the dominant Cl⁻ channel in the TAL is CLC-K2. First, CLC-K1 is highly expressed at the apical and basolateral membranes of the thin ascending limb, and CLC-K1 knockout mice display primary dysfunction of thin ascending limbs, rather than the TAL (see NaCl transport in the thin ascending limb).^{3,138,153,154} Second, loss-of-function mutations in CLC-Kb are associated with Bartter syndrome, indicating a dominant role of this channel in TAL NaCl transport.²⁰² A common T481S polymorphism in human *CLCNKB* that increases channel activity 20-fold has been associated with hypertension, suggesting that gain of function in CLC-Kb increases NaCl transport by the TAL and/or other segments of the distal nephron.²⁰³⁻²⁰⁵ Finally, CLC-K2 protein is highly expressed at the basolateral membrane of mouse TAL, with further expression in DCT, connecting tubule (CNT), and type A intercalated cells.²⁰⁶ *Clcnk2* knockout in mice ablates the ~10 pS chloride channel in TAL cells and leads to salt-wasting, resembling Bartter syndrome.²⁰⁷ These mice also failed to respond to furosemide, indicating the central role of CLC-K2 in transepithelial NaCl transport. Furthermore, immunolabeling of CLC-K2 knockout kidney with antibodies that cross-react with CLC-K1 and CLC-K2 revealed signal extending beyond the thin ascending limb into the medullary TAL, suggesting that CLC-K1 plays a role there. More on Bartter syndrome is discussed in Chapter 71.

A key advance was the characterization of the barttin subunit of CLC-K channels, which is coexpressed with

CLC-K1 and CLC-K2 in several nephron segments including TAL.^{155,156} Unlike rat CLC-K1, the rat CLC-K2, human CLC-Ka, and human CLC-Kb paralogs are not functional in the absence of barttin coexpression.^{156,157} CLC-Kb coexpressed with barttin is highly selective for Cl^- .^{14,155,156} CLC-Kb/barttin channels are activated by increases in extracellular Ca^{2+} and an alkaline extracellular pH but inhibited at an acidic pH.¹⁵⁶ CLC-Ka/barttin channels have similar pH and calcium sensitivities but exhibit higher permeability to Br^- .¹⁵⁶ Despite the considerable homology between CLC-Ka and CLC-Kb, they also differ considerably in pharmacologic sensitivity to various Cl^- channel blockers, potential lead compounds for the development of paralog-specific inhibitors.²⁰⁸

Correlation between functional characteristics of CLC-K proteins with native Cl^- channels in TAL has been problematic, but studies in knockout mice discussed earlier have begun to provide clarity. Wide variation in single-channel conductance has been reported for basolateral Cl^- channels in the TAL, most likely due to experimental conditions.²⁰⁹ Single-channel conductance has not been reported for CLC-Kb/barttin channels because of poor expression in heterologous systems, complicating the comparison of CLC-Kb/barttin to native Cl^- channels. Single-channel conductance has, however, been reported for the rat CLC-K1 mutant V166E mutant, which alters channel gating without expected effects on single-channel amplitude—coexpression with barttin increases the single-channel conductance of V166E CLC-K1 from about 7 pS to 20 pS.¹⁵⁷ Therefore part of the reported variability in native single-channel conductance may reflect heterogeneity in CLC-Kb and/or CLC-Ka interactions with barttin. Regardless, a study using whole-cell recording suggests that CLC-K2 (CLC-Kb in humans) is the dominant Cl^- channel in TAL and other segments of the distal nephron.²⁰⁹ Like CLC-Kb/barttin, this native channel is highly Cl^- -selective.^{14,155,156,209} TAL cells from wild-type mice exhibited a dominant basolateral chloride conductance of 8 pS, which was entirely absent in *Clnk2* knockout mice. Coupled with the strong Bartter syndrome phenotype, these results support the key role of CLC-K2 in driving transepithelial NaCl transport along the TAL.

Electroneutral $\text{K}^+\text{-Cl}^-$ cotransport has also been implicated in transepithelial NaCl transport in the TAL (see Fig. 6.14), functioning in K^+ -dependent Cl^- exit at the basolateral membrane.¹⁴ The $\text{K}^+\text{-Cl}^-$ cotransporter KCC4 is expressed at the basolateral membrane of medullary and cortical TAL and in macula densa.^{210,211} TAL cells have been reported to contain a Cl^- -dependent NH_4^+ transport mechanism that is sensitive to millimolar concentrations of furosemide and barium (Ba^{2+}).²¹² NH_4^+ ions have the same ionic radius as K^+ and are transported by KCC4 and other KCCs; KCC4 is also sensitive to Ba^{2+} and millimolar furosemide, consistent with the pharmacology of $\text{NH}_4^+\text{-Cl}^-$ cotransport in the TAL.²¹²⁻²¹⁴ Furthermore, consistent with basolateral membrane KCC4 expression, the basolateral membrane of TAL also contains a Ba^{2+} -sensitive $\text{K}^+\text{-Cl}^-$ transporter, accounting for the effects of basolateral Ba^{2+} and/or increased K^+ on the transmembrane PD.^{14,210,211,214}

There is thus considerable evidence for basolateral $\text{K}^+\text{-Cl}^-$ cotransport in the TAL, mediated by KCC4.^{210,211} However, direct confirmation of a role for basolateral $\text{K}^+\text{-Cl}^-$ cotransport

in transepithelial NaCl transport is lacking. Indeed, KCC4 knockout mice do not have a prominent defect in function of the TAL but exhibit instead a renal tubular acidosis.²¹⁰ This has been attributed to defects in acid extrusion by $\text{H}^+\text{-ATPase}$ in type A intercalated cells but could also result from reduced medullary NH_4^+ reabsorption by the TAL due to the loss of basolateral NH_4^+ exit through KCC4.^{210,213,215}

Finally, there is also evidence for the existence of Ba^{2+} -sensitive K^+ channel activity at the basolateral membrane of the TAL, providing an alternative exit pathway for K^+ to that mediated by KCC4.²¹⁶⁻²¹⁸ This may help stabilize the basolateral membrane potential above the equilibrium potential for Cl^- , thus maintaining a continuous driving force for Cl^- exit via CLC-Kb/barttin Cl^- channels.²¹⁸ Patch-clamp experiments have identified two types of K^+ channels in the basolateral membrane of the TAL: a 40-pS inwardly rectifying K^+ channel and a Na^+ - and Cl^- -activated, 80 to 150 pS K^+ channel (KCa4.1 or slo2.2).^{217,219} The 40 pS K^+ channel was absent in the TAL of *Kcnj10*^{-/-} mice, suggesting that the 40 pS K^+ channel is a Kir4.1 and Kir4.5 heterotetramer.²¹⁹ Although Kir4.1 is also detected in human TAL, loss-of-function *KCNJ10* mutations do not cause Bartter syndrome, suggesting Kir4.1 disruption has no significant effect on transport function in the TAL.²²⁰ This may reflect secondary activation of alternative K^+ channels along TAL, as *Kcnj10*^{-/-} mice demonstrate vasopressin-induced stimulation of a 80 to 150 pS K^+ channel.²²¹ Basolateral K^+ channels may also attenuate increases in intracellular K^+ that are generated by the basolateral $\text{Na}^+\text{-K}^+\text{-ATPase}$, thus maintaining transepithelial NaCl transport.²¹⁶⁻²¹⁸

REGULATION OF NaCl TRANSPORT BY THE THICK ASCENDING LIMB

ACTIVATING INFLUENCES

Transepithelial NaCl transport by the TAL is regulated by a complex blend of competing neurohumoral influences, which are required to maintain the urinary concentrating capacity and modulate salt balance. In particular, increases in intracellular cAMP tonically stimulate ion transport in the TAL; the stimulatory hormones and mediators that increase cAMP in this segment include vasopressin, PTH, glucagon, calcitonin, and β -adrenergic activation (see Fig. 6.10). These overlapping cAMP-dependent stimuli are thought to result in maximal baseline stimulation of transepithelial NaCl transport.⁹¹ For example, to characterize the vivo effect of these hormones requires prior suppression or absence of circulating vasopressin, PTH, calcitonin, and glucagon.⁹¹ This baseline activation is, in turn, modulated by a number of negative influences, most prominently prostaglandin E_2 (PGE_2) and extracellular Ca^{2+} (see Fig. 6.10). Other hormones and autacoids working through cGMP-dependent signaling, including NO, have potent negative effects on NaCl transport within the TAL.²²² In contrast, AngII has a stimulatory effect on NaCl transport within the TAL.²²³

Vasopressin is perhaps the most extensively studied positive modulator of transepithelial NaCl transport in the TAL. The TAL, with the exception of macula densa cells,²²⁴ expresses type 2 vasopressin receptors (V_2Rs), and microdissected TALs respond to the hormone with an increase in intracellular cAMP.²²⁵ Vasopressin activates apical NKCC2

within minutes in perfused mouse TAL segments and also exerts a longer-term influence on NKCC2 expression and function. The acute activation of NKCC2 is achieved partly through insertion of NKCC2 from subapical vesicles to the plasma membrane.²²⁶ This trafficking-dependent activation is abrogated by treatment of perfused tubules with tetanus toxin, which cleaves the vesicle-associated membrane proteins VAMP-2 and VAMP-3.²²⁶ A rat model expressing a dominant-negative V₂R mutant under the control of the uromodulin promoter (active in TAL and the early part of the DCT²²⁷) has provided some insights into NKCC2 activation by vasopressin. These rats demonstrated polyuria and defective urinary concentration, as well as hypercalciuria, reminiscent of mild Bartter syndrome.²²⁴ The absence of V₂R expression in macula densa cells was suggested to maintain TGF independent of vasopressin signaling.

Activation of NKCC2 by vasopressin is also associated with phosphorylation of several of N-terminal threonines in the transporter protein. Treatment of rats with the V₂ agonist desmopressin (DDAVP) induces phosphorylation of these residues *in vivo*.²²⁶ These threonine residues are substrates for SPAK (STE20/SPS1-related proline/alanine-rich kinase) and OSR1 (oxidative stress-responsive kinase 1), first identified as key regulatory kinases for NKCC1 and other cation-chloride cotransporters.²²⁸ Phosphorylation of these sites is required to maximally activate NKCC1 and NKCC2.^{229,230} SPAK and OSR1 are in turn activated by upstream WNK (With No lysine [K]) kinases, such that SPAK or OSR1 require coexpression with WNK4 to activate NKCC1 fully, at least in *in vitro* systems.²²⁸ Consistent with a role for WNK4 in activation of NKCC2, WNK4 knockout mice display lower levels of phosphorylated NKCC2, a blunted natriuretic response to furosemide, hypokalemia, and normocalciuria.²³¹

Similarly, mice expressing a mutant SPAK that cannot be activated by upstream WNK kinases have a marked reduction in N-terminal phosphorylation of both NKCC2 and the thiazide-sensitive NaCl cotransporter (NCC), with associated salt-sensitive hypotension.²³² SPAK also requires the sorting receptor SORLA (sorting protein-related receptor with A-type repeats) for proper trafficking within TAL cells, such that targeted deletion of SORLA results in a marked reduction in N-terminal NKCC2 phosphorylation.²³³ The WNK kinases appear to regulate SPAK and NKCC2 in a chloride-dependent fashion, phosphorylating and activating SPAK and the transporter in response to a reduction in intracellular chloride concentration (see “Integrated NaCl and K⁺ Transport in the Distal Nephron”).²³⁴

Of the two kinases, SPAK and OSR1, OSR1 appears more critical for NKCC2 function in the TAL, since nephron-specific deletion of OSR1 leads to decreased N-terminal phosphorylation of NKCC2 and a Bartter syndrome-like phenotype.²³⁵ Several groups reported an increase in NKCC2 N-terminal phosphorylation and an increased response to furosemide in SPAK knockout mice, suggesting overcompensation by OSR1.^{236–238} However, it has been determined that antibodies against phosphorylated NKCC2 also bind to phosphorylated NCC in the C57BL/6 mouse background used in these earlier studies.^{239,240} Reanalysis of several mouse models with a specific antibody revealed disruption of activation of WNK-SPAK/OSR1 has little

effect on baseline NKCC2 phosphorylation.²⁴⁰ However, baseline Na⁺ absorption by isolated perfused TAL segments from SPAK-deficient mice is profoundly impaired, though this may reflect an absence of activating factors present *in vivo*.²⁴¹ However, the SPAK/OSR1 pathway may be important for NKCC2 activation in response to hormonal signaling. Further complexity arises from the influence of the adaptor protein, calcium-binding protein 39 (CAB39, also called mouse protein-25, MO25²⁴²), which can both increase SPAK/OSR1-driven phosphorylation of NKCC2 and activate SPAK/OSR1 independently of WNK kinases by promoting kinase dimerization.^{242,243} WNK4 is also capable of direct interaction with CAB39, promoting activation of NKCC2 in the absence of SPAK or OSR1 expression.²⁴⁴ In support of SPAK/OSR1-independent activation of NKCC2, mice in which both SPAK and OSR1 were disrupted in the kidney still retain significant N-terminal phosphorylation of NKCC2. While this could reflect a direct effect of WNK4 on NKCC2, it could also result from direct phosphorylation by other pathways, such as vasopressin-protein kinase A or an as-yet unidentified kinase.²⁴⁵ Therefore there appear to be multiple potential pathways for NKCC2 activation—a WNK4-dependent SPAK/OSR1 pathway, a WNK4-independent SPAK/OSR1 pathway, a SPAK/OSR1-independent WNK4 pathway, and a WNK4-SPAK/OSR1 independent pathway.²⁴⁴ It should be noted that NKCC2 phosphorylation has a smaller effect on activity than it does for NKCC1 and NCC.²⁴⁶ Thus changes in NKCC2 phosphorylation may not always reflect large changes in NKCC2 activity.²⁴⁷

Though the relevance to humans is unclear, studies in mouse medullary TAL cells have shown that vasopressin switches furosemide-sensitive apical Cl[−] transport in the TAL from a K⁺-independent NaCl mode to the classic Na⁺-K⁺-2Cl[−] mode.³ Underscoring the metabolic advantages of paracellular Na⁺ transport, which is critically dependent on the apical entry of K⁺ via Na⁺-K⁺-2Cl[−] cotransport (see earlier), vasopressin accomplishes a doubling of transepithelial NaCl transport without affecting cellular ²²Na⁺ uptake (an indicator of the transcellular component of NaCl transport). This doubling in transepithelial absorption occurs without an increase in O₂ consumption, highlighting the energy efficiency of ion transport by the TAL.³ The mechanism of this shift in transport mode involves splice variants of mouse NKCC2 with a novel, shorter C-terminus that confer sensitivity to cAMP when coexpressed with full-length NKCC2.²⁴⁸ These shorter splice variants appear to encode furosemide-sensitive, K⁺-independent NaCl cotransporters when expressed alone in *Xenopus* oocytes.²⁴⁹ However, the *in vivo* relevance of these phenomena is not clear, and it is unknown whether similar splice variants exist in species other than mouse.

In addition to its acute effects on NKCC2, vasopressin increases transepithelial NaCl transport by activating apical K⁺ channels and basolateral Cl[−] channels in the TAL.^{91,225} Details have yet to emerge of the regulation of the basolateral CLC-K2/barttin Cl[−] channel complex by vasopressin, cAMP, and related pathways. However, the apical K⁺ channel ROMK is directly phosphorylated by protein kinase A on three serine residues (S25, S200, and S294 in the ROMK2 isoform). Phosphorylation of at least two of these three serines is required for detectable K⁺ channel activity in *Xenopus* oocytes. Mutation of all three serines to

alanine abolishes phosphorylation and transport activity, and all three serines are required for full channel activity.²⁵⁰ Phosphorylation of S25 by the kinase SGK1 stimulates trafficking of the channel to the cell membrane, without affecting channel gating.²⁵¹ In contrast, phosphorylation of S200 and S294 modulates P_O via effects on pH-dependent gating and activation by binding of phosphatidylinositol 4,5-bisphosphate (PIP2) to the C-terminal domain of the channel.^{252,253}

Vasopressin also has considerable long-term effects on transepithelial NaCl transport by the TAL. Sustained increases in circulating vasopressin result in hypertrophy of medullary TAL cells, accompanied by a doubling in baseline active NaCl transport.²²⁵ Water restriction or treatment with DDAVP also results in an increased NKCC2 protein abundance in rat TAL cells. Consistent with a direct effect of vasopressin-dependent signaling, expression of NKCC2 is reduced in mice with a heterozygous deletion of the G_s stimulatory G protein, through which the V_2R activates cAMP generation.²²⁵ Increases in cAMP are thought to induce transcription of the *SLC12A1* gene that encodes NKCC2 directly, given the presence of a cAMP-response element in the 5' promoter.^{225,226} Abrogation of the tonic negative effect of PGE_2 on cAMP generation with indomethacin also results in a considerable increase in abundance of the NKCC2 protein.²²⁵ Finally, in addition to these effects on NKCC2 expression, water restriction or DDAVP treatment increases abundance of the ROMK protein at the apical membrane of TAL cells.²⁵⁴

A role for β_3 -adrenergic receptors in activation of NKCC2 has also been proposed, with administration of the selective agonist BRL37344 increasing NKCC2 phosphorylation *ex vivo* in wild-type but not β_3 -adrenergic receptor knockout mice, which also displayed a mild Na^+ - and K^+ -wasting phenotype.²⁵⁵

INHIBITORY INFLUENCES

Stimulation of transepithelial NaCl transport by cAMP-generating hormones (e.g., vasopressin and PTH) is modulated by several negative neurohumoral influences (see Fig. 6.10).⁹¹ In particular, extracellular Ca^{2+} and PGE_2 exert dramatic inhibitory effects on ion transport by TAL and distal nephron segments through a plethora of synergistic mechanisms. Extracellular Ca^{2+} and PGE_2 both activate the G_i inhibitory G protein in TAL cells, opposing the stimulatory, G_s -dependent effects of vasopressin on intracellular levels of cAMP.^{256,257} Extracellular Ca^{2+} exerts its effect through the calcium-sensing receptor (CaSR), which is expressed at the basolateral membrane of TAL cells; PGE_2 primarily signals through EP_3 prostaglandin receptors.^{91,257,258} Increases in intracellular Ca^{2+} due to the activation of the CaSR and other receptors directly inhibit cAMP generation by a Ca^{2+} -inhibitable adenylate cyclase that is expressed in the TAL, accompanied by an increase in phosphodiesterase-dependent degradation of cAMP (Fig. 6.15).^{257,259} These negative stimuli likely inhibit baseline transport in the TAL; abrogation of the negative effect of PGE_2 with indomethacin increases NKCC2 protein abundance,²⁶⁰ whereas targeted deletion of the CaSR in mouse kidney activates NKCC2 via increased N-terminal phosphorylation.²⁶¹

Activation of the CaSR and other receptors in the TAL also results in the downstream generation of arachidonic acid metabolites, with potent negative effects on NaCl transport (see Fig. 6.15). Extracellular Ca^{2+} thus activates phospholipase A_2 (PLA₂) in TAL cells, generating arachidonic acid, which, in response to activation of the CaSR in TAL, is metabolized by cytochrome P450 ω -hydroxylase to 20-HETE.²⁵⁷ 20-HETE has potent negative effects on apical Na^+ - K^+ -2Cl⁻ cotransport, apical K^+ channels, and the basolateral Na^+ - K^+ -ATPase.^{91,257} PLA₂-dependent generation of 20-HETE also underlies in part the negative

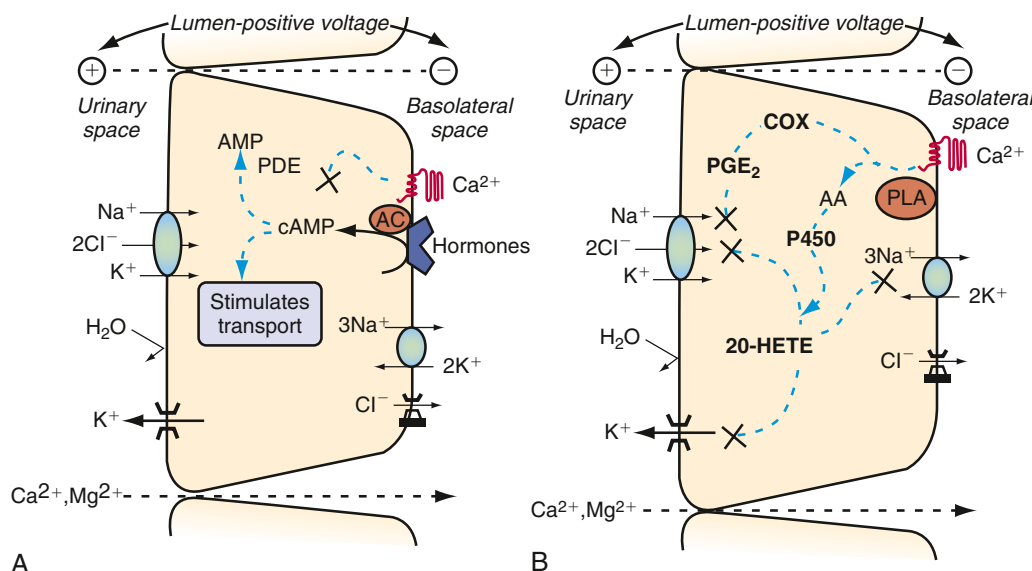


Fig. 6.15 Inhibitory effects of the calcium-sensing receptor (CaSR) on transepithelial NaCl transport in the thick ascending limb. (A) Activation of the basolateral CaSR inhibits the generation of cAMP in response to vasopressin and other hormones (see text for details). (B) Stimulation of phospholipase A_2 by the CaSR leads to liberation of arachidonic acid, which is in turn metabolized to 20-HETE (20-hydroxyeicosatetraenoic acid) or prostaglandin E_2 (PGE_2). 20-HETE inhibits apical Na^+ - K^+ -2Cl⁻ cotransport, apical K^+ channels, and the basolateral Na^+ - K^+ -ATPase. (From Hebert SC. Calcium and salinity sensing by the thick ascending limb: a journey from mammals to fish and back again. *Kidney Int Suppl.* 2004;91:S28–S33.)

effect of bradykinin and AngII on NaCl transport.^{91,257} Activation of the CaSR also induces tumor necrosis factor- α (TNF- α) expression in the TAL, which activates COX-2 and thus generation of PGE₂ (see Fig. 6.15); this PGE₂ in turn results in additional inhibition of NaCl transport.²⁵⁷ Reduction of TNF and NKCC2 isoform expression in vivo via shRNA silencing revealed that TNF promotes natriuresis in response to salt-loading through NKCC2A.²⁶²

The relative importance of the CaSR in the regulation of NaCl transport by the TAL is strikingly illustrated by the phenotype of a subset of patients with CaSR gain-of-function mutations. In addition to the typical suppressed PTH and hypocalcemia caused by gain-of-function mutations in the CaSR (autosomal dominant hypoparathyroidism), these patients manifest a hypokalemic alkalosis, polyuria, and increases in circulating renin and aldosterone.^{263,264} Therefore the persistent inhibition of NaCl transport in the TAL by these overactive mutants of the CaSR causes a Bartter-like syndrome.²⁵⁷

UROMODULIN

TAL cells express high levels of the membrane-bound, glycosylphosphatidylinositol (GPI)-anchored protein, uromodulin (Tamm-Horsfall glycoprotein), which is not expressed by macula densa cells but has been shown to be expressed in the early part of the DCT.²²⁷ Uromodulin is released by proteolytic cleavage at the apical membrane and is secreted as the most abundant protein in normal human urine (20 to 100 mg/day).²⁶⁵ Uromodulin has a host of emerging roles in the physiology and biology of the TAL. A high-salt diet increases uromodulin expression, suggesting a role in ion transport.²⁶⁵ In this regard, uromodulin facilitates membrane trafficking and function of the NKCC2 protein, with similar effects on apical ROMK protein.^{266,267}

Mutations in the *UMOD* gene encoding uromodulin are associated with autosomal dominant tubulointerstitial kidney disease, type 1. This syndrome includes progressive tubulointerstitial damage and chronic kidney disease (CKD), variably penetrant hyperuricemia, gout, and renal cysts.²⁶⁵ Common (≈ 0.8) genetic variants in the *UMOD* promoter are associated with a 20% higher risk of CKD and a 15% higher risk for hypertension.^{265,268} These polymorphisms are associated with higher renal uromodulin expression and higher urinary uromodulin excretion.^{268,269}

Uromodulin-transgenic mice also manifest salt-sensitive hypertension due to activation of the SPAK kinase and activating N-terminal phosphorylation of NKCC2. Human hypertensive subjects homozygous for susceptibility variants in *UMOD* appear to have an analogous phenotype, with exaggerated natriuresis in response to furosemide compared with those who are homozygous for protective variants.²⁶⁸ These findings are compatible with the stimulatory effects of uromodulin on NKCC2 and ROMK—that is, a net gain of function in TAL transport.^{266,267}

ANATOMY OF THE DISTAL NEPHRON

The distal nephron that extends beyond the TAL is the final arbiter of urinary NaCl excretion and a critical target for natriuretic and antinatriuretic stimuli. The understanding of the cellular organization and molecular phenotype of

the distal nephron continues to evolve, and the advent of single-cell and nucleus transcriptomics techniques is beginning to provide powerful refinements.²⁷⁰ The DCT begins at a variable distance after the macula densa, with an abrupt transition between NKCC2-positive cortical TAL cells and DCT cells that express the thiazide-sensitive NaCl cotransporter NCC. Phenotypic analysis of cell types based on expression of marker proteins has revealed considerable differences in the organization of the DCT, CNT, and CCD in rodents, rabbits, and human kidneys.²⁷¹ Rabbit kidneys are unique in the axial demarcation of DCT, CNT, and CCD segments, at both a molecular and morphologic level; the organization of the DCT to CCD is considerably more complex in other species, with boundaries that are much less absolute.²⁷¹ Notably, however, the overall repertoire of transport proteins expressed does not vary among these species.

The early DCT (DCT1) of mouse kidney expresses NCC and a specific marker, parvalbumin, which also distinguishes the DCT1 from the adjacent cortical TAL (Fig. 6.16).²⁷² Cells of the late DCT (DCT2) in rodents coexpress NCC with proteins involved in transcellular Ca²⁺ transport including the apical calcium channel, TRPV5, the cytosolic calcium-binding protein calbindin D_{28K}, and the basolateral Na⁺-Ca²⁺ exchanger NCX1.²⁷² NCC is coexpressed with ENaC in the late DCT2 of mouse, where the two proteins physically and perhaps functionally interact,²⁷³ with robust expression of ENaC continuing in the downstream CNT and CCD.²⁷² In contrast, rabbit kidneys do not have a DCT1 or DCT2 and exhibit abrupt transitions between NCC- and ENaC-positive DCT and CNT segments, respectively.²⁷¹ Human kidneys express calbindin D_{28K} all along the DCT and CNT, extending into the CCD; however, the intensity of expression varies at these sites. Approximately 30% of cells in the distal convolution of human kidneys express NCC, with 70% expressing ENaC (CNT cells); ENaC and NCC overlap in expression at the end of the human DCT segment. Finally, cells of the early CNT of human kidneys express ENaC in

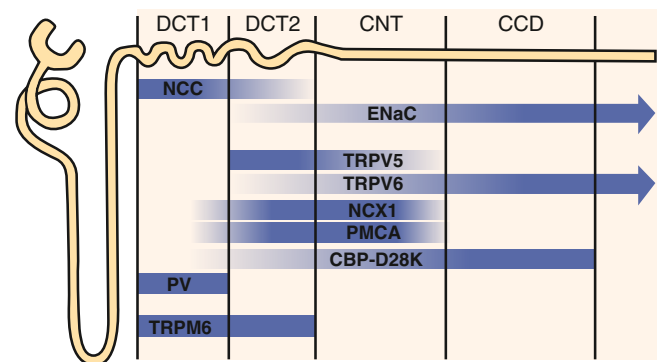


Fig. 6.16 Schematic representation of the segmentation of the mouse distal nephron and distribution and abundance of Na⁺, Ca²⁺, and Mg²⁺-transporting proteins. CBP-D28K, Calbindin-D28K; CCD, cortical collecting duct; CNT, connecting tubule; DCT1, DCT 2, distal convoluted tubules 1 and 2; ENaC, epithelial Na⁺ channel; NCC, thiazide-sensitive NaCl cotransporter; NCX1, Na⁺-Ca²⁺ exchanger; PMCA, plasma membrane Ca²⁺-ATPase; PV, parvalbumin; TRPM6, apical Mg²⁺ entry channel; TRPV5 and TRPV6, apical Ca²⁺ entry channels.^{272,619,620}

the absence of aquaporin-2, the apical vasopressin-sensitive water channel.²⁷¹

Although primarily contiguous with the DCT, CNT cells share several traits with principal cells of the CCD including apical expression of ENaC and ROMK, the K⁺ secretory channel; the capacity for NaCl reabsorption and K⁺ secretion in this nephron segment is as much as 10 times higher than that of the CCD.²⁷⁴ Intercalated cells are the minority cell type in the distal nephron, emerging within the DCT and CNT and extending into the early inner medullary collecting duct (IMCD).²⁷⁵ Three subtypes of intercalated cells have been defined on the basis of differences in the subcellular distribution of the H⁺-ATPase and the presence or absence of the basolateral Cl⁻-HCO₃⁻ exchanger, anion exchanger 1 (AE1). Type A intercalated cells extrude protons via an apical H⁺-ATPase in series with basolateral AE1; type B intercalated cells secrete HCO₃⁻ and OH⁻ via an apical anion exchanger (SLC26A4 or pendrin) in series with basolateral H⁺-ATPase.²⁷⁵ In rodents, the most prevalent subtype of intercalated cells in the CNT is the non-A, non-B intercalated cell, which possesses an apical Cl⁻-HCO₃⁻ exchanger (SLC26A4 or pendrin) along with apical H⁺-ATPase.²⁷⁵ Although intercalated cells play a dominant role in acid-base homeostasis, Cl⁻ transport by type B intercalated cells performs an increasingly appreciated role in distal nephron NaCl transport (see the subhead “Cl⁻ Transport” under “Connecting Tubules and the Cortical Collecting Duct” later).

The OMCD encompasses two separate subsegments corresponding to the outer and inner stripes of the outer medulla, OMCD_o and OMCD_i, respectively. OMCD_o and OMCD_i contain principal cells with apical amiloride-sensitive Na⁺ channels (ENaC); however, the primary role of this nephron segment is renal acidification, with a particular dominance of type A intercalated cells in OMCD_i.^{6,276} The

OMCD may also play a role in K⁺ reabsorption via the activity of apical H⁺-K⁺-ATPase pumps,²⁷⁷⁻²⁷⁹ although deletion of this transport protein does not alter the ability to conserve K⁺ substantially.²⁸⁰

Finally, the inner medullary collecting duct begins at the boundary between the outer and inner medulla and extends to the tip of the papilla. The IMCD is arbitrarily separated into three equal zones, denoted IMCD1, IMCD2, and IMCD3.⁶ The IMCD plays a particularly prominent role in vasopressin-sensitive water and urea transport.⁶ The early IMCD contains principal cells and intercalated cells; all three subsegments (IMCD1-3) express apical ENaC, albeit less than in the CNT and CCD.²⁸¹ The roles of the IMCD and OMCD in NaCl homeostasis have been more elusive than those of the CNT and CCD. However, because ENaC is expressed in the IMCD and OMCD, homologous mechanisms are expected to function in NaCl reabsorption by the CNT, CCD, OMCD, and IMCD segments.

DISTAL CONVOLUTED TUBULE

MECHANISMS OF NaCl TRANSPORT IN THE DISTAL CONVOLUTED TUBULE

The DCT reabsorbs about 10% of filtered NaCl.^{5,282} The apical absorption of Na⁺ and Cl⁻ by the DCT is electroneutral²⁸³ and inhibited by thiazide diuretics.²⁸⁴ Apical absorption is mediated by NCC, which has high affinities for both Na⁺ and Cl⁻^{283,285} and measured Hill coefficients of about 1 for each ion, consistent with electroneutral cotransport.²⁸⁵

NCC expression is the defining characteristic of the DCT (Fig. 6.17).²⁷¹ The human *SLC12A3* gene encodes three isoforms (NCC1, NCC2, and NCC3), but only NCC3 has been studied extensively because NCC1 and NCC2 are not expressed in rats or mice.²⁸⁶ Human isoforms may undergo differential regulation because NCC1 and NCC2 contain a

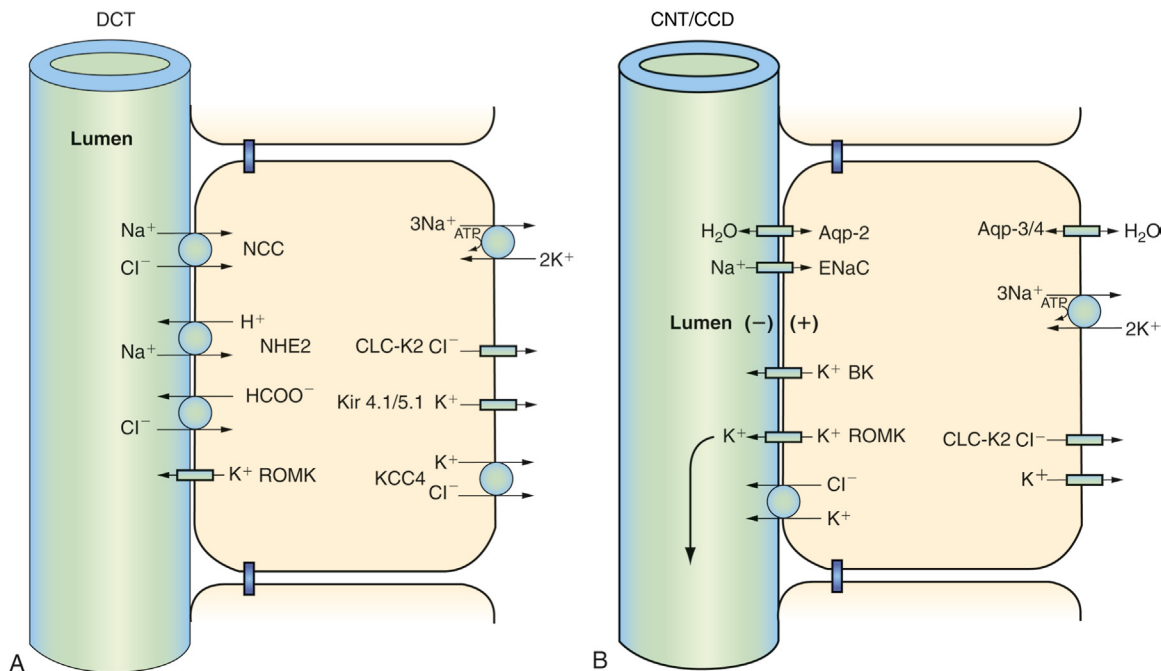


Fig. 6.17 Transport pathways for NaCl and K⁺ in distal convoluted tubule (DCT) cells (A) and principal cells of the connecting tubule (CNT) and cortical collecting duct (CCD) (B). Aqp-2, 3/4, Aquaporin-2, aquaporin-3/4; ENaC, epithelial Na⁺ channel; KCC4, K⁺-Cl⁻ cotransporter-4; NCC, thiazide-sensitive NaCl cotransporter; NHE-2, Na⁺-H⁺ exchanger-2; ROMK, renal outer medullary K⁺ channel.

region in their carboxyl-termini that is absent from NCC3 and contains a serine (S811), which undergoes phosphorylation and contributes to cotransporter activity.²⁸⁷ Loss-of-function mutations in the *SLC12A3* gene cause Gitelman syndrome, familial hypokalemic alkalosis with hypomagnesemia, and hypocalciuria. Mice with homozygous deletion of the *Slc12a3* gene encoding NCC exhibit marked morphologic defects in the early DCT, with a reduction in the absolute number of DCT cells and changes in ultrastructural appearance.²⁸⁸⁻²⁹⁰ Similarly, thiazide treatment promotes marked apoptosis of the proximal part of DCT, suggesting that thiazide-sensitive NaCl cotransport plays an important role in modulating growth and regression of this nephron segment.²⁹¹

Coexpression of NCC and ENaC occurs in the “late DCT” (DCT2) of many species, either in the same cells or in adjacent cells in the same tubule.²⁷¹ While ENaC is primarily the Na⁺ transport pathway of CNT and CCD cells, there is evidence for other Na⁺ and Cl⁻ entry pathways in DCT cells. The Na⁺-H⁺ exchanger NHE2 (SLC9A2) is coexpressed with NCC at the apical membrane of rat DCT cells.²⁹² As in the PT, perfusion of the DCT with formate and oxalate stimulates DIDS-sensitive NaCl transport that is distinct from the thiazide-sensitive transport mediated by NCC.²⁹³ Therefore a parallel arrangement of Na⁺-H⁺ exchange and Cl⁻ anion exchangers may contribute to electroneutral NaCl absorption by the DCT (see Fig. 6.17). The Na⁺-H⁺ antiporter NHE2 (SLC9A2) and the anion exchanger SLC26A6 may also be expressed in DCT cells.^{294,295}

At the basolateral membrane, as in other nephron segments, Na⁺ exits via Na⁺-K⁺-ATPase and the DCT appears to have the highest Na⁺-K⁺-ATPase activity of the entire nephron (see Fig. 6.3).^{17,271} Although CLC-K1 mRNA can also be detected in microdissected DCT segments,^{155,206} several lines of evidence indicate that Cl⁻ primarily exits the basolateral membrane via the CLC-K2 channel. First, the basolateral membrane of rabbit DCT contains Cl⁻ channel activity with functional characteristics similar to those of CLC-K2.^{209,296} Second, CLC-K2 protein is expressed at the basolateral membrane of DCT and CNT cells. Third, CLC-K2 knockout mice display a loss of response to furosemide and a markedly blunted response to thiazide, implicating this channel in chloride reabsorption along both the TAL and DCT.²⁰⁷ Finally, loss-of-function mutations in CLC-Kb, the human ortholog of CLC-K2, cause classic Bartter syndrome (Bartter syndrome type 3). This Bartter subtype has a phenotype that is typically intermediate between Bartter syndrome type I and Gitelman syndrome, consistent with loss of function in DCT.^{202,207,297}

Basolateral membrane K⁺ channels play a critical role in DCT function. Cell-attached patches in basolateral membranes of microdissected DCTs detect an inward-rectifying K⁺ channel, with characteristics similar to those of heteromeric KIR4.1/KIR5.1.^{220,298,299} Basolateral membranes of the DCT express immunoreactive KIR4.1 and KIR5.1 protein.^{220,298,299} While KIR4.2 mRNA has been detected in DCT cells,³⁰⁰ one study detected no KIR4.2 mRNA or protein in DCT, validated by loss of detection along the PT in KIR4.2 knockout mice.⁶³ Patients with loss-of-function mutations in the *KCNJ10* gene that encodes KIR4.1 develop a syndrome including epilepsy, ataxia, sensorineural deafness, and tubulopathy (EAST or SeSAME syndrome).^{220,301} The associated tubulopathy includes hypokalemia, metabolic

alkalosis, hypocalciuria, and hypomagnesemia.^{220,301} Mice in which *Kcnj10* has been deleted following development demonstrate hypokalemic alkalosis with hypocalciuria and suppression of NCC abundance,³⁰² indicating a key role of this channel in supporting transepithelial NaCl transport.^{220,302,303} While KIR4.1 activity is detected in the TAL, KIR4.1 disruption in mice has no significant effect on TAL membrane potential or NKCC2 expression, so its physiologic relevance is unclear.³⁰⁴ In contrast, *Kcnj10* disruption markedly depolarizes the basolateral cell membrane of DCT cells, indicating KIR4.1/KIR5.1 channels play a key role in setting the membrane potential in the DCT.^{302,305} Humans with mutations in *KCNJ16* (encoding KIR5.1) present with hypokalemia, but in contrast to EAST syndrome, they display metabolic acidosis rather than metabolic alkalosis, possibly due to effects on proximal tubule ammoniagenesis.³⁰⁶ The roles of KIR5.1 in renal ion transport are likely to be complex and species dependent because *Kcnj16* disruption in mice increases NCC activity³⁰⁷ but causes a Gitelman-like phenotype in Dahl salt-sensitive rats.³⁰⁸ In addition to sensing the membrane potential, the KIR4.1/KIR5.1 channels at the basolateral membrane of DCT cells are hypothesized to function in basolateral K⁺ recycling, maintaining adequate Na⁺-K⁺-ATPase activity for NaCl absorption and other aspects of DCT function.

REGULATION OF NaCl TRANSPORT IN THE DISTAL CONVOLUTED TUBULE

Early studies showed that dietary NaCl deprivation activates thiazide-sensitive NaCl transport along the DCT.⁵ When distal salt delivery was increased further, by administering loop diuretics continuously and administering saline, increases in transport capacity were observed, together with considerable hypertrophy of DCT cells. These effects involve components of the renin-angiotensin-aldosterone system, as discussed later.

Aldosterone has also been suggested as a factor that modulates NCC, which would make the DCT part of the aldosterone-sensitive distal nephron. Administration of aldosterone to adrenalectomized rats activated NCC.³⁰⁹ Dietary salt restriction and exogenous mineralocorticoids were also shown to increase the abundance of NCC and phosphorylated, active NCC.³¹⁰ Rat DCT cells were shown to express 11-beta hydroxysteroid dehydrogenase type 2 (11βHSD2)³¹¹ at low levels, and its deletion from mice resulted in hypertrophy of DCT cells and higher abundance of phosphorylated NCC.³¹²

The more recent recognition that dietary potassium intake is a powerful NCC and DCT regulatory factor has modified the view of aldosterone's role. Vallon and colleagues³¹³ documented that dietary potassium deprivation increased the abundance of phosphorylated NCC in mice. Two groups subsequently showed that low potassium intake can increase NCC abundance and activity even in the setting of high-salt intake,^{314,315} and another showed that high potassium intake can suppress NCC, even when dietary salt intake is low.³¹⁶ Because high potassium diets are associated with high aldosterone secretion and low NCC activity,³¹⁶ the effects of potassium thus predominate over those of sodium and aldosterone. It is now apparent that DCT cells, and NCC, are exquisitely sensitive to plasma [K⁺]. Effects of K⁺ occur rapidly, as a short-term gavage with high potassium solution

leads to rapid NCC dephosphorylation.³¹⁷ Similarly, raising the plasma potassium concentration, either by potassium infusion,³¹⁸ by administering the ENaC blocker amiloride,³¹⁴ or by deleting Na^+ channels³¹⁹ strikingly reduces the abundance of phosphorylated NCC. These effects on the abundance of phosphorylated NCC are functionally relevant, as directional changes in phosphorylated NCC abundance, in the setting of potassium challenge, are associated with concordant changes in thiazide-sensitive natriuresis.³²⁰ The mechanisms involving WNK kinases in the potassium effect are discussed later.

The recognition that potassium plays a dominant role in regulating NCC and the DCT suggested that some effects observed during aldosterone infusion might be secondary to induced potassium imbalance. While mice with nephron-specific disruption of the mineralocorticoid receptor (MR) exhibit low NCC activity, as would be expected if aldosterone stimulates NCC directly, NCC could be stimulated to normal levels by dietary K^+ restriction, showing effects of MR deletion on potassium balance were responsible for changes in NCC.^{320,321} Similarly, mosaic deletion of MR in renal epithelial cells revealed no differences in NCC abundance and phosphorylation between DCT cells that did and did not express MRs, and dietary Na^+ restriction upregulated NCC to a similar extent in all cells.³²² Finally, in mice with deletion of the MR target $\alpha\text{-ENaC}$, switching mice from normal diet to a high- Na^+ /low- K^+ diet normalized plasma $[\text{K}^+]$ and the reduced abundances of total and phosphorylated NCC.³¹⁹ In summary, the preponderance of evidence suggests that the effects of plasma $[\text{K}^+]$ are dominant and that aldosterone plays only a modifying role in regulating NCC.

A central role in the regulation of DCT Na^+ and Cl^- transport is played by WNK1 and WNK4, key regulatory kinases in the distal nephron that were initially identified as two of the causative genes for familial hyperkalemic hypertension (FHHt; also known as pseudohypoaldosteronism type II or Gordon syndrome). FHHt is in every respect the mirror image of Gitelman syndrome, encompassing hypertension, hyperkalemia, hyperchloremic metabolic acidosis, suppressed plasma renin activity and aldosterone, and hypercalciuria.³²³ Furthermore, FHHt behaves like a gain of function in NCC and/or the DCT in that treatment with thiazides typically results in resolution of the entire syndrome.³²³ However, simple transgenic overexpression of NCC in DCT does not replicate the phenotype in mice, indicating specific effects of the mutant WNK1 and WNK4 alleles.^{323,324}

According to a highly simplified scheme (Fig. 6.18), WNK kinases activate the downstream kinases SPAK and/or OSR1 by binding to them along their conserved carboxyl terminal domains and directly phosphorylating them. Activated SPAK and OSR1, enhanced by interactions with MO25 (Cab39),²⁴² then bind to and activate NCC by phosphorylation key residues in the NCC amino terminal cytoplasmic domain.³²⁵

Mutations in *WNK1* and *WNK4* cause FHHt by increasing WNK protein abundance along the DCT. Intronic mutations in *WNK1* enhance the expression of a full-length kinase-active form of WNK1 along the DCT, where it is normally expressed only at low levels.³²⁶ Mutations in the *WNK4* coding region cluster around an acid-rich conserved region of the protein or near the carboxyl terminal domain.³²⁷

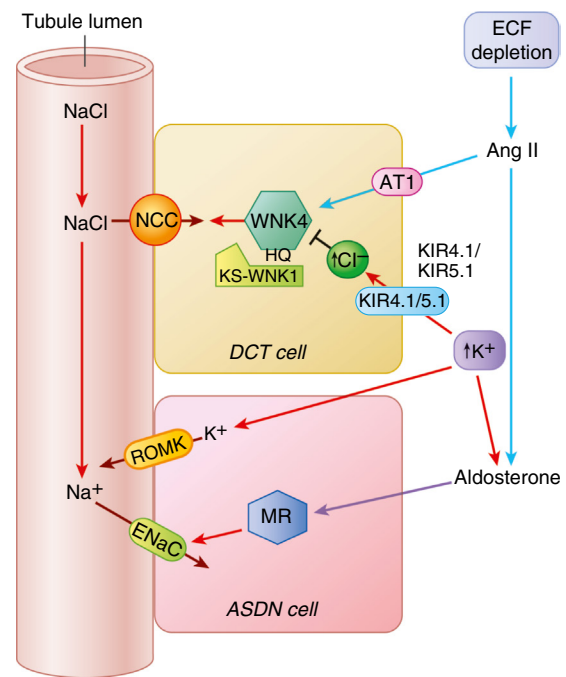


Fig. 6.18 Role of WNK kinases in the differential effects of extracellular fluid (ECF) volume depletion and hyperkalemia ($[\text{K}^+]$) on transport along the distal nephron. In the lumen, Na^+ delivery to the aldosterone-sensitive distal nephron (ASDN, depicted as lower part of tubule) is determined by the extent of Na^+ entry into the DCT (depicted as upper part of tubule) and reabsorption by the thiazide-sensitive NaCl cotransporter (NCC). WNK4, likely together with kidney-specific (KS) WNK1, activates NCC.⁶²¹ This process is stimulated by angiotensin II (Ang II), which also enhances aldosterone secretion by the adrenal gland to stimulate the epithelial Na^+ channel, ENaC. Sodium reabsorption along both the DCT and ASDN corrects ECF volume depletion. During hyperkalemia, NCC is turned off via increased intracellular chloride (modulated by K^+ flux through KIR4.1/KIR5.1 channels), increasing Na^+ delivery to the ASDN, where it is reabsorbed primarily by ENaC, promoting K^+ secretion.

These mutations disrupt WNK4 binding to proteins that are essential for WNK4 degradation (see later).³²⁸⁻³³⁰ The *WNK1* gene has at least three separate promoters and a number of alternative splice forms. The predominant intrarenal WNK1 isoform is generated via a promoter that bypasses the N-terminal exons that encode the kinase domain, yielding a kinase-deficient short form of the protein (kidney-specific WNK1, “KS-WNK1”), while the full-length WNK1 is referred to as long-WNK1 (L-WNK1).³³¹

Several lines of evidence suggest WNK4 rather than L-WNK1 is the major NCC-activating WNK kinase in vivo. First, disruption of WNK4 in mice dramatically reduces NCC activity.^{332,333} Second, when mice with an FHHt-like phenotype induced by mutation of the protein KLHL3 (see later) are interbred with WNK4 knockout mice, increased WNK1 abundance does not compensate for the loss of WNK4, suggesting a key role for WNK4.³³⁴ Third, transgenic mice expressing the FHHt-causing WNK4 mutant Q562E display DCT hyperplasia; when bred with NCC knockout mice, this hyperplasia was completely suppressed.³³⁵ In addition, these data support other studies suggesting that changes in NaCl entry via NCC can modulate hyperplasia or regression of the DCT.^{271,288,289,335,336} Generation of

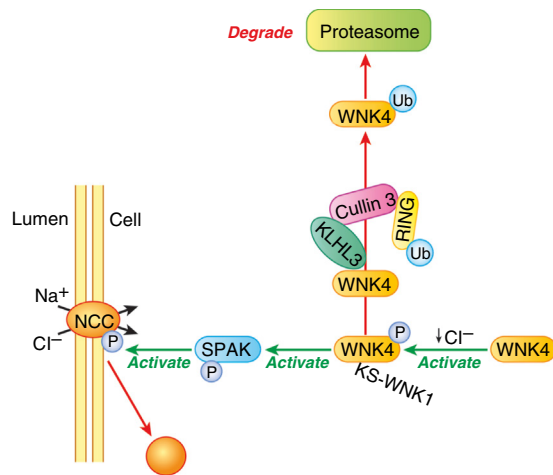


Fig. 6.19 Detailed scheme of WNK kinase regulation. NCC is phosphorylated and activated by SPAK, which is phosphorylated and activated by WNK4, which is activated by low intracellular chloride concentration. WNKs can interact with a cullin-ring ligase, via the adaptor KLHL3, leading to WNK ubiquitination and subsequent degradation by the proteasome. When dephosphorylated, NCC can be removed from the apical plasma membrane.

mice with L-WNK1 deletion specifically in the kidney may shed more light on its contribution to NCC regulation in vivo, but evidence suggests that KS-WNK1 plays a major role in NCC activation. While early in vitro studies suggested KS-WNK1 inhibits WNK4 and thus NCC, more recent studies suggest KS-WNK1 may play a critical role in assembly of cytoplasmic protein complexes called “WNK bodies” that facilitate activation of the WNK4-SPAK/OSR1 pathway.^{337,338}

WNKs, especially WNK4, appear to be sensitive to inhibition by chloride (Fig. 6.19), which directly binds to the catalytic site and inhibits autophosphorylation and activation.^{339,340} Thus when intracellular chloride concentration is low, WNK kinases are maximally activated, and when it is higher, WNK kinases are less active.³⁴¹ This chloride binding is critical for the potassium-sensing function of DCT cells. Reduction in potassium intake and/or hypokalemia leads to reduced basolateral K⁺ concentration in the DCT and subsequent KIR4.1/KIR5.1-dependent hyperpolarization.^{302,305,314} Hyperpolarization has been proposed to lead to chloride exit via basolateral CLC-K2 channels; the resulting reduction in intracellular chloride activates the WNK-SPAK/OSR1 cascade, resulting in NCC phosphorylation and activation.^{314,342} This model helps explain the activating effect of potassium depletion on NCC, the inhibitory effect of potassium loading on NCC, and critical role of the DCT and NCC in potassium homeostasis.³¹⁴

WNK kinases are regulated by Cullin 3 (CUL3) and Kelch-like 3 (KLHL3), components of an E3 ubiquitin ligase complex that targets the WNKs for degradation (see Fig. 6.19).^{330,343,344} Mutations in the *CUL3* and *KLHL3* genes also cause FHHt and account for the majority of cases. Disease-associated mutations in KLHL3 abrogate binding to WNK4 and vice versa.³³⁰ In turn, disease-associated mutations in *CUL3* may deplete levels of both CUL3 and KLHL3, preventing WNK degradation.^{345,346} Physiologically, phosphorylation of KLHL3 by protein

kinase C, downstream of AngII, also prevents the interaction between KLHL3 and WNK4, leading to NCC activation.³⁴⁷ More information about the mechanisms of FHHt and the roles of regulatory proteins can be found in Chapters 16 and 44. More recently, similar mutations in WNK1 have been identified, but these mutations cause a renal tubular acidosis type IV phenotype rather than FHHt and appear to preferentially affect degradation of KS-WNK1 rather than L-WNK1.³⁴⁸

Contributions and interactions with other kinases including WNK3, SGK1, and a truncated form of WNK1 (see later) also contribute to the complexity of NCC activation by WNK4 and SPAK/OSR1 kinases, as do CUL3 and KLHL3.^{330,343-345,347,349-352} Furthermore, signals that activate or inhibit NCC must be integrated with the WNK4-SPAK/OSR1 pathway. For example, the activation of NCC by the AngII receptor type 1 (AT₁R) appears to require the downstream activation of SPAK by WNK4.^{353,354} Increased delivery of glucose or fructose to the DCT was shown to activate NCC through WNK4 and SPAK via activation of the CaSR.³⁵⁵ Furthermore, local synthesis of uromodulin by the DCT or increased delivery of shed uromodulin from the TAL appears to activate NCC in the DCT1.²²⁷ Changes in circulating and local levels of AngII, aldosterone, vasopressin, K⁺, uromodulin, and delivery of solutes from upstream are thus expected to have different and often opposing effects on the activity of NCC in the DCT (see also Fig. 6.18 and “Integrated NaCl and K⁺ Transport in the Distal Nephron”).^{349,353,354,356-359}

CONNECTING TUBULES AND THE CORTICAL COLLECTING DUCT

APICAL Na⁺ TRANSPORT

The apical membrane of CNT cells and principal cells contain prominent Na⁺ and K⁺ conductances, without a measurable apical conductance for Cl[−] (Fig. 6.17).^{209,274,360,361} Entry of Na⁺ occurs via the highly selective epithelial Na⁺ channel (ENaC), which is sensitive to micromolar concentrations of amiloride (Fig. 6.20).³⁶² This selective absorption of positive charge generates a lumen-negative PD, the magnitude of which varies considerably as a function of mineralocorticoid status and other factors. This lumen-negative PD serves to drive the following critical processes: 1. K⁺ secretion via apical K⁺ channels; 2. paracellular Cl[−] transport through the adjacent tight junctions; and/or 3. electrogenic H⁺ secretion via adjacent type A intercalated cells.³⁶³

ENaC is a heteromeric channel complex formed by the assembly of separate, homologous subunits, denoted α -, β -, and γ -ENaC.¹⁴ These channel subunits share a common structure, with intracellular N- and C-terminal domains, two transmembrane segments, and a large glycosylated extracellular loop.¹⁴ *Xenopus* oocytes expressing α -ENaC alone have detectable Na⁺ channel activity (see Fig. 6.20), which facilitated the initial identification of this subunit by expression cloning; functional complementation of this modest activity was then used to clone the other two subunits by expression cloning.¹⁴ Full channel activity requires the coexpression of all three subunits, which causes a dramatic increase in expression of the channel complex at the plasma membrane (see Fig. 6.20).³⁶⁴ The single-channel characteristics of heterologously expressed

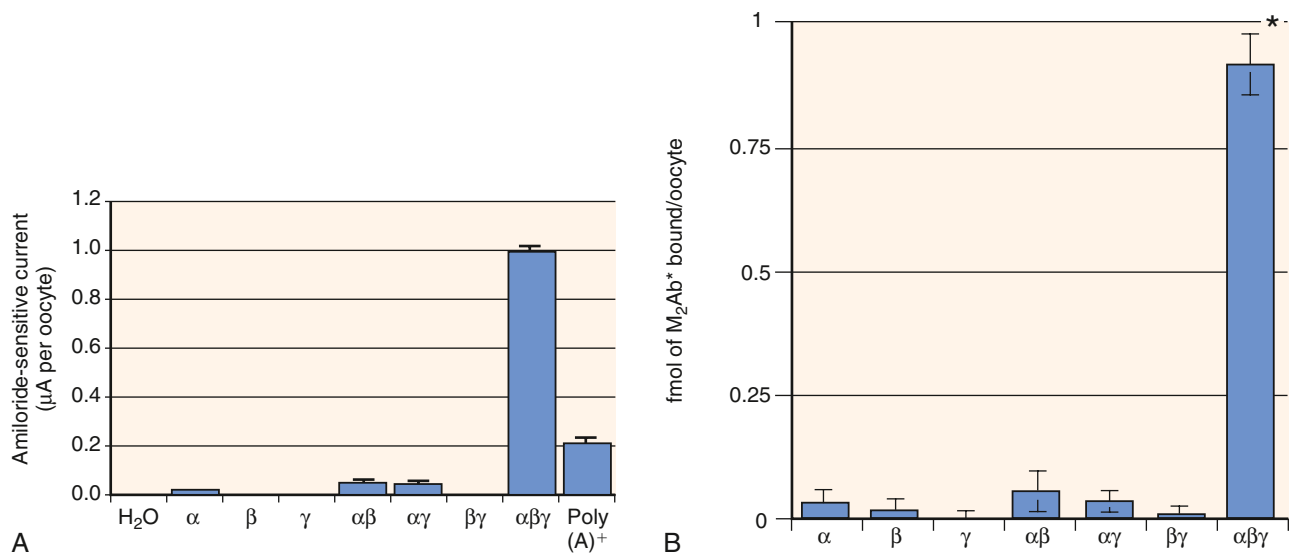


Fig. 6.20 Maximal expression of the amiloride-sensitive epithelial Na⁺ channel (ENaC) at the plasma membrane requires the coexpression of all three subunits (α-, β-, and γ-ENaC). (A) Amiloride-sensitive current in *Xenopus* oocytes expressing the individual subunits and various combinations thereof. (B) Surface expression is markedly enhanced in *Xenopus* oocytes that coexpress all three subunits. The individual cDNAs were engineered with an external epitope tag; expression of the channel proteins at the cell surface is measured by binding of a monoclonal antibody (M₂Ab⁺) to the tag. Poly A⁺, Polyadenylated mRNA. (A from Canessa CM, Schild L, Buell G, et al. Amiloride-sensitive epithelial Na⁺ channel is made of three homologous subunits. *Nature*. 1994;367:463–467. B from Firsov D, Schild L, Gautschi I, et al. Cell surface expression of the epithelial Na channel and a mutant causing Liddle syndrome: a quantitative approach. *Proc Natl Acad Sci U S A*. 1996;93:15370–15375.)

ENaC are essentially identical to those of the amiloride-sensitive channel detectable at the apical membrane of CCD cells.^{14,362} Determination of the structure of the assembled channel by cryo-electron microscopy revealed that ENaC assembles with a stoichiometry of 1α:1β:1γ subunits.³⁶⁵

Recessive loss-of-function mutations in the three subunits of ENaC are a cause of pseudohypoaldosteronism type I.^{14,366} Patients with this syndrome typically present with severe neonatal salt-wasting, hypotension, acidosis, and hyperkalemia. This dramatic phenotype underscores the critical roles of ENaC activity in renal NaCl reabsorption, maintenance of the extracellular fluid volume, K⁺ secretion, and H⁺ secretion. In contrast, gain-of-function mutations in all three ENaC subunits cause Liddle syndrome, an autosomal dominant hypertensive syndrome accompanied by suppressed aldosterone and variable hypokalemia.³⁶⁷ Most mutations associated with Liddle syndrome disrupt interactions between a PPxY motif in the C-terminus of channel subunits with the NEDD4-2 ubiquitin ligase leading to increased apical membrane expression of the channel. Mutations in the extracellular loops of α-ENaC and γ-ENaC have also been identified,^{368,369} with the α-ENaC mutation increasing intrinsic activity of the channel.³⁶⁹

ENaC protein is detectable at the apical membrane of CNT cells and principal cells in the CCD, OMCD, and IMCD (Fig. 6.17).^{276,281} However, several lines of evidence support the hypothesis that the CNT makes the dominant contribution to amiloride-sensitive Na⁺ reabsorption by the distal nephron:

1. Amiloride-sensitive Na⁺ currents in the CNT are twofold to fourfold higher than in the CCD; the maximal capacity of the CNT for Na⁺ reabsorption is estimated to be about 10 times higher than that of the CCD.²⁷⁴

2. Targeted deletion of α-ENaC in the collecting duct abolishes amiloride-sensitive currents in CCD principal cells but does not affect Na⁺ or K⁺ homeostasis; the residual ENaC expression in the late DCT and CNT of these knockout mice easily compensates for the loss of the channel in CCD cells.³⁷⁰
3. Na⁺-K⁺-ATPase activity in the CCD is considerably less than that of the DCT (see also Fig. 6.3), suggesting a greater capability for transepithelial NaCl absorption by the DCT and CNT.¹⁷
4. Apical recruitment of ENaC subunits in response to dietary Na⁺ restriction begins in the CNT, with progressive recruitment of subunits in the downstream CCD at lower levels of dietary Na⁺; although the CNT plays a dominant role in ENaC-mediated sodium transport, it does so primarily in an aldosterone-independent mechanism, with aldosterone-mediated sodium transport in the CCD involved in the finely tuned regulation of sodium transport.^{371,372}
5. Patch-clamp analysis of knock-in mice homozygous for a Liddle syndrome ENaC mutant showed that the primary site of increased Na⁺ reabsorption is the DCT2/CNT rather than the CNT/CCD.³⁷³

Cl⁻ TRANSPORT

There are two major pathways for Cl⁻ absorption in the CNT and CCD—paracellular transport across the tight junction and transcellular transport across type B intercalated cells (Fig. 6.21).^{275,374} The CNT and CCD are “tight” epithelia, with comparatively low paracellular permeability that is not selective for Cl⁻ over Na⁺. However, voltage-driven paracellular Cl⁻ transport in the CCD may play a considerable role in transepithelial NaCl absorption.³⁷⁵ The CNT, DCT, and collecting duct express claudin-3, claudin-4,

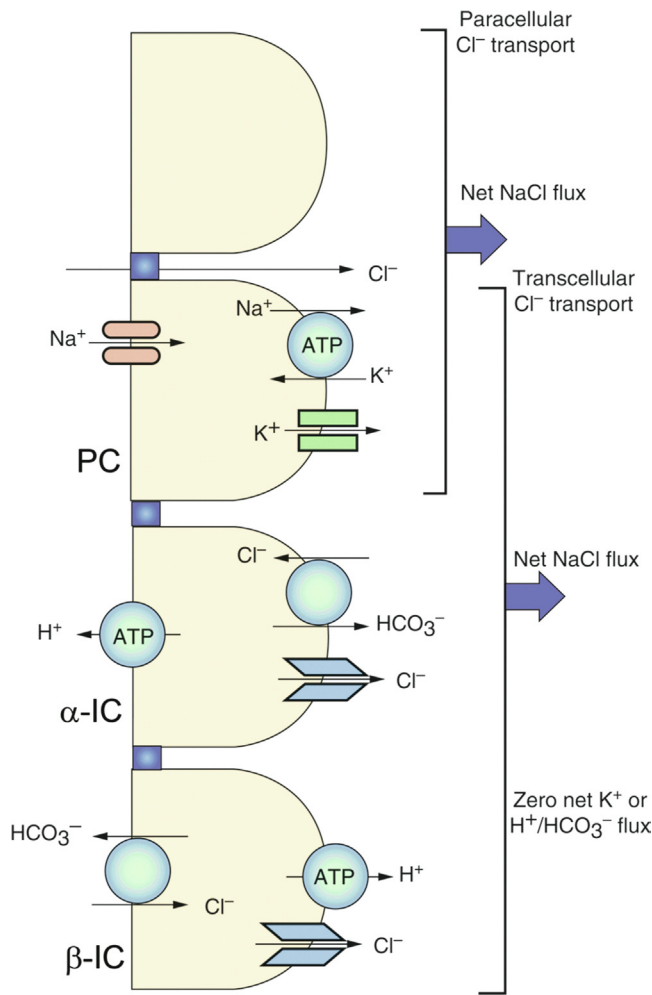


Fig. 6.21 Transepithelial Cl⁻ transport by principal (PC) and α- (α-IC) and β- (β-IC) intercalated cells. The lumen-negative potential difference generated by principal cells drives paracellular Cl⁻ absorption. Alternatively, transepithelial transport occurs in β-intercalated cells via apical Cl⁻-HCO₃⁻ exchange (SLC26A4/pendrin) and basolateral Cl⁻ exit via CLC-K2. (Modified from Moe OW, Baum M, Berry CA, Rector FC Jr. Renal transport of glucose, amino acids, sodium, chloride, and water. In Brenner BM, ed. *Brenner and Rector's the Kidney*. Philadelphia: WB Saunders; 2004. p. 413–452.)

and claudin-8; claudin-8 in particular may function as a paracellular cation barrier that prevents backleak of Na⁺, K⁺, and H⁺ in the distal nephron.^{14,376} Claudin-4 and claudin-8 have been proposed to interact to form a paracellular pathway for Cl⁻ in the collecting duct, thus mediating transepithelial Cl⁻ absorption via the paracellular pathway.³⁷⁷ CCD-specific knockout of claudin-4 in mice leads to NaCl wasting and hypotension,³⁷⁸ while CCD-specific disruption of claudin-8 causes hypotension, hypokalemia, and metabolic alkalosis.³⁷⁹ This model proposes that claudin-4 and claudin-8 are functionally interdependent, but one study showed that claudin-4 levels did not change under conditions where claudin-8 expression changed.³⁸⁰ Rather, claudin-8 appears to functionally interact with ENaC, possibly to prevent solute and water backflow and couple paracellular and transcellular permeability. Regulated changes in paracellular permeability may also contribute to Cl⁻ absorption by the CNT and CCD. In particular,

WNK4 increases paracellular Cl⁻ permeability in transfected MDCK II cell lines.³⁸¹ FHHt-associated WNK4 also increases paracellular permeability, due perhaps to an associated hyperphosphorylation of claudin proteins.^{381,382} Claudin-4-mediated chloride conductance is also negatively regulated by cleavage in its second extracellular loop by channel-activating protease 1 (cap1).³⁷⁸ Claudin-8 was shown to be a target of KLHL3-mediated proteasomal degradation, with FHHt-associated KLHL3 (see “Regulation of NaCl Transport in the Distal Convolute Tubule”) displaying impaired interaction.³⁷⁹

Transcellular Cl⁻ absorption across intercalated cells is thought to play a quantitatively greater role in the CNT and CCD than paracellular transport.³⁷⁴ In the simplest scheme, this process requires the concerted function of type A and type B intercalated cells, achieving net electrogenic Cl⁻ absorption without affecting HCO₃⁻ or H⁺ excretion (see also Fig. 6.21).³⁷⁴ Chloride enters type B intercalated cells via apical Cl⁻-HCO₃⁻ exchange, followed by exit from the cell via basolateral Cl⁻ channels. Recycling of Cl⁻ at the basolateral membrane of adjacent type A intercalated cells also results in HCO₃⁻ absorption and extrusion of H⁺ at the apical membrane. The net effect is electrogenic Cl⁻ absorption.

At the basolateral membrane, intercalated cells display robust Cl⁻ conductance, with transport characteristics similar to those of CLC-K2/barttin.^{209,383} CLC-K2 protein is also detected at the basolateral membrane of type A intercalated cells, and CLC-K2 activity has been observed in type B cells.^{206,384} The basolateral Na⁺-K⁺-2Cl⁻ cotransporter NKCC1 in adjacent type A intercalated cells also plays an evident role in transepithelial Cl⁻ absorption by the CCD.³⁸⁵ At the apical membrane, the SLC26A4 exchanger (also known as pendrin) has been conclusively identified as the elusive Cl⁻-HCO₃⁻ exchanger of type B and non-A, non-B intercalated cells; this exchanger functions as the apical entry site during transepithelial Cl⁻ transport by the distal nephron.²⁷⁵ SLC26A4 mutations cause Pendred syndrome, which manifests as sensorineural hearing loss and goiter; these patients do not have an appreciable renal phenotype.²⁷⁵ However, pendrin knockout mice develop hypotension during severe dietary NaCl restriction³⁸⁶ and are resistant to mineralocorticoid-induced hypertension.³⁸⁷ Pendrin has indirect effects on ENaC abundance and activity by modulating luminal ATP and HCO₃⁻ concentrations; pendrin and ENaC are also both coactivated by AngII,³⁸⁸⁻³⁹¹ and pendrin expression is also induced by aldosterone.³⁹² Overexpression of pendrin in intercalated cells in transgenic mice thus causes hypertension, with an increase in ENaC activity and electroneutral NaCl absorption (see later).³⁹³ Conversely, disruption of pendrin in mice decreased ENaC-mediated Na⁺ absorption by reducing channel P₀ and channel density at the apical membrane.³⁹⁴ Finally, replacement of dietary NaCl with Na⁺-HCO₃⁻ results in Cl⁻ wasting in pendrin knockout mice and increased apical pendrin expression in type B intercalated cells of normal littermate controls.³⁹⁵ Several groups have reported that pendrin expression is exquisitely responsive to changes in distal chloride delivery.³⁹⁶ Importantly, mice with deletion of the MR specifically in type B intercalated cells infused with aldosterone display lower Cl⁻ reabsorption.³⁹⁷ Therefore pendrin plays a critical role in distal nephron Cl⁻ absorption,

underlining the particular importance of transcellular Cl^- transport in this process. Of broader relevance, these studies underline the important role of Cl^- homeostasis in the maintenance of extracellular volume and pathogenesis of hypertension.³⁹⁶

ELECTRONEUTRAL NaCl COTRANSPORT

Although thiazide-sensitive NaCl cotransport is considered the exclusive provenance of the DCT, which expresses the canonical thiazide-sensitive transporter NCC (see earlier, “Mechanisms of NaCl Transport in the Distal Convoluted Tubule”), approximately 50% of NaCl transport in rat CCD is electroneutral, amiloride resistant, and thiazide sensitive,^{398,399} and this has also been demonstrated in mouse CCD.⁴⁰⁰ This transport activity remains in CCDs from mice with genetic disruption of NCC and ENaC. This thiazide-sensitive, electroneutral NaCl transport has been proposed to be mediated by the parallel activity of the Na^+ -driven $\text{SLC4A8 Cl}^-/\text{HCO}_3^-$ exchanger and $\text{SLC26A4 Cl}^-/\text{HCO}_3^-$ exchanger (pendrin; see earlier).⁴⁰⁰ SLC4A8 knockout mice display only a mild perturbation of NaCl and water balance due to compensation by NCC; combined SLC4A8 and NCC disruption caused intravascular volume contraction and hypokalemia.⁴⁰¹ However, a newer study demonstrated complete absence of SLC4A8 mRNA from the cortex including the CCD;⁴⁰² furthermore, the activity of heterologously expressed SLC4A8 is completely resistant to HCTZ inhibition.⁴⁰⁰ Rather, electroneutral NaCl transport along CCD may involve coupling of pendrin with NHE2.⁴⁰²

The apical entry of Na^+ requires a basolateral exit of Na^+ in type B intercalated cells, evidently mediated by the basolateral $\text{Na}^+/\text{HCO}_3^-$ transporter SLC4A9 .² Finally, a series of elegant experiments has revealed that electroneutral NaCl transport in type B intercalated cells is uniquely among mammalian renal epithelial cells energized by and thus dependent on the activity of the basolateral H^+/ATPase rather than $\text{Na}^+/\text{K}^+/\text{ATPase}$ activity.²

REGULATION OF NaCl TRANSPORT IN THE CONNECTING TUBULE AND CORTICAL COLLECTING DUCT

ALDOSTERONE

The DCT, CNT, and collecting ducts have generally been considered to constitute the aldosterone-sensitive distal nephron, expressing the MR and 11β -hydroxysteroid dehydrogenase-2 (11β -HSD2) enzyme that protects against illicit activation by glucocorticoids.²⁷¹ However, data suggest that aldosterone does not directly act on the DCT and CNT, with MR-dependent effects likely to be stimulated by glucocorticoids.^{403,404} Aldosterone is a dominant activator of distal nephron NaCl transport, with a plethora of mechanisms and transcriptional targets.⁴⁰⁵ For example, aldosterone increases the expression of the $\text{Na}^+/\text{K}^+/\text{ATPase } \alpha_1$ - and β_1 -subunits in the CCD, in addition to inducing SLC26A4 , the apical $\text{Cl}^-/\text{HCO}_3^-$ exchanger of intercalated cells.^{387,406} Aldosterone may also affect paracellular permeability of the distal nephron via posttranscriptional modification of claudins and other components of the tight junction.⁴⁰⁷ Impressive progress has been made in understanding the downstream effects of aldosterone on synthesis, trafficking,

and membrane-associated activity of ENaC subunits. A detailed discussion of aldosterone's actions may be found in Chapter 12; here we summarize the major findings of relevance to NaCl transport.

Aldosterone increases abundance of α -ENaC via a glucocorticoid-response element in the promoter of the *SCNN1A* gene.⁴⁰⁸ Aldosterone also relieves tonic inhibition of the *SCNN1A* gene by a complex that includes the DOT1A (disruptor of telomere silencing splicing variant *a*) and AF9 and AF17 transcription factors.⁴⁰⁹ An aldosterone-dependent reduction in promoter methylation is also involved.⁴¹⁰ This transcriptional activation results in an increased abundance of α -ENaC protein in response to exogenous aldosterone or dietary NaCl restriction (Fig. 6.22); this response is blunted by spironolactone, indicating involvement of the MR.⁴¹¹⁻⁴¹³ At baseline, α -ENaC transcripts in the kidney are less abundant than those encoding β - and γ -ENaC⁴¹⁴ (see Fig. 6.22), so induction of α -ENaC is thought to relieve a major bottleneck in the processing and trafficking of active ENaC complexes.⁴¹⁴

Aldosterone also plays a role in trafficking of ENaC subunits to the plasma membrane by regulating accessory proteins that interact with preexisting ENaC subunits. Aldosterone rapidly induces expression of serum and glucocorticoid-induced kinase-1 (SGK1); coexpression of SGK1 with ENaC subunits in *Xenopus* oocytes potently activates ENaC activity due to increased plasma membrane expression.^{413,415,416} Importantly, an analogous redistribution of ENaC subunits from the cytoplasm to the apical membrane occurs in the CNT and early CCD, in response to aldosterone infusion or NaCl restriction (see Fig. 6.22).^{371,411,413} This redistribution of ENaC in the CNT correlates with induction of SGK1 protein.⁴¹³

SGK1 modulates membrane expression of ENaC by interfering with regulated endocytosis of its channel subunits. This is mediated by the ubiquitin ligase NEDD4-2, which is thought to ubiquitinate ENaC subunits, resulting in their removal from the cell membrane and degradation in lysosomes and the proteasome.⁴¹⁴ PPxY domains in the C-termini of all three ENaC subunits bind to WW domains of NEDD4-2.⁴¹⁷ These PPxY domains are deleted, truncated, or mutated in patients with Liddle syndrome, leading to a gain of function in channel activity.^{364,367} Coexpression of NEDD4-2 with wild-type ENaC markedly inhibits channel activity due to removal from the cell membrane, whereas channels bearing Liddle syndrome mutations are resistant. Phosphorylation of NEDD4-2 by SGK1, which also possesses a PPxY motif, abrogates the ability of NEDD4-2 to inhibit ENaC.^{418,419} Aldosterone also stimulates NEDD4-2 phosphorylation in vivo.⁴²⁰ NEDD4-2 phosphorylation in turn results in ubiquitin-mediated degradation of SGK1, indicating feedback regulation in this system.⁴²¹ Aldosterone also reduces NEDD4-2 protein expression in cultured CCD cells, and acetylation of ENaC antagonizes ENaC ubiquitination, suggesting additional levels of regulation.^{422,423}

Induction of SGK1 by aldosterone thus appears to stimulate redistribution of ENaC subunits from the cytoplasm to the apical membrane of CNT and CCD cells. There is considerable axial heterogeneity in the recruitment of ENaC to the plasma membrane, which begins in the CNT and only extends into the CCD and OMCD in NaCl -restricted

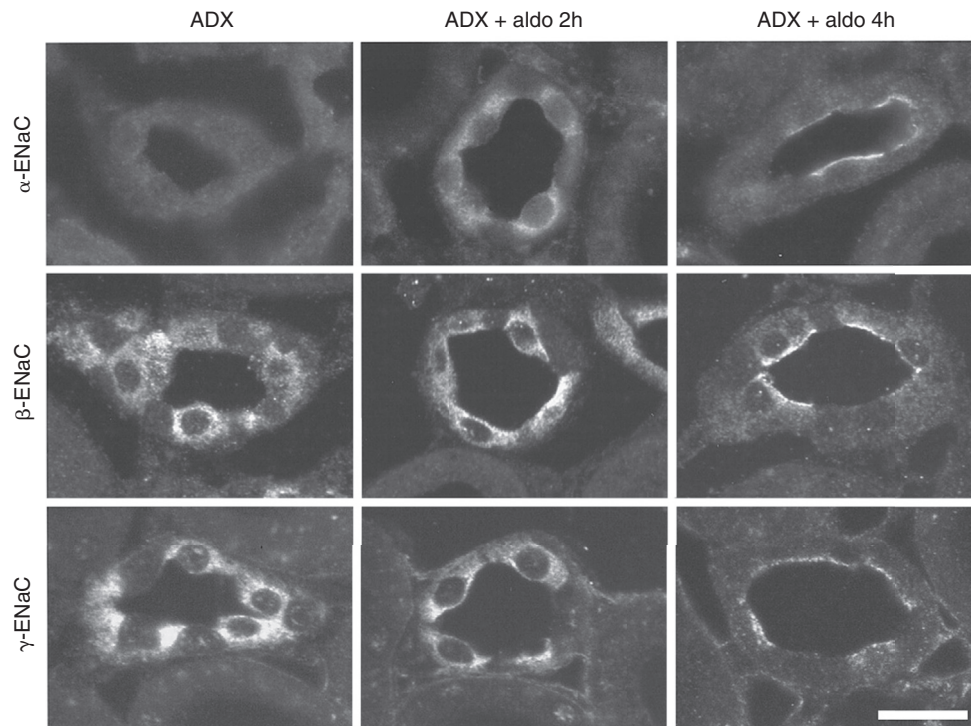


Fig. 6.22 Immunofluorescence images of connecting tubule (CNT) profiles in kidneys from adrenalectomized rats (ADX) and from ADX rats 2 and 4 h after aldosterone (aldo) injection. Antibodies against the α -, β -, and γ -subunits of ENaC reveal absent expression of the former in ADX rats, with progressive induction by aldosterone. All three subunits traffic to the apical membrane in response to aldosterone. This coincides with rapid aldosterone induction of the SGK kinase in the same cells; SGK is known to increase the expression of ENaC at the plasma membrane (see text for details). Bar $\equiv$ 15 μ m. (From Loffing J, Zecevic M, Féraille E, et al. Aldosterone induces rapid apical translocation of ENaC in early portion of renal collecting system: possible role of SGK. *Am J Physiol Renal Physiol*. 2001;280:F675–F682.)

or aldosterone-treated animals.^{271,413} The underlying causes of this progressive axial recruitment are not yet clear, but NEDD4-2 expression is inversely related to the apical distribution of ENaC, with low expression in the CNT and increased expression levels in the CCD.²⁷¹ Therefore the relative balance among SGK1, ENaC, and NEDD4-2 likely figures prominently in membrane recruitment of ENaC subunits.⁴²²

NEDD4-2 and ENaC are part of a larger regulatory complex that includes the signaling protein RAF-1, stimulatory aldosterone-induced chaperone GILZ1 (glucocorticoid-induced leucine zipper-1), and scaffolding protein CNK3.^{424,425} The mTORC2 (mammalian target of rapamycin complex 2) kinase complex is another component, catalyzing upstream activation of SGK1 and thus inducing activation of ENaC.^{426,427}

Despite altered apical trafficking of ENaC in SGK1 knockout mice, ENaC activity is normal even after aldosterone administration, suggesting that other aldosterone-induced proteins contribute to ENaC activation via MRs.^{428,429} For example, another aldosterone-induced protein, ankyrin G, a cytoskeletal protein involved in vesicular trafficking, increases ENaC activity in cultured CCD cells by promoting its plasma membrane insertion from recycling endosomes.⁴³⁰

Finally, aldosterone indirectly activates ENaC channels through the induction of channel-activating proteases, which increase open channel probability by cleavage of the extracellular domains of α - and γ -ENaC. Western blotting

of kidney lysates from rats subjected to NaCl restriction or aldosterone treatment has revealed α - and γ -ENaC subunits of lower molecular mass than those detected in control animals, consistent with proteolytic cleavage.^{412,431} Proteases that have been implicated in the processing of ENaC include furin, elastase, and three membrane-associated proteases denoted CAP1-3 (channel-activating protease-1, protease-2, and protease-3).⁴³²⁻⁴³⁴ Filtered proteases such as plasmin may also contribute to ENaC activation in nephrotic syndrome.⁴³⁴ CAP1 was initially identified from *Xenopus* A6 cells as an ENaC-activating protease; the mammalian ortholog is an aldosterone-induced protein in principal cells.^{435,436} Urinary excretion of CAP1, also known as prostaticin, is increased in hyperaldosteronism, with a reduction after adrenalectomy.⁴³⁶ CAP1 is tethered to the plasma membrane by a GPI linkage, whereas CAP2 and CAP3 are transmembrane proteases.^{433,435} All three proteases activate ENaC by increasing the P_o of the channel, without increasing membrane expression.⁴³³ However, kidney-specific CAP1⁴³⁷ and global CAP2⁴³⁸ knockout mice display normal sodium balance in vivo, which may reflect compensation by other CAPs. Proteolytic cleavage of ENaC appears to activate the channel by removing the self-inhibitory effect of external Na^+ . In the case of furin-mediated proteolysis of α -ENaC, this appears to involve the removal of an inhibitory domain from within the extracellular loop.^{433,439} Extracellular Na^+ appears to interact with a specific acidic cleft in the extracellular loop of α -ENaC, causing inhibition of the

channel.⁴⁴⁰ Unprocessed channels at the plasma membrane are thought to function as a reserve pool, capable of rapid protease-mediated activation.⁴³²

VASOPRESSIN AND OTHER FACTORS

Although not typically considered an antinatriuretic hormone, vasopressin has well-characterized stimulatory effects on NaCl transport by the CCD.^{91,441} Vasopressin directly activates ENaC in murine CCD, increasing channel P_o .⁴⁴² In perfused rat CCD, vasopressin and aldosterone can have synergistic effects on Na^+ transport.⁴⁴¹ Furthermore, water and Na^+ -restriction synergistically increase ENaC P_o in murine CCDs.⁴⁴² Prostaglandins inhibit this effect of vasopressin, particularly in the rabbit CCD, in part through reductions in vasopressin-generated cAMP.^{91,441} There are, however, considerable species-dependent differences in the interactions between vasopressin and negative modulators of NaCl transport in the CCD including prostaglandins, bradykinin, endothelin, and α_2 -adrenergic tone.^{91,441} Regardless, cAMP rapidly increases the Na^+ conductance of apical membranes in the CCD by increasing ENaC localization to the plasma membrane⁴⁴³ and open channel probability.^{442,444} Notably, cAMP inhibits retrieval of ENaC subunits from the plasma membrane via PKA-dependent phosphorylation of the phosphoacceptor sites in NEDD4-2 that are targeted by SGK1. Therefore both aldosterone and vasopressin converge on NEDD4-2 in the regulation of ENaC activity in the distal nephron.⁴⁴⁵ Analogous to the effect on trafficking of aquaporin-2 in principal cells, cAMP also seems to stimulate exocytosis of ENaC subunits to the plasma membrane.⁴⁴⁴ Finally, similar to the long-term effects of vasopressin on aquaporin-2 and NKCC2 expression, chronic treatment with DDAVP increases abundances of the β - and γ -ENaC subunits.^{225,446}

Activation of ENaC by vasopressin directly affects water homeostasis. Hypernatremic mice treated with the ENaC inhibitor benzamil thus exhibit further increases in serum tonicity due to a reduction in urinary osmolality.⁴⁴⁷ In adrenalectomized mice, which lack circulating aldosterone, vasopressin maintains ENaC activity in the distal nephron.⁴⁴⁸ In addition to inducing secretion of aldosterone, which activates ENaC, AngII itself directly activates amiloride-sensitive Na^+ transport in perfused CCDs; blockade by losartan or candesartan suggests that this activation is mediated by AT_1Rs .⁴⁴⁹ Perfusion of tubules with AngI stimulates ENaC, an effect blocked by angiotensin-converting enzyme (ACE) inhibition with captopril, suggesting that intraluminal conversion of AngI to AngII can occur in the CCD.⁴⁵⁰ Importantly, luminal AngII has a greater effect on ENaC activity than bath AngII does.⁴⁴⁹ In contrast to PT cells that express extremely low levels of renin, CNT cells express high levels of renin.⁴⁵¹ Angiotensinogen secreted by the PT may thus be converted to AngII in the CNT via locally generated renin and ACE and/or related proteases.⁴⁵¹ AngII also activates chloride absorption across intercalated cells via a pendrin (SLC26A4) and H^+ -ATPase-dependent mechanism,⁴⁵² as well as via activation of ClC-K2 .⁴⁵³

Luminal perfusion with ATP or uridine triphosphate (UTP) inhibits amiloride-sensitive Na^+ transport and reduces ENaC P_o in the CCD via activation of luminal P2Y_2 purinergic receptors.^{454,455} P2Y_2 receptor knockout mice display salt-resistant hypertension due in part to

upregulation of NKCC2 activity in the TAL. Resting ENaC activity is also increased, but suppressed aldosterone and downregulation of the α -subunit of ENaC blunt the role of amiloride-sensitive transport.^{455,456} Administration of exogenous mineralocorticoid to clamp mineralocorticoid activity at higher levels reveals that P2Y_2 receptor activation may be a major mechanism for the modulation of ENaC P_o in response to changes in dietary NaCl.³¹³ Increased dietary NaCl thus leads to increased urinary ATP and UTP excretion in mice.⁴⁵⁵ Endogenous ATP from principal cells inhibits ENaC, and ENaC activity is not responsive to increased dietary NaCl in P2Y_2 receptor knockout mice.^{313,455} In addition, the activation of apical ionotropic purinergic receptors, likely P2X_4 and/or $\text{P2X}_4/\text{P2X}_6$, can inhibit or activate ENaC, depending on luminal Na^+ concentration; these receptors may also participate in fine-tuning ENaC activity in response to dietary NaCl.⁴⁵⁷ ATP-derived adenosine may also contribute to lower NaCl reabsorption along the CCD by selectively inhibiting ClC-K2 activity in intercalated cells via the adenosine 1 receptor (A_1R).⁴⁵⁸

As in other nephron segments, NaCl transport by the CNT and CCD is modulated by metabolites of arachidonic acid generated by cytochrome P450 monooxygenases. In particular, arachidonic acid inhibits ENaC channel activity in the rat CCD via generation of the epoxygenase product 11,12-EET (epoxyeicosatrienoic acid) by the CYP2C23 enzyme.⁴⁵⁹ Disruption of Cyp4a10, another P450 monooxygenase, in mice, results in salt-sensitive hypertension.⁴⁶⁰ Urinary excretion of 11,12-EET is reduced in these knockout mice, with a blunted effect of arachidonic acid on ENaC channel activity in the CCD.⁴⁶⁰ These mice also became normotensive after treatment with amiloride, indicative of *in vivo* activation of ENaC. It appears that deletion of Cyp4a10 reduces activity of the murine ortholog of rat CYP2C23 (CYP2C44 in mouse) and/or related epoxygenases via reduced generation of a ligand for PPAR α (peroxisome proliferator-activated receptor α) that induces epoxygenase activity.⁴⁶⁰ The mechanism(s) whereby 11,12-EET inhibits ENaC are unknown as yet. However, renal 11,12-EET production is known to be salt sensitive, suggesting that generation of this mediator may serve to reduce ENaC activity during high dietary NaCl intake.⁴⁵⁹

Finally, activation of PPAR γ by thiazolidinediones results in amiloride-sensitive hypertension, suggesting *in vivo* activation of ENaC.^{461,462} Thiazolidinediones (TZDs; e.g., rosiglitazone, pioglitazone, and troglitazone) are insulin-sensitizing drugs used for the treatment of type II diabetes. Treatment with these agents is frequently associated with fluid retention, suggesting an effect on renal NaCl transport. Indeed, selective deletion of the murine PPAR γ gene in principal cells abrogates the increase in amiloride-sensitive transport seen in response to TZDs.^{461,462} This effect by itself does not explain edema since primary ENaC activation leads to hypertension and hypokalemia, not to edema; therefore the mechanisms of TZD-induced edema are complex.⁴⁶³ Conversely, mice with disruption of α -ENaC specifically along CNT/CCD display blunted increases in total body water and extracellular fluid volume in response to rosiglitazone administration, suggesting the effects of PPAR γ are mediated through ENaC.⁴⁶⁴ TZDs appear to induce transcription of the genes encoding γ -ENaC and SGK1; SGK1 knockout mice display attenuated

TZD-associated edema.^{461,465,466} Notably, however, other studies have failed to detect an effect of TZDs on ENaC activity, which may instead activate a nonspecific cation channel within the IMCD.^{467,468} Regardless, the beneficial effect of spironolactone in patients with type 2 diabetes with TZD-associated volume expansion is consistent with *in vivo* activation of NaCl absorption in the distal nephron.⁴⁶⁹ In addition, the risk of peripheral edema is increased considerably in patients treated with both TZDs and insulin therapy. Insulin appears to activate ENaC via SGK1-dependent mechanisms; PPAR γ is required for the full activating effect of insulin on ENaC, such that this clinical observation may reflect synergistic activation of ENaC by insulin and TZDs.^{467,470,471}

POTASSIUM TRANSPORT

Maintenance of K⁺ balance is important for a multitude of physiologic processes. Changes in intracellular K⁺ affect cell volume regulation, regulation of intracellular pH, enzymatic function, protein synthesis, DNA synthesis, and apoptosis.¹⁴ Changes in the ratio of intracellular to extracellular K⁺ affect the resting membrane potential, leading to depolarization in hyperkalemia and hyperpolarization in hypokalemia. Thus disorders of extracellular K⁺ have a dominant effect on excitable tissues, chiefly heart and muscle. A growing body of evidence has also implicated hypokalemia and/or reduced dietary K⁺ in the pathobiology of hypertension, heart failure, and stroke; these and other clinical consequences of K⁺ disorders are reviewed in Chapter 16.

Potassium is predominantly an intracellular cation, with only 2% of total body K⁺ residing in the extracellular fluid. Extracellular K⁺ is maintained within a narrow range by three primary mechanisms. First, the distribution of K⁺ between the intracellular and extracellular space is determined by the activity of the Na⁺-K⁺-ATPase, the Na⁺-K⁺-2Cl⁻ cotransporter NKCC1, the four K⁺-Cl⁻ cotransporters, and a plethora of K⁺ channels. Skeletal muscle contains as much as 75% of total body potassium and exerts considerable influence on extracellular K⁺. Short-term and long-term regulation of muscle Na⁺-K⁺-ATPase plays a dominant role in determining the distribution of K⁺ between the intracellular and extracellular spaces; regulation of the uptake of K⁺ by skeletal muscle is reviewed in Chapter 16. Second, the colon absorbs and secretes K⁺, with considerable mechanistic and regulatory similarities to renal K⁺ secretion. K⁺ secretion in the distal colon is increased after dietary loading and in end-stage kidney disease (ESKD).^{14,472,473} However, the colon has a relatively limited capacity for K⁺ excretion. Therefore changes in renal K⁺ excretion play the dominant role in responding to changes in K⁺ intake, and hyperkalemia is most commonly observed in individuals with impaired kidney function. Regulated K⁺ secretion by the CNT and CCD plays a critical role in the response to hyperkalemia and K⁺ loading, while increases in K⁺ reabsorption by intercalated cells of the CCD and OMCD counteract hypokalemia or K⁺ deprivation.

This section reviews the mechanisms and regulation of transepithelial K⁺ transport along the nephron. Transport pathways for K⁺ also play important roles in renal NaCl transport, particularly within the TAL. Furthermore, Na⁺

absorption via ENaC in the aldosterone-sensitive distal nephron generates a lumen-negative PD that drives distal K⁺ excretion. These pathways are discussed in the previous section on renal NaCl transport; related issues relevant to K⁺ homeostasis *per se* are addressed here.

PROXIMAL TUBULE

The PT reabsorbs some 50% to 70% of filtered K⁺ (Fig. 6.23). PTs generate minimal transepithelial K⁺ gradients, and fractional reabsorption of K⁺ is similar to that of Na⁺.²⁷⁷ K⁺ absorption follows that of fluid, Na⁺, and other solutes, such that this nephron segment does not play a direct role in regulated renal excretion.^{474,475} Notably, however, changes in NaCl reabsorption by the PT have considerable effects on distal tubular flow and distal tubular Na⁺ delivery, with attendant effects on the excretory capacity for K⁺ (see later, “K⁺ Secretion by the Distal Convolved Tubule, Connecting Tubule, and Cortical Collecting Duct”).

The mechanisms involved in transepithelial K⁺ transport by the PT are not completely clear, although active transport does not appear to play a major role.^{475,476} Luminal barium has modest effects on transepithelial K⁺ transport, suggesting a component of transcellular transport via barium-sensitive K⁺ channels.⁴⁷⁷ However, the bulk of K⁺ transport is thought to occur via the paracellular pathway, driven by the lumen-positive PD in the mid to late PT (see Fig. 6.2).^{477,478} The total K⁺ permeability of the PT is thus rather high due to features of the paracellular pathway.^{477,478} The combination of luminal K⁺ concentrations that are about 10% higher than that of plasma, a lumen-positive PD of about 2 mV (see Fig. 6.2), and high paracellular permeability, leads to considerable paracellular absorption in this segment. This absorption has been proposed to occur via convective

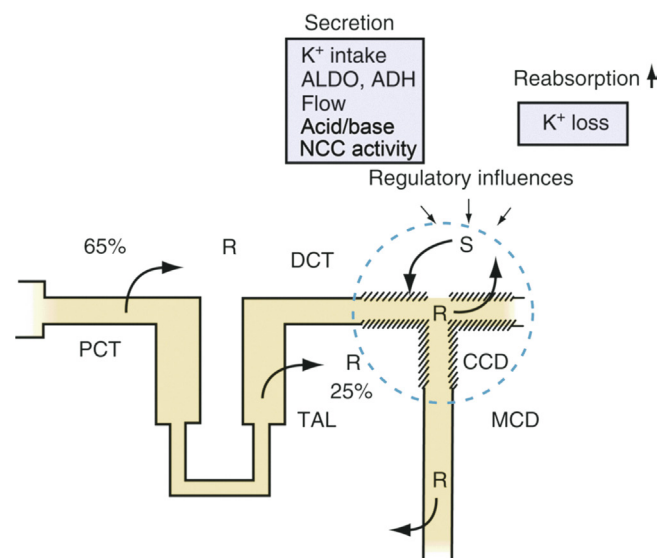


Fig. 6.23 K⁺ transport along the nephron. Approximately 90% of filtered K⁺ is reabsorbed by the PT and the loop of Henle. K⁺ is secreted along the initial and cortical collecting ducts. Net reabsorption occurs in response to K⁺ depletion, primarily within the medullary collecting duct. *ADH*, Antidiuretic hormone; *ALDO*, aldosterone; *CCD*, cortical collecting duct; *DCT*, distal convoluted tubule; *MCD*, medullary collecting duct; *PCT*, proximal convoluted tubule; *R*, reabsorption; *S*, secretion; *TAL*, thick ascending limb.

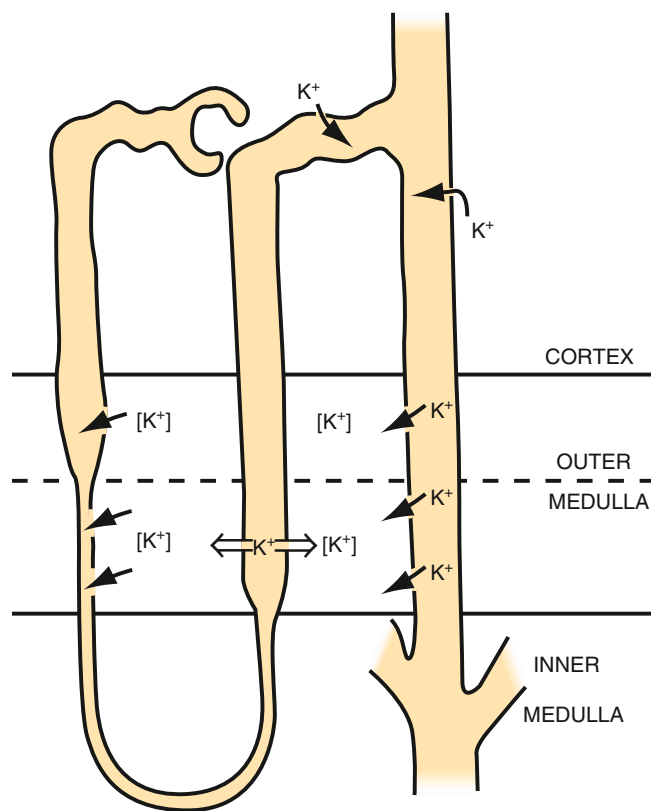


Fig. 6.24 Schematic representation of medullary K⁺ recycling. Medullary interstitial K⁺ increases considerably after dietary K⁺ loading due to the combined effects of secretion in the cortical collecting duct, absorption in the outer medullary collecting duct, thick ascending limb, and inner medullary collecting duct, and secretion in the descending thin limb. See text for details. (From Stokes JB. Consequences of potassium recycling in the renal medulla. Effects of ion transport by the medullary thick ascending limb of Henle's loop. *J Clin Invest.* 70:219–229, 1982.)

transport—solvent drag due to frictional interactions between water and K^+ —rather than diffusional transport.⁴⁷⁹ However, the primary pathway for water movement in the PT is transcellular, via aquaporin-1 and aquaporin-7 water channels in the apical and basolateral membrane.^{19,32,34} Therefore the apparent convective transport of K^+ likely constitutes so-called pseudosolvent drag, with hypothetical uncharacterized interactions between water traversing the transcellular route and diffusion of K^+ along the paracellular pathway.⁴⁷⁹

LOOP OF HENLE

Transport by the loop of Henle plays a critical role in medullary K^+ recycling (Fig. 6.24). Several lines of evidence have indicated that a considerable fraction of K^+ secreted by the CCD is reabsorbed by the medullary collecting ducts and then secreted into the late PT and/or descending thin limbs of long-looped nephrons.⁴⁸⁰ In K^+ -loaded rats, there is thus a doubling of luminal K^+ in the terminal thin descending limbs, with a sharp drop after inhibition of CCD K^+ secretion by amiloride.⁴⁸¹ Enhancement of CCD K^+ secretion by treatment with DDAVP also increases luminal K^+ in the descending thin limbs.⁴⁸² This recycling

pathway (secretion in CCD, absorption in OMCD and IMCD, secretion in descending thin limb) is associated with a marked increase in medullary interstitial K^+ . Passive transepithelial K^+ absorption by the thin ascending limb and active absorption by the TAL also contribute to this increase in interstitial K^+ (see Fig. 6.24).⁴⁸³ Specifically, the absorption of K^+ by the ascending thin limb, TAL, and OMCD exceeds the secretion by the descending thin limbs, thus trapping K^+ in the interstitium.

The physiologic significance of medullary K^+ recycling is not completely clear. However, an increase in interstitial K^+ concentration from 5 to 25 mmol/L dramatically inhibits Cl^- (and to a lesser extent, Na^+) transport by perfused TALs.⁴⁸³ This may contribute to the well-documented diuretic effects of a high- K^+ diet⁴⁸⁴ and would increase Na^+ delivery to the CNT and CCD, thus enhancing the lumen-negative PD in these tubules and increasing K^+ secretion.⁴⁸³ Alternatively, the marked increase in medullary interstitial K^+ after dietary K^+ loading may limit the difference between luminal and peritubular K^+ in the CCD, thus minimizing passive K^+ loss from the collecting duct.

K⁺ is secreted into the descending thin limbs by passive diffusion, driven by the high medullary interstitial K⁺ concentration. Descending thin limbs thus have a high-K⁺ permeability, without evidence for active transepithelial K⁺ transport.⁴⁸⁵ Transepithelial K⁺ absorption by ascending thin limbs has not to our knowledge been measured, but similar to NaCl transport (see earlier, “NaCl Transport by the Thin Ascending Limb”), it is presumably passive. Active transepithelial K⁺ transport across the TAL includes a transcellular component, via apical Na⁺-K⁺-2Cl⁻ cotransport mediated by NKCC2, and a paracellular pathway (see Fig. 6.14). Luminal K⁺ channels play a critical role in generating the lumen-positive PD in the TAL, as summarized earlier (see “NaCl Transport by the Thick Ascending Limb: Apical K⁺ Channels”). Secretion of K⁺ through these may also play a role in the response to high dietary K⁺. Patch-clamp analysis of split-open TALs revealed that 70 pS ROMK exhibited a higher P_o in mice placed on a low-Na⁺/high-K⁺ diet.⁴⁸⁶ This may be dependent on NKCC2 activity since micropuncture showed that furosemide increased K⁺ secretion in the early distal tubule in mice on a normal diet but decreased it in mice on the low-Na⁺/high-K⁺ diet.

DISTAL NEPHRON

K⁺ SECRETION

Approximately 90% of filtered K^+ is reabsorbed by the PT and loop of Henle (see Fig. 6.23); the fine-tuning of renal K^+ excretion occurs in the remaining distal nephron. The bulk of regulated secretion occurs in the DCT2 and principal cells within the CNT and CCD, whereas K^+ reabsorption primarily occurs in the OMCD (see later). Generally, the CCD is considered the primary site for distal K^+ secretion, partially due to the greater ease with which this segment is perfused and studied. However, as is the case for NaCl absorption (see “Connecting Tubules and the Cortical Collecting Duct: Apical Na^+ Transport”), the bulk of distal K^+ secretion appears to occur before the CCD, within the DCT2 and CNT under basal conditions.^{277,361}

In principal cells, apical Na^+ entry via ENaC generates a lumen-negative PD, which drives passive K^+ exit through

apical K^+ channels. Distal K^+ secretion is therefore dependent on delivery of adequate luminal Na^+ to the CNT and CCD, essentially ceasing when luminal Na^+ drops below 8 mmol/L.⁴⁸⁷⁻⁴⁸⁹ At steady state, urinary K^+ excretion equals intake, but Na^+ intake can affect K^+ excretion over the short term, such that excretion is enhanced by excess Na^+ intake and reduced by Na^+ restriction.^{487,488} Secreted K^+ enters principal cells via the basolateral $Na^+-K^+-ATPase$, which also generates the gradient that drives apical Na^+ entry via ENaC (see Fig. 6.23).

Two major subtypes of apical K^+ channels function in K^+ secretion by the CNT and CCD; a small-conductance (SK), 30 pS channel and a large-conductance, Ca^{2+} -activated, 150 pS (maxi-K or BK) channel (Fig. 6.17).^{177,361,490} The density and high P_o of the SK channel are sufficient to mediate the bulk of K^+ secretion in the CCD under baseline conditions.⁴⁹¹ Notably, SK channel density is considerably higher in the CNT than CCD, consistent with the greater capacity for Na^+ absorption and K^+ secretion in the CNT.³³ The SK has been identified as ROMK. ROMK protein has been localized at the apical membrane of principal cells,^{176,492} and SK channel activity is absent from apical membranes of the CCD in ROMK (encoded by *Kcnj1*) knockout mice.¹⁷⁷ ROMK knockout mice are normokalemic with increased excretion of K^+ , illustrating the considerable redundancy in distal K^+ secretory pathways; distal K^+ secretion in these mice is mediated by apical BK channels (see later).^{177,493} However, dietary K^+ loading induced hyperkalemia in ROMK1 knockout mice, consistent with a role in K^+ secretion along the CCD.¹⁷⁹ Of interest, loss-of-function mutations in human *KCNJ1* genes are associated with Bartter syndrome; ROMK expression is critical for the 30 pS and 70 pS channels that generate the lumen-positive PD in the TAL (see Fig. 6.14).^{177,185} These patients typically have slightly higher serum K^+ levels than those with other forms of Bartter syndrome. Patients with severe neonatal hyperkalemia have also been described, presumably the result of a transient developmental deficit in apical BK channel activity.^{14,182} Note that Bartter syndrome due to *KCNJ1* disruption may specifically reflect a defect in ROMK2 and/or ROMK3 function since, consistent with absence of ROMK1 along TAL, ROMK1 knockout mice do not display a Bartter syndrome phenotype.¹⁷⁹

The apical Ca^{2+} -activated BK channel plays a critical role in flow-dependent K^+ secretion by the CNT and CCD.⁴⁹⁰ BK channels have a heteromeric structure, with α -subunits that form the channel pore and modulatory β -subunits that affect the biophysical, regulatory, and pharmacologic characteristics of the channel complex.⁴⁹⁰ BK α -subunit transcripts are expressed in multiple segments, and channel protein is detectable at the apical membrane of principal and intercalated cells in the CCD and CNT.⁴⁹⁰ The β -subunits are differentially expressed within the distal nephron. Thus β_1 -subunits are restricted to the CNT, with no expression in intercalated cells, whereas β_4 -subunits are detectable at the apical membranes of the TAL, DCT, and intercalated cells.^{490,494} Increased distal flow has a well-established stimulatory effect on K^+ secretion, due in part to enhanced delivery and absorption of Na^+ and to increased removal of secreted K^+ .^{487,488} The pharmacology of flow-dependent K^+ secretion in the CCD is consistent with dominant involvement of BK channels, and flow-dependent K^+ secretion is reduced in mice with targeted

deletion of the α_1 - and β_1 -subunits.^{490,495-497} Both mice strains develop hyperaldosteronism that is exacerbated by a high- K^+ diet, leading to hypertension in the α_1 -subunit knockout.⁴⁹⁷ Disruption of the β_2 subunit also leads to hyperaldosteronism, but flow-induced K^+ secretion is normal, suggesting compensation by other isoforms.⁴⁹⁸ Ca^{2+} -dependence of BK activation involves TRPV4. A high- K^+ diet increases TRPV4 expression and leads to its redistribution to the apical membrane in CCD, and TRPV4 knockout mice display decreased BK activity in CCD and hyperkalemia after dietary K^+ loading.⁴⁹⁹⁻⁵⁰¹

The greater density of BK channels in intercalated cells in both the CCD and CNT^{502,503} suggested a major role for intercalated cells in K^+ secretion. However, the much lower density of $Na^+-K^+-ATPase$ activity in intercalated cells is considered inadequate to support K^+ secretion across the apical membrane.⁵⁰⁴ The evidence supports a role for the basolateral $Na^+-K^+-2Cl^-$ cotransporter NKCC1 in K^+ secretion mediated by apical BK channels. NKCC1 expression at the basolateral membrane of intercalated cells provides an alternative entry pathway for basolateral K^+ secreted at the apical membrane.^{505,506} A basolateral Na^+ pump, the ouabain-insensitive furosemide-sensitive $Na^+-ATPase$, whose activity has been detected in cell culture models of intercalated cells, may recycle Na^+ across the basolateral membrane in the absence of significant $Na^+-K^+-ATPase$ activity.⁵⁰⁵ At the apical membrane, BK-mediated K^+ secretion is only partially dependent on luminal Na^+ ; K^+ secretion would eventually hyperpolarize the membrane in the absence of apical Na^+ entry.⁵⁰⁷ An intriguing possibility is that apical Cl^- channels allow for the parallel secretion of K^+ and Cl^- in intercalated cells.⁵⁰⁸

Mice with a targeted deletion of the β_4 -subunit exhibit normal K^+ excretion on a normal diet⁵⁰⁴ but develop hyperkalemia with a blunted increase in K^+ excretion and urinary flow rates when fed a high- K^+ diet, which increases urinary and tubular flow rates and tubular shear stress. Intercalated cells protrude into the lumen of distal tubules; flow-activated BK channels reduce the cell volume of intercalated cells after K^+ loading, reducing tubular resistance, increasing tubular flow rates, and increasing distal K^+ secretion.⁵⁰⁴ Intercalated cells from β_4 -knockouts fail to significantly decrease cell volume in response to high- K^+ diet.

The physiologic rationale for the presence of two apical secretory K^+ channels, ROMK and BK channels, is not completely clear. However, the high density and higher P_o of ROMK channels are perhaps better suited for a role in basal K^+ secretion, with additional recruitment of the higher capacity, flow-activated BK channels when additional K^+ secretion is required.⁴⁹⁰ BK channels may function in partially Na^+ -independent K^+ secretion by intercalated cells, with ROMK functioning in ENaC- and Na^+ -dependent K^+ excretion by DCT, CNT, and CCD cells. Regardless, at the whole-organ level, the two K^+ channels can substitute for one another, with BK-dependent K^+ secretion in ROMK knockout mice and an upregulation of ROMK in the distal nephron of α_1 -subunit BK knockouts.^{493,496}

Other K^+ channels reportedly expressed at the luminal membranes of the CNT and CCD include voltage-sensitive channels such as KV1.3, the calcium-activated, small-conductance SK3 channel; and double-pore K^+ channels,

such as TWIK-1 and KCNQ1.^{509,512} KCNQ1 is expressed at the apical membrane of principal cells in the CCD, whereas TWIK-1 is expressed at the apical membrane of intercalated cells.^{512,513} The roles of these channels in renal K⁺ handling are unknown. However, KV1.3 may play a role in distal K⁺ secretion since specific blockade with margatoxin reduces K⁺ secretion in CCDs of rat kidneys from animals on a high-K⁺ diet.⁵¹⁴ Other apical K⁺ channels in the distal nephron may subserve other physiologic functions. The apical KV1.1 channel may contribute to Mg²⁺ transport by the DCT by hyperpolarizing the apical membrane, thus increasing the driving force for Mg²⁺ influx via TRPM6 (transient receptor potential cation channel 6); missense mutations in KV1.1 are a cause of genetic hypomagnesemia.⁵¹⁵

Basolateral membrane K⁺ channels appear to set the resting potential of the basolateral membrane of principal cells and function in K⁺ secretion and Na⁺ absorption (by recycling K⁺ to maintain Na⁺-K⁺-ATPase activity) at the apical membrane. A single predominant basolateral activity has been identified in principal cells from the rat CCD using whole-cell recording techniques under conditions in which ROMK is inhibited (low intracellular pH or presence of the ROMK inhibitor tertiapin-Q).⁵¹⁶ This basolateral current is tetraethylammonium (TEA) insensitive, barium sensitive, and acid sensitive (pK_a ≈ 6.5), with a conductance of about 17 pS and weak inward rectification. Candidate inward-rectifying K⁺ channel subunits localized at the basolateral membrane of the CCD include KIR4.1, KIR5.1, KIR7.1, and KIR2.3.⁵¹⁶ KIR4.1 and KIR5.1 channels generate a predominant 40 pS basolateral K⁺ channel in murine principal cells,⁵¹⁷ with both KIR4.1 and KIR5.1 participating in generating the membrane potential that permits Na⁺ entry through ENaC.^{219,308} Notably, basolateral K⁺ channel activity increases on a high-K⁺ diet, suggesting a role in transepithelial K⁺ secretion.⁵¹⁶ Disruption of KIR5.1 in rats reveals it plays a critical role in collecting duct function, particularly with respect to maintenance of K⁺ homeostasis.³⁰⁸ Activation of KIR4.1/KIR5.1 by insulin and IGF-1 may also facilitate Na⁺ reabsorption along the CCD by hyperpolarizing the basolateral membrane.⁵¹⁸

Apical K⁺-Cl⁻ cotransport (or functionally equivalent pathways) is also implicated in distal K⁺ secretion.^{66,487,519,520} In rat distal tubules, anion-dependent K⁺ secretion that is not influenced by luminal Ba²⁺ has been recorded, suggesting that it does not involve apical K⁺ channel activity.⁵²¹ Cl⁻-coupled K⁺ secretion is detectable in the DCT and early CNT.⁵²² In addition, similar pathways are detectable in rabbit CCD, where a decrease in luminal Cl⁻ concentration from 112 to 5 mmol/L increases K⁺ secretion by 48%.⁵²³ A reduction in basolateral Cl⁻ also decreases K⁺ secretion without an effect on transepithelial voltage or Na⁺ transport, and a lumen to bath Cl⁻ gradient reverses K⁺ flux to cause K⁺ absorption.⁵²³ In perfused CCDs from rats treated with mineralocorticoid, vasopressin increases K⁺ secretion; because this increase in K⁺ secretion is resistant to luminal Ba²⁺ (2 mmol/L), vasopressin may stimulate Cl⁻-dependent K⁺ secretion.^{14,524} Pharmacologic study results of perfused tubules are consistent with K⁺-Cl⁻ cotransport mediated by the KCCs. A study using an antibody validated in knockout mice suggests that KCC3a, encoded by the *Slc12a9* gene, may mediate electroneutral secretion of KCl

in type B or type-non-A/non-B intercalated cells along the CNT.⁵²⁵ Other functional possibilities for Cl⁻-dependent K⁺ secretion include parallel operation of apical H⁺-K⁺-exchange and Cl⁻-HCO₃⁻ exchange in type B intercalated cells.⁵¹⁹

A provocative study by Frindt and Palmer serves to underline the importance of ENaC-independent K⁺ excretion (see also “Integrated NaCl and K⁺ Transport in the Distal Nephron”).⁵²⁶ Rats were infused with sufficient amiloride expected to inhibit more than 98% of ENaC activity. Whereas amiloride almost abolished K⁺ excretion in rats on a normal K⁺ intake, acute and long-term high-K⁺ diets led to an increasing fraction of K⁺ excretion that was independent of ENaC activity (≈50% after 7–9 days on a high-K⁺ diet).

K⁺ REABSORPTION BY THE COLLECTING DUCT

In addition to K⁺ secretion, the distal nephron is capable of considerable reabsorption, primarily during restriction of dietary K⁺.²⁷⁷⁻²⁷⁹ This reabsorption is accomplished largely by intercalated cells in the OMCD via the activity of apical H⁺/K⁺-ATPase pumps. Under K⁺-replete conditions, apical H⁺/K⁺-ATPase activity recycles K⁺ with an apical K⁺ channel, without reabsorption. Under K⁺-restricted basolateral conditions, transepithelial K⁺ transport appears to occur, with K⁺ transported by the H⁺/K⁺-ATPase exiting intercalated cells via a K⁺ channel.⁵²⁷

H⁺-K⁺-ATPase holoenzymes are members of the P-type family of ion transport ATPases, which also include subunits of the basolateral Na⁺-K⁺-ATPase.⁵²⁸ HKα₁ and HKα₂ are also referred to as the gastric and colonic subunits, respectively; humans also have an HKα₄-subunit.^{528,529} A specific HKβ-subunit interacts with the HKα-subunits to ensure delivery to the cell surface and complete expression of H⁺-K⁺-ATPase activity; HKα₂- and HKα₄-subunits are also capable of interaction with Na⁺-K⁺-ATPase β-subunits.^{14,530} Within the kidney, the HKα₁-subunit is expressed at the apical membrane of at least a subset of type A intercalated cells.⁵²⁹ HKα₂-subunit distribution in the distal nephron is more diffuse, with robust expression at the apical membrane of types A and B intercalated cells and connecting segment cells and lesser expression in principal cells.⁵³¹⁻⁵³³ The human HKα₄-subunit is reportedly expressed in intercalated cells.⁵²⁹

HKα₁- and HKα₂-subunits are both constitutively expressed in the distal nephron, but perfusion studies suggest holoenzymes containing the HKα₁-subunit predominate at baseline, based on omeprazole/SCH-28080-sensitive and ouabain-resistant H⁺-K⁺-ATPase activity.⁵³⁴ K⁺ deprivation increases the overall activity of H⁺-K⁺-ATPase in the collecting duct, with the emergence of ouabain-sensitive H⁺-K⁺-ATPase activity consistent with activation of HKα₂-subunits.¹⁴ K⁺ restriction also induces a dramatic upregulation of HKα₂-subunit expression in the outer and inner medulla during K⁺ depletion; HKα₁-subunit expression is unaffected.¹⁴ HKα₂-subunit knockout mice exhibit lower plasma and muscle K⁺ when K⁺ is deprived. However, this appears to reflect loss of K⁺ in the colon because renal K⁺ excretion is appropriately reduced.²⁸⁰ The lack of an obvious renal phenotype in HKα₁- or HKα₂-subunit knockout mice may reflect redundancy in the expression of HKα-subunits in the distal nephron.^{280,535} However, data from HKα₁- and HKα₂-subunit knockout

mice suggest ATPase-type-independent compensatory mechanisms.⁵³⁶

The importance of K^+ reabsorption mediated by the collecting duct is dramatically illustrated by the phenotype of transgenic mice with overexpression of a gain-of-function mutation in H^+-K^+ -ATPase in which a tyrosine-to-alanine mutation within the C-terminal tail of the $HK\beta$ -subunit abrogates regulated endocytosis from the plasma membrane. These mice have higher plasma K^+ than wild-type littermates, with approximately half the fractional excretion of K^+ .⁵³⁷

The H^+-K^+ -ATPase may also serve to recycle K^+ , facilitating downregulation of ENaC by the purinergic signaling system during high dietary Na^+ intake. In contrast to wild-type mice, ENaC activity in $HK\alpha_1$ knockout mice is uncoupled from dietary Na^+ intake, and urinary [ATP] is not increased by a high Na^+ diet.⁵³⁸

REGULATION OF DISTAL K^+ TRANSPORT

MODULATION OF RENAL OUTER MEDULLARY POTASSIUM CHANNEL ACTIVITY

ROMK and other Kir channels are inward rectifying—that is, K^+ flows into the cell more readily than out of the cell (Kir, inward-rectifying renal K^+ channel). While outward conductance is usually less than inward conductance, K^+ efflux through ROMK predominates in the CNT and CCD because the membrane potential is more positive than the equilibrium potential for K^+ . Intracellular magnesium (Mg^{2+}) and polyamines play key roles in inward rectification, with binding and blocking of the channel pore, mediated by a single transmembrane residue, asparagine-171 in ROMK1.⁵³⁹⁻⁵⁴¹ Intracellular Mg^{2+} in the TAL, DCT, CNT, and principal cells is thought to have a significant effect on ROMK activity because it inhibits outward ROMK-dependent currents in principal cells.⁵⁴² The blocking affinity of Mg^{2+} is enhanced at lower extracellular K^+ concentrations, which should aid in reducing K^+ secretion during hypokalemia and K^+ deficiency.⁵⁴² Reduced intracellular Mg^{2+} block may explain the hypokalemia associated with hypomagnesemia, wherein distal K^+ secretion is enhanced.^{541,542} Increased Na^+ delivery due to lower reabsorption through NCC may also contribute.⁵⁴³

In addition to inward rectification, the endogenous ROMK channels in the TAL and principal cells exhibit a high P_o . This is maintained by the combined effects of binding of PIP2 to the channel, direct channel phosphorylation by PKA, ATP binding to the ROMK-CFTR complex, and cytoplasmic pH. PIP2 binding to ROMK maintains the channel in an open state, whereas cytoplasmic acidification inhibits the channel.⁵⁴⁴ PKA-mediated phosphorylation of several serines (S25, S200, and S294) in the ROMK2 isoform is required for full channel function.²⁵⁰ Phosphorylation of the N-terminal S25 site increases channel localization at the apical membrane.⁵⁴⁵ Phosphorylation of S200 and S294 maintains the channel in a high P_o state, in part by modulating the effects of PIP2, ATP, and pH.^{187,252,253}

Because ROMK channels exhibit such a high P_o , physiologic regulation of the channel is primarily achieved by regulated changes in the number of active channels on the plasma membrane. The associated mechanisms are

discussed in the context of the adaptation to K^+ loading and hyperkalemia and K^+ deprivation and hypokalemia.

ALDOSTERONE AND K^+ LOADING

Aldosterone plays an important role in K^+ homeostasis. Aldosterone release by the adrenal gland is induced by hyperkalemia and/or a high- K^+ diet,⁵⁴⁶ and it exerts a potent kaliuretic effect.⁵⁴⁷ There is a clear relationship at all levels of serum K^+ between circulating levels of the hormone and the ability to excrete K^+ .

Aldosterone has no effect on the density of apical ROMK channels in the CCD but increases the density of ENaC in the CNT and CCD.⁵⁴⁸ Aldosterone activates ENaC via interrelated effects on the synthesis, trafficking, and membrane-associated activity of the channel subunits (see “Regulation of NaCl Transport in the Connecting Tubule and Cortical Collecting Duct”). The more lumen-negative PD resulting from stimulation of ENaC activity thus promotes K^+ secretion by principal cells.

A component of the adaptation to a high- K^+ intake is aldosterone independent. For example, a high- K^+ diet increases apical Na^+ reabsorption and K^+ secretion in the CCD in adrenalectomized animals.⁵⁴⁹ This is most likely through activation of the Na^+-K^+ -ATPase inducing secondary activation of apical Na^+ and K^+ channels.⁵⁵⁰ Increased dietary K^+ increases the apical densities of ROMK and ENaC in the CCD.^{548,551} Notably, this increase in ENaC and ROMK density in the CCD occurs within hours of consuming a high- K^+ diet, with minimal increases in circulating aldosterone (Table 6.1).⁵⁵² In contrast, a week of low NaCl intake, with almost a thousand-fold increase in aldosterone, or 2 days of aldosterone infusion sufficient to induce hypokalemia, have no effect on ROMK channel density (see Table 6.1).⁵⁵² Measurement of whole-cell currents using the ROMK inhibitor tertiapin-Q indicates an upregulation of ROMK activity in the CNT by a high- K^+ diet.⁵⁵³ More rapid effects promoting K^+ secretion through ROMK in response to acute K^+ intake may involve activation of mTORC2. A combination of in vitro and in vivo experiments showed that independent of aldosterone secretion, K^+ directly acts on principal cells to activate ENaC via mTORC2-SGK1.^{554,555}

Flow-stimulated K^+ secretion through BK channels along the CCD of mice and rats is enhanced on a high- K^+ diet, with an absence of flow-dependent K^+ secretion in rats on a low- K^+ diet.^{493,556} This is accompanied by the appropriate changes in transcript levels for α - and β_{2-4} -subunits of the BK channel in microdissected CCDs (β_1 -subunits are restricted to the CNT).⁴⁹⁰ The α -subunit protein is largely intracellular in K^+ -restricted rodents but strongly apical in K^+ -loaded rodents.^{499,556} Aldosterone does not contribute to BK channel activity or expression in response to a high- K^+ diet.⁵⁵⁷

The changes in trafficking and/or activity of the ROMK channel that are induced by dietary K^+ appear in large part to involve tyrosine phosphorylation and dephosphorylation (see later). However, WNK1 and WNK4 kinases, initially identified as causative genes for FHHt (see also “Regulation of NaCl Transport in the Distal Convoluted Tubule”) may also contribute. WNK4 coexpression dramatically lowers ROMK membrane expression in *Xenopus* oocytes; FHHt-associated mutations dramatically increase this effect, suggesting direct inhibition of K^+ secretion in FHHt.⁵⁵⁸

Table 6.1 Effect of High-K⁺ Diet, Aldosterone, and/or NaCl Restriction on Small-Conductance Channel Density in the Rat Cortical Collecting Duct

Parameter	K ⁺ Channel Density (μm ²)	Plasma Aldosterone (ng/dL)	Plasma K (mmol/L)
Control	0.41	15	3.68
High-K ⁺ diet, 6h	1.51	36	NM
High-K ⁺ diet, 48h	2.13	98	4.37
Low-Na ⁺ diet, 7 days	0.48	1260	NM
Aldosterone infusion, 48h	0.44	550	2.44
Aldosterone + high-K ⁺ diet	0.32	521	3.80

NM, Not measured.

Modified from Palmer LG, Frindt G: Regulation of apical K channels in rat cortical collecting tubule during changes in dietary K intake. *Am J Physiol* 277:F805-F812, 1999.

Consistent with this, a patch-clamp study that showed DCT2/CNT isolated from transgenic mice expressing FHHt-causing mutant WNK4 have lower ENaC and ROMK activity.⁵⁵⁹ The aldosterone-induced SGK1, which phosphorylates WNK4, is activated by the upstream kinase mTORC2. Disruption of mTORC2 in renal epithelia in mice causes profound hyperkalemia when they are placed on a high-K⁺ diet. Patch-clamp analysis showed that in CNT/CCD, Ba²⁺-sensitive K⁺ currents were almost absent. SGK1 phosphorylation was ablated, suggesting that hyperkalemia may result from unchecked WNK4-mediated ROMK endocytosis.⁵⁶⁰ While this study found ENaC activity was unaltered, a more recent study, using a different model of renal epithelial mTORC2 disruption and different experimental conditions, suggested that lower ENaC activity may contribute to the observed hyperkalemia.⁵⁵⁴

The full-length L-WNK1 inhibits ROMK activity by inducing endocytosis of the channel protein, but whether kinase activity is required for this effect is controversial.⁵⁶¹⁻⁵⁶³ L-WNK1 and WNK4 induce endocytosis of ROMK via interaction with intersectin, a multimodular endocytic scaffold protein,⁵⁶⁴ and the clathrin adaptor protein, autosomal recessive hypercholesterolemia (ARH), is required for basal and L-WNK1-stimulated ROMK endocytosis.⁵⁶⁵

The shorter KS-WNK1 isoform, which lacks the kinase domain, appears to inhibit the effect of L-WNK1 on ROMK.^{562,563} The ratio of KS-WNK1 to L-WNK1 transcripts is reduced by K⁺ restriction (greater endocytosis of ROMK) and increased by K⁺ loading (reduced endocytosis of ROMK), suggesting that this ratio between KS-WNK1 and L-WNK1 functions as a type of switch to regulate distal K⁺ secretion.^{562,563,566} The inhibitory effect of KS-WNK1 tracks to the first 253 amino acids of the protein, encompassing the initial 30 amino acids unique to this isoform and an adjacent autoinhibitory domain.⁵⁶⁷ Consistent with an important inhibitory effect of KS-WNK1 on ROMK, transgenic mice overexpressing this region of KS-WNK1 have lower serum K⁺, higher fractional excretion of K⁺, and increased ROMK expression at the apical membrane of CNT and CCD cells.⁵⁶⁷

The BK channel is also regulated by the WNK kinases. WNK4 inhibits BK channel activity and protein expression, whereas FHHt-associated mutations in WNK4 also enhance the inhibitory effect via ubiquitination.⁵⁶⁸⁻⁵⁷⁰ A

high-K⁺ diet increases L-WNK1 expression selectively in intercalated cells.⁵⁷¹ Intercalated cell-specific L-WNK1 knockout mice display lower BK activity and higher plasma K⁺ on a high-K⁺ diet compared with wild type.⁵⁷² L-WNK1 mediated activation of SPAK, which then inhibits ERK1/2 signaling-mediated degradation of BK channels, is likely to contribute.^{573,574}

K⁺ DEPRIVATION

A reduction in dietary K⁺ leads to a dramatic drop in urinary K⁺ excretion within 24 hours.^{566,575} This is due to both an induction of reabsorption by intercalated cells in the OMCD and a reduction in ROMK activity in principal cells.^{14,278,279} The mechanisms involved in K⁺ reabsorption by intercalated cells are discussed earlier; notably, H⁺/K⁺-ATPase activity in the collecting duct does not appear to be regulated by aldosterone.⁵⁷⁶

Considerable progress has been made in defining the signaling pathways that regulate ROMK activity in response to changes in dietary K⁺. ROMK apical abundance is markedly lower along the CCD in K⁺-depleted animals and higher in animals on a high-K⁺ diet.^{551,577} ROMK membrane insertion and hence activity in response to dietary K⁺ are modulated by tyrosine phosphorylation. Phosphorylation of tyrosine residue 337 stimulates endocytosis, and dephosphorylation induces exocytosis.⁵⁷⁸⁻⁵⁸⁰ Levels of protein tyrosine phosphatase-1D do not vary with K⁺ intake, but intrarenal activity of the cytoplasmic tyrosine kinases c-src and c-yes are inversely related to dietary K⁺ intake, with a decrease under high-K⁺ conditions and a marked increase after several days of K⁺ restriction.^{14,581} c-Src colocalizes with ROMK in the TAL and principal cells of the CCD.⁵⁵¹ Moreover, inhibition of protein tyrosine phosphatase activity dramatically increases the proportion of intracellular ROMK in the CCD of animals on a high-K⁺ diet.⁵⁵¹

The neurohumoral factors that induce the K⁺-dependent trafficking and expression of apical ROMK and BK channels have only come into focus rather recently.^{551,556,577} Several studies have implicated the intrarenal generation of superoxide anions in the activation of cytoplasmic tyrosine kinases.⁵⁸²⁻⁵⁸⁴ Potential candidates for the upstream kaliuretic factor include AngII and insulin-like growth factor-1 (IGF-1).⁵⁸² AngII inhibits ROMK activity in K⁺-restricted rats but not rats on a normal K⁺ diet.⁵⁸⁵ This inhibition involves

downstream activation of superoxide production and c-src activity; induction of AngII by a low-K⁺ diet thus appears to play a major role in reducing distal tubular K⁺ secretion.⁵⁸⁶

Reports of transient postprandial kaliuresis in sheep, independent of changes in plasma K⁺ or aldosterone, have suggested that an enteric or hepatoporal K⁺ sensor controls kaliuresis via a sympathetic reflex; tissue kallikrein has recently emerged as a candidate mediator for this postprandial kaliuresis (see later).⁵⁸⁷ Such a feed-forward control has the theoretical advantage of greater stability because it operates before changes in plasma K⁺.⁵⁸⁸ Changes in ROMK phosphorylation status and insulin-sensitive muscle uptake can be seen in K⁺-deficient animals in the absence of a change in plasma K⁺, suggesting that upstream activation of the major mechanisms that serve to reduce K⁺ excretion (reduced K⁺ secretion in the CNT and CCD, decreased peripheral uptake, and increased K⁺ reabsorption in the OMCD) does not require changes in plasma K⁺.⁵⁸⁹ Consistent with this hypothesis, moderate K⁺ restriction, without an associated drop in plasma K⁺, is sufficient to induce AngII-dependent superoxide generation and c-src activation, leading to inhibition of ROMK activity.⁵⁸⁶

VASOPRESSIN

Vasopressin has a well-characterized stimulatory effect on K⁺ secretion by the distal nephron.^{482,590} From an evolutionary viewpoint, this serves to preserve K⁺ secretion during dehydration and extracellular volume depletion, when circulating levels of vasopressin are high and tubular delivery of Na⁺ and fluid is reduced. Stimulation of basolateral V₂Rs activates ENaC, increasing the driving force for K⁺ secretion by principal cells; the relevant mechanisms have been discussed earlier in this chapter (see “Regulation of NaCl Transport in the Connecting Tubule and Cortical Collecting Duct: Vasopressin and Other Factors”). In addition, vasopressin activates ROMK directly in the CCD, as does cAMP.^{491,591} PKA-mediated phosphorylation of three serine residues (S25, S200, and S294 in the ROMK2 isoform) is required for full activity in *Xenopus* oocytes (see “Regulation of NaCl Transport by the Thick Ascending Limb: Activating Influences”). Finally, stimulation of luminal V₁ receptors also stimulates K⁺ secretion in the CCD via activation of BK channels.⁵⁹²

TISSUE KALLIKREIN

The serine protease tissue kallikrein (TK) is involved in the generation of kinins, ultimately stimulating the formation of bradykinin.⁵⁹³ Within the kidney, TK is synthesized in CNT cells and released into the tubular lumen and peritubular interstitium. Data have revealed a role for TK in postprandial kaliuresis.⁵⁹³ Oral K⁺-Cl⁻ loading leads to a spike in urinary K⁺ and TK excretion in rats, mice, and humans.⁵⁹³ The increase in urinary TK after K⁺ loading is not accompanied by changes in urinary aldosterone and can be detected in aldosterone synthase knockout mice.⁵⁹⁴ Mice deficient in TK demonstrate postprandial hyperkalemia, indicating a role in postprandial kaliuresis. This transient hyperkalemia is accompanied by a marked increase in K⁺ reabsorption by perfused CCDs due to higher HKα₂-subunit expression and H⁺/K⁺-ATPase activity. Luminal but not basolateral TK administration lowers CCD H⁺/K⁺-ATPase activity in TK knockout mice, consistent with direct proteolytic activation. There is also a marked increase

in Na⁺ reabsorption by perfused CCDs from TK knockout mice, without development of a lumen-negative PD; this is consistent with an increased activity of the electroneutral NaCl cotransport mediated by NHE2 and the SLC26A4 Cl⁻-HCO₃⁻ exchanger (see later also⁴⁰²). This electroneutral transport pathway is inhibited by bradykinin, so activation by TK deletion presumably reflects loss of tonic inhibition by TK-generated bradykinin.³⁹⁸ TK also mediates proteolytic cleavage of the γ-subunit of ENaC, with reduced ENaC activity in TK-deficient mice; net Na⁺ balance is thus neutral in these mice.⁵⁹⁵

In summary, TK secretion from CNT cells is induced by oral K⁺-Cl⁻ loading, causing proteolytic activation of ENaC and thus an increase in ENaC-driven K⁺ secretion and bradykinin-dependent inhibition of electroneutral NaCl cotransport in the CCD.^{398,400,595} There is consequently a further augmentation of electrogenic Na⁺ transport (favoring K⁺ secretion) and direct luminal inhibition of H⁺/K⁺-ATPase activity and thus a decrease or tonic inhibition of K⁺ reabsorption. TK is likely the postprandial factor functioning in feed-forward control of plasma K⁺.^{587,588}

INTEGRATED NaCl AND K⁺ TRANSPORT IN THE DISTAL NEPHRON

Evidence from in vivo studies in mice with disruption of one or more genes on control diets, modified diets, or after administration of drugs or hormones has clearly demonstrated that effects in one segment exert compensatory effects to maintain homeostasis. For example, pendrin knockout mice with combined genetic ablation of NCC⁵⁹⁶ or administration of hydrochlorothiazide⁵⁹⁷ display salt-wasting and volume contraction. Similarly, ENaC blockade with amiloride leads to enhanced natriuresis in NCC knockout mice.⁵⁹⁸ In mice with disruption of MRs specifically in renal epithelia, while activities of both ENaC and NCC are diminished, the effect on NCC is secondary to the hyperkalemia induced by lower ENaC activity.^{320,321} These mice are hypotensive, but normalization of plasma [K⁺] by dietary K⁺ restriction stimulates NCC activity resulting in normalization of blood pressure. These interactions between segments are likely to be like those seen with diuretic treatment and may contribute to diuretic resistance (see Chapter 50 for discussion on diuretics). Other examples of this intersegment crosstalk are described in more depth later.

In the classical model of renal K⁺ secretion, the lumen-negative PD generated by Na⁺ entry via ENaC induces the exit of K⁺ via apical K⁺-selective channels. This scheme explains much of the known physiology and pathophysiology of renal K⁺ secretion yet has several key consequences that bear emphasis. First, enhanced NaCl reabsorption upstream of the CNT and CCD will reduce the delivery of luminal Na⁺ to the CNT and CCD, decrease the lumen-negative PD, and thus decrease K⁺ secretion; K⁺ secretion by the CCD essentially stops when luminal Na⁺ drops below 8 mmol/L.⁴⁸⁷⁻⁴⁸⁹ The increasingly refined phenotypic understanding of FHHt, caused by kinase-induced gain of function of the DCT, underscores that variation in NCC-dependent NaCl absorption has profound effects on the ability to excrete dietary K⁺ (see Fig. 6.18).³³⁵ Second, aldosterone is a kaliuretic hormone, induced by hyperkalemia. However,

under certain circumstances associated with induction of aldosterone, such as dietary Na^+ restriction, Na^+ balance is maintained without effects on K^+ homeostasis. How the kidney independently regulates NaCl and K^+ handling by the aldosterone-sensitive distal nephron, the “aldosterone paradox,” has just been unraveled. The major factors in the integrated control of NaCl and K^+ transport include ENaC -independent K^+ excretion within the distal nephron and the differential regulation of various signaling pathways by aldosterone, AngII , and dietary K^+ ; electroneutral thiazide-sensitive NaCl transport within the CCD may also contribute.^{361,398-400,599,600}

Thiazide-sensitive electroneutral NaCl transport within the CCD may be mediated by the parallel activities of NHE2 and the $\text{SLC26A4 Cl}^-/\text{HCO}_3^-$ -exchanger, but regulatory influences are not fully characterized.^{400,402} However, electroneutral NaCl transport within the CCD is induced by volume depletion and mineralocorticoid treatment.³⁹⁸⁻⁴⁰⁰ This mechanism mediates about 50% of Na^+ reabsorption in the CCD under these conditions without affecting the luminal PD and thus without direct effect on K^+ excretion. Therefore electroneutral, thiazide-sensitive NaCl transport affords the ability to increase Na^+ reabsorption by the CCD without affecting K^+ excretion. The converse occurs after several days of high- K^+ diet, which increases the fraction of ENaC -independent, amiloride-resistant K^+ excretion to about 50%. Again, this presumptively electroneutral, aldosterone-independent pathway for K^+ excretion uncouples distal tubular Na^+ and K^+ excretion.⁵²⁶

WNK -dependent signaling constitutes a major pathway for integrating NaCl and K^+ transport within the distal nephron and is regulated by the differential influence of K^+ intake on circulating AngII , ROMK activity, ratio of WNK1 isoforms, and activity of NCC in the DCT. Thus AngII activates NCC via the WNK4-SPAK pathway, reducing delivery of Na^+ to the CNT and limiting K^+ secretion.^{354,601,602} In contrast, AngII inhibits ROMK activity via several mechanisms including downstream activation of c-src tyrosine kinases.⁵⁸⁴⁻⁵⁸⁶ Whereas K^+ restriction induces renin and circulating AngII , increases in dietary K^+ are suppressive.^{586,603} A decrease in circulating and local AngII partially explains why NCC phosphorylation and activity are downregulated by a high- K^+ diet; teleologically, this serves to increase delivery of Na^+ to the CNT, thus increasing K^+ secretion.³⁵⁹

The DCT functions as a $[\text{K}^+]$ sensor, directly responding to changes in circulating K^+ , rather than to maintain plasma $[\text{Na}^+]$ (see Fig. 6.19), as demonstrated by the observation that acute K^+ -loading dephosphorylates NCC even in Na^+ -restricted mice.⁶⁰⁴ Reduction in potassium intake and/or hypokalemia reduce basolateral $[\text{K}^+]$ in the DCT; the subsequent hyperpolarization of DCT cells is dependent on basolateral KIR4.1 -containing K^+ channels.^{302,314} Hyperpolarization induces Cl^- exit via basolateral CLC-K2 ; the resulting decrease in intracellular Cl^- disinhibits WNK kinases, resulting in phosphorylation of NCC and activation of the transporter by SPAK/OSR1 .³¹⁴ Microperfusion and kidney slice studies have shown that phosphorylation of NCC stimulated by lower extracellular $[\text{K}^+]$ is dependent on decreased intracellular Cl^- .⁶⁰⁵ Along the DCT, WNK4 appears to be the major WNK activating NCC ³³³ and WNK4 is also more sensitive to changes in intracellular chloride than WNK1 or WNK3 .³⁴⁰ The essential role

of this pathway in mediating responses to dietary K^+ restriction is clearly demonstrated by the development of hypokalemia in mice lacking KIR4.1 ,³⁰⁵ WNK4 ,⁶⁰⁶ SPAK/OSR1 ,⁶⁰⁷ and NCC .⁶⁰⁸ (see Fig. 6.18 for summary of model) An alternative mechanism has been proposed in which KLHL3 , part of the ubiquitin ligase complex that degrades WNK4 , is phosphorylated in response to K^+ depletion. This prevents WNK4 binding to the complex, decreasing WNK4 degradation, which promotes NCC activation.⁶⁰⁹

NCC undergoes rapid dephosphorylation in response to K^+ loading^{605,610} through a process mediated by protein phosphatases. Evidence supports primary roles for PP3 (calcineurin) and PP1 .⁶¹⁰⁻⁶¹² Glucocorticoid-induced leucine zipper protein (GILZ) may also play a role in the response to high dietary K^+ by inhibiting SPAK activity and hence NCC . Compared with wild-type mice, GILZ knockout mice are more sensitive to NCC inhibition by thiazides and have greater abundance of phosphorylated NCC . GILZ knockout mice display elevated plasma $[\text{K}^+]$ at baseline, with dietary Na^+ restriction increasing it further. When plasma $[\text{K}^+]$ increases, GILZ -mediated inhibition of NCC may serve to maintain distal Na^+ delivery to ENaC to drive kaliuresis.⁶¹³ Most studies have used diets that are almost completely deficient in K^+ , or with very high levels (e.g., 5% KCl to examine NCC activation status). Relatively small changes in plasma $[\text{K}^+]$ also alter the degree of NCC phosphorylation in vivo, suggesting the role of NCC to maintain plasma $[\text{K}^+]$ is homeostatically important under dietary K^+ levels that do not show extreme deviation from normal.³⁴⁰

Altering NCC activity may affect distal K^+ secretion by inducing remodeling of the CNT, not just by altering Na^+ delivery to ENaC .³³⁶ Mice expressing a constitutively active SPAK mutant specifically along DCT1 display higher NCC activity, and hyperkalemia, which was associated with lower volume of the CNT and lower ROMK abundance, which may limit K^+ secretion. Chronic blockade of NCC increased CNT volume and ROMK abundance and normalized plasma K^+ , suggesting altered Na^+ delivery may induce CNT remodeling.

Finally, in principal cells, increases in aldosterone induce SGK1 , which phosphorylates WNK4 and attenuates the effect of WNK4 on ROMK , while activating ENaC via NEDD4-2 -dependent effects.⁶¹⁴ However, when dietary K^+ intake is reduced, increased AngII leads to stimulation of c-src tyrosine kinase, causing direct inhibition of ROMK activity via tyrosine phosphorylation.^{578,580,615} Higher c-src tyrosine kinase activity also abrogates the effect of SGK1 on WNK4 .^{600,616} While NEDD4-2 can regulate NCC during K^+ restriction,⁶¹⁷ ENaC regulation may be more important. In NEDD4-2 knockout mice, chronic K^+ restriction led to hypokalemia and urinary K^+ wasting that was reversed with the ENaC blocker benzamil.⁶¹⁸ While both insulin and IGF-1 activate ENaC , insulin promotes kaliuresis, whereas IGF-1 exerts an antikaluretic response. Patch-clamp analysis of isolated mouse CCD cells showed that this differential effect on K^+ secretion may be due to differential effects on CLC-K2 (which is expressed in intercalated cells).³⁸³ IGF-1 stimulates CLC-K2 , which may promote net NaCl reabsorption, thus reducing the driving force for K^+ secretion by the CCD. In contrast, insulin inhibits CLC-K2 , which would enhance the generation of the electrogenic drive for K^+ secretion.

Independently of aldosterone, higher K^+ intake rapidly activates mTORC2-SGK1 signaling to activate ENaC.^{554,555} This may involve depolarization of the basolateral membrane by higher interstitial K^+ that promotes WNK1/4 scaffolding and subsequent mTORC activation. mTORC2 may also stimulate ROMK more directly through PKC to sustain K^+ secretion.

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